



Functional polymer–ceramic hybrid coatings: Status, progress, and trend

Zhenqiang Zhang^a, Yinjie Huang^a, Qingyi Xie^a, Guojun Liu^{b,*}, Chunfeng Ma^{a,*}, Guangzhao Zhang^a

^a Faculty of Materials Science and Engineering, South China University of Technology, Guangzhou 510640, PR China

^b Department of Chemistry, Queen's University, 90 Bader Lane, Kingston, Ontario K7L 3N6, Canada

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ABSTRACT

The pursuit of achieving both ceramic-like hardness and polymer-like flexibility in a coating, known as a polymer–ceramic hybrid coating, is a challenging yet highly desirable goal. The application of these coatings spans various domains such as foldable displays, wearable devices, maritime industries, and biomedical engineering. Particularly, endowing polymer–ceramic hybrid coatings with functions such as transparency, anti-liquid adhesion, anti-biofouling, and self-healing expand their potential in fields necessitating highly protective performance, which have gained significant attention in recent years. In this comprehensive review, our main objective is to provide interested readers with a clear framework for assessment and future exploration of this topic. We systematically outline the fundamentals of functional polymer–ceramic hybrid coatings, explaining their fabrication intricacies. Additionally, we explore their practical applications, intricately tailored to the unique requirements of each field. Concluding our review, we address the key challenges facing modern functional polymer–ceramic coatings and propose potential paths for future advancements.

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1. Introduction

Protective coatings are crucial for safeguarding both devices and industrial equipment, finding widespread utility across aerospace, transportation, construction, and electronics [1–4]. Generally, pro-

tective coatings include hard coatings (e.g., metals, ceramics, and composites) and soft coatings (e.g., elastomers and gels). Hard coatings resist physical damage through sufficient hardness and strength, and soft coatings protect coated materials through buffering energy absorption. However, the limitations of hard coatings,

Abbreviations: APMS, γ -aminopropylmethyldiethoxysilane; ATSB, aluminum tri-secapms-butoxide; B₂O₃, boron trioxide; BN, boron nitride; C18, *n*-octadecyltrimethoxysilane; C3-TMOS, *n*-propyltrimethoxysilane; C8-TEOS, octyltriethoxysilane; CA, citric acid; CaCO₃, calcium carbonate; CCl₄, tetrachloromethane; Cou-DCMC, coumaric-*N,N'*-dicyclohexyl-4-morpholinecarboxamidine; COVID-19, coronavirus disease 2019; Cu₂O, cuprous oxide; CV, crystal violet; DA reaction, Diels–Alder reaction; DBP, dibutyl phthalate; DMDEOS, dimethoxydimethylsilane; DPSD, diphenylsilanediol; *E*^{*}, $E^* = E/(1-\nu^2)$; *E*, is the Young's modulus and ν is the Poisson's ratio (calculated from nanoindentation results); *E. coli*, *Escherichia coli*; ECTS, 2-(3,4-epoxycyclohexyl)ethyl-trimethoxysilane; EMSH, cycloaliphatic epoxy-functionalized oligosiloxanes; EPOSS, epoxycyclohexyl polyhedral oligomeric silsesquioxane; EPSQ, ladder-like epoxycyclohexyl polysilsesquioxane; FC₂-COOH, pentafluoropropionic acid; FC₆-COOH, tridecafluoroheptanoic acid; Fe₃O₄, iron oxide; FP, silane-bearing amphiphilic fluorocarbon acrylate-PEG telomers; GAA, glacial acetic acid; GPOSS, octakis(3-glycidylpropyl) polyhedral oligomeric silsesquioxane; GPTMS, 3-glycidylpropyltrimethoxysilane; *H*, hardness (calculated from nanoindentation results); H₂SO₄, sulfuric acid; ITO, indium tin oxide; K₂CO₃, potassium carbonate; KH550, 3-aminopropyltriethoxysilane; KOH, potassium hydroxide; LBL, layer-by-layer; LED, light emitting diode; MHHPA, methylhexahydrophthalic anhydride; MMT, montmorillonite; MOPTMS, methacryloxypropyltrimethoxysilane; MPOSS, octakis(3-methacryloxypropyl); MSCs, mesoporous silica antireflection coatings; MTMS, methyltrimethoxysilane; NIPU, non-isocyanate polyurethane; ODA, 4,4'-diaminodiphenyl ether; OLED, organic light-emitting diode; *P. sp.*, *Pseudomonas sp.*; PAA, polyanion poly(acrylic acid); PDMS, polydimethylsiloxane; PEG, poly(ethylene glycol); PEI, polyethylenimine; PEMA, Poly[ethylene-alternating-(maleic anhydride)]; PET, polyethylene terephthalate; PFOTES, 1*H*,1*H*,2*H*,2*H*-perfluorooctyltriethoxysilane; PHPS, perhydropolysilazane; PI, polyimide; PMDA, pyromellitic dianhydride; PMMA, polymethyl methacrylate; POSS, cage-like polysilsesquioxane; PTFPMS, poly[(3,3,3-trifluoropropyl)methylsiloxane]; PTMS, phenyltrimethoxysilane; PTSA, poly(3-(trimethoxysilyl)propyl acrylate); PVA, poly(vinyl alcohol); PVDFFP, poly(vinylidene fluoride-co-hexafluoropropylene); *S. aureus*, *Staphylococcus aureus*; *S. mutans*, *Streptococcus mutans*; SBS, poly(styrene-*b*-butadiene-*b*-styrene); Si-CPDs, Si-doped carbonized polymer dots; SiO₂, silicon dioxide; TEA, triethylamine; TEOS, tetraethyl orthosilicate; THF, tetrahydrofuran; TMOS, tetramethoxysilane; TMPTA, trimethylolpropane triacrylate; TPDA, 3-(trimethoxysilyl)propyldimethyloctadecyl ammonium chloride; TPOZ, zirconium alkoxide; UPy, 2-Ureido-4[1*H*]-pyrimidinone; UV, ultraviolet; VUV, vacuum ultraviolet; *W_e*, elastic recovery; ZP, epoxy–zirconium nanoparticle; ZrO₂, zirconium dioxide.

* Corresponding authors.

E-mail addresses: gliu@chem.queensu.ca (G. Liu), msmcf@scut.edu.cn (C. Ma).

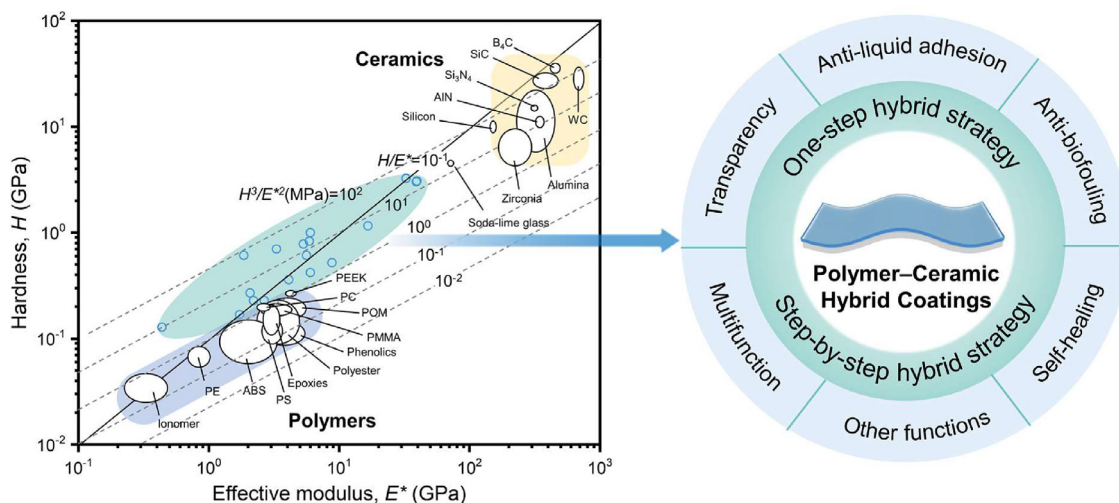


Fig. 1. Classification of polymer-ceramic hybrid coatings and their diverse functions.

including poor deformation ability leading to local stress concentration and susceptibility to cracking, hinder their use in flexible substrates. By contrast, soft coatings exhibit excellent deformation ability, making them suitable for many flexible substrates such as rubber, polyimide (PI), and polyethylene terephthalate (PET), albeit with compromised hardness and strength [5–8]. The simultaneous achievement of ceramic-like hardness and polymer-like flexibility in a coating is inherently challenging but indispensable for foldable displays, wearable devices, maritime industries, biomedical engineering, and other applications [9–11]. Particularly, endowing flexible hard coatings with functions (transparency, anti-liquid adhesion, anti-biofouling, self-healing, etc.) can expand their application in fields requiring highly protective performance. In recent decades, the interest in designing functional flexible hard coatings has grown significantly due to their potential in prolonging the lifespan of coated materials and devices.

Based on the type of the matrix used, numerous flexible hard coatings such as inorganic or metal coatings, organic polymeric coatings, and polymer-ceramic hybrid coatings have been developed [12–29]. For inorganic or metal coatings, the introduction of the soft phase into the hard phase is an effective approach for overcoming intrinsic brittleness [12–16]. However, most coatings are not sufficiently flexible and involve complex processing (e.g., magnetron sputtering and thermal spraying) [17]. Organic polymers feature excellent mechanical flexibility and structural tunability [18–20]. Thus, flexible hard organic polymeric coatings can be prepared by high cross-linking degrees of the network [21–25]. The desired function can be achieved readily by the introduction of functional groups in the backbone structure or side chain [26–29]. Compared with metal alloy or inorganic coatings, organic polymeric coatings are not sufficiently hard. Another promising approach is the organic-inorganic hybrid strategy, which has been widely observed in biological systems. Among them, hard yet tough bone and nacre are familiar examples. The bone consists of organic collagenous proteins and inorganic mineral nanoparticles to form a hierarchical structure [30]. The nacre is a “brick-and-mortar” hybrid structure using organic biopolymer and inorganic aragonite platelets [31]. Inspired by this, the coating encompassing the ceramic-like high hardness and mechanical endurance of inorganic components with the polymer-like low modulus and tunability of organic components (i.e., polymer-ceramic hybrid coating) can be fabricated by using the organic-inorganic hybrid strategy. As shown in Fig. 1, hard coatings, such as ceramic coatings, are typically accompanied by high elastic modulus, which causes

mismatched elastic behaviors when coated on flexible substrates with low elastic modulus, especially polymeric substrates, causing the coating to peel off or crack from the substrate [32]. A high hardness (higher hardness than most engineering plastics) and polymer-like low modulus, that is, a high hardness/elastic modulus (H/E^*) value (diagonal solid lines in Fig. 1) can be attained in the coating using the organic-inorganic hybrid strategy to obtain the co-elastic deformation with the flexible substrate. The value H/E^* represents the selection criterion, which is related to elastic strain resistance to failure as well as resilience [33]. The coating with high H/E^* (close to or greater than 0.1) can distribute the load applied to the coating over a wider area, exhibiting excellent crack resistance [13,34]. Another selection criterion, H^3/E^{*2} is directly related to a surface's resistance against yield pressure [35,36]. Notably, these coatings exhibit a high H^3/E^{*2} value (diagonal dashed lines in Fig. 1) similar to ceramic materials, i.e., excellent resistance to plastic deformation [37]. In addition, the good elastic recovery rate (W_e) of these coatings allows them to undergo reversible deformation. In brief, polymer-ceramic hybrid coatings overcome the intrinsic shortcomings of inorganic/polymeric coatings and exhibit a combination of ceramic-like hardness and wear resistance with polymer-like flexibility and resilience. Furthermore, the molecular architecture of organic components can be tuned in a broad range to obtain polymer-ceramic hybrid coatings with tailor-design functions.

Note that not all organic-inorganic hybrid materials simultaneously display ceramic-like hardness and polymer-like flexibility. The introduction of organic phases can increase the flexibility of inorganic materials, but they are susceptible to a significant decrease in hardness due to the uneven distribution of organic phases or the reduction of cross-linking degree. Judicious designs are needed to combine these apparently incompatible mechanical properties. First, the organic-inorganic hybrid materials require an appropriate volume fraction of inorganic phase to organic one. Second, the organic and inorganic phases need to be hybridized at the nano-scale or even at the molecular-scale, and efficient polymerization is required at the intra-/interphases to achieve a highly cross-linking network, which endows the material with a certain deformability to allow the high stresses to dissipate locally. Therefore, the combination of ceramic-like hardness and polymer-like flexibility is difficult to be combined and requires judicious design.

In the last few decades, considerable effort has been made to develop functional polymer-ceramic hybrid coatings, and many

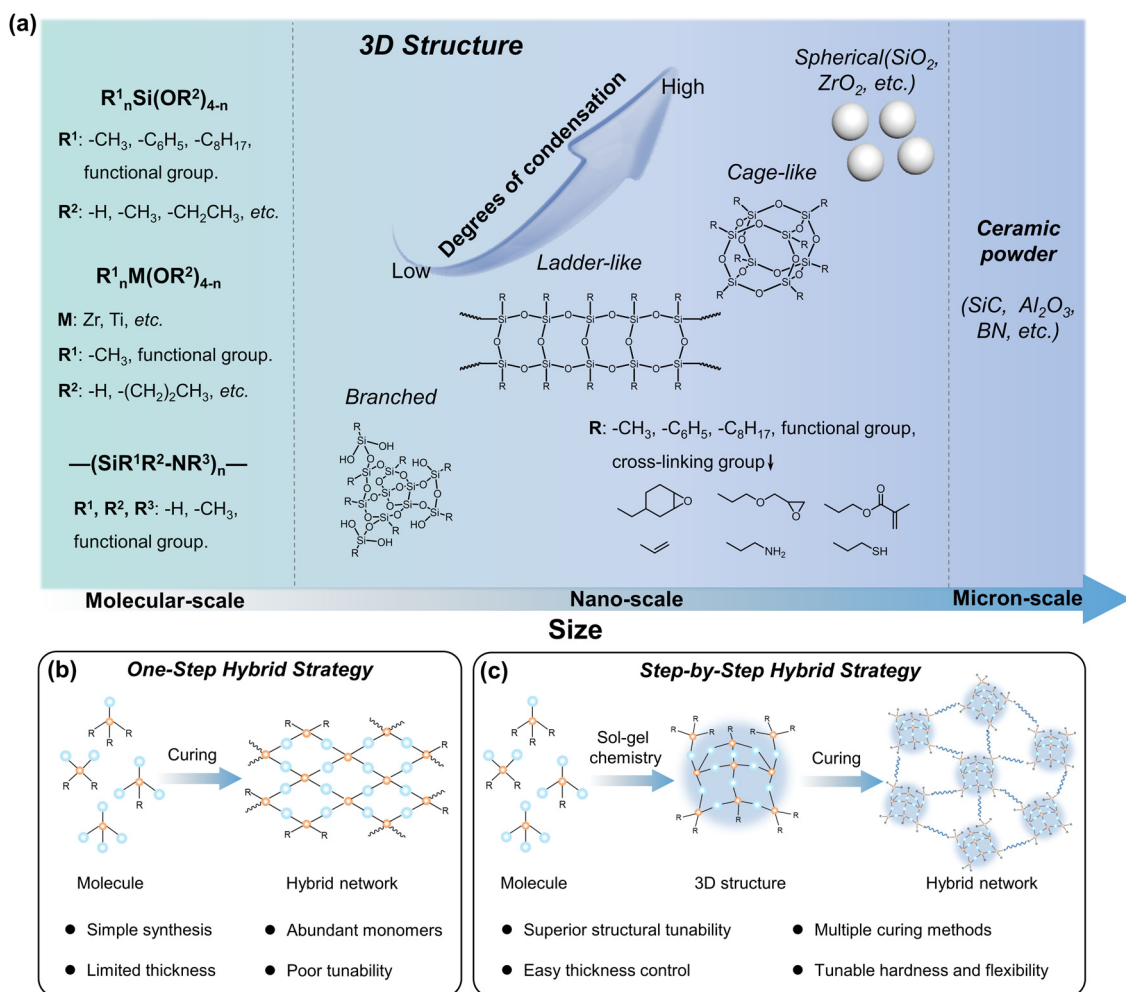


Fig. 2. A summary of (a) the building blocks and representative fabrication strategies of polymer–ceramic hybrid coatings including (b) one-step hybrid strategy and (c) step-by-step hybrid strategy.

excellent works about polymer–ceramic hybrid coatings applied in foldable displays, flexible and wearable electronics, maritime industries, and biomedical engineering, among others, have been published. However, there remains a lack of a comprehensive review on the coating design and the relationships between structure and performance so far. Moreover, the relationship between the coating structure and functions is yet to be systematically defined. Therefore, the design principles of these coatings are urgently needed to be fully discussed to guide their application. In this review, the basic building blocks of the coatings and their preparation techniques are presented. Next, their functions mainly in transparency, anti-liquid adhesion, anti-biofouling, self-healing, and multifunction are systematically reviewed. Additionally, the requisite functions related to each application are discussed comprehensively. Finally, major challenges facing modern functional polymer–ceramic coatings are discussed, and future development directions are proposed.

2. Methodology for the synthesis of polymer–ceramic hybrid coatings

As shown in Fig. 2a, the building blocks can be categorized into the following three primary types according to their size: molecular-, nano-, and micron-scale. Different building blocks correspond to different strategies for fabricating coatings. The general fabrication routes of polymer–ceramic hybrid coatings are clas-

sified into the following two major categories: i) one-step hybrid strategy and ii) step-by-step hybrid strategy (Fig. 2b and c). Molecular-scale building blocks including organosilanes, silicon alkoxides, metal alkoxides, metal chlorides, cyclosiloxanes, or polysilazanes can be used to prepare polymer–ceramic hybrid coatings through a one-step hybrid strategy. Nano-scale building blocks including 3D-structured polysiloxanes (branched, cage-like, ladder-like, or spherical) and some other nanomaterials can be used to prepare polymer–ceramic hybrid coatings through a step-by-step hybrid strategy. Micron-scale particles, such as ceramic powders, can be used as fillers for formulations to prepare flexible hard coatings. Some international companies have commercialized several ceramic-powders-based formulations for obtaining flexible ceramic coatings, such as CeRam-Kote 54® (Freecom, USA) and Atometal (B&B KOREA Co., Ltd., Korea), which have been used for the protection of expensive and critical equipment in petroleum and petrochemical, natural gas, and maritime industries [38,39]. However, in these coatings, a combination of various kinds of micron-scale particles is used. Micron-scale particles easily agglomerate and exhibit poor structural tunability, which are merely involved in some product introductions. There are few relevant scientific research papers using micron-scale particles as building blocks. Actually, the organic–inorganic hybrid materials need to have some kind of bonding (covalent bond or supramolecular assembly) between the polymer and the inorganic part. Physically filling with inorganic particles into a polymer cannot be called a

“polymer–ceramic hybrid coating”, and this type of coating is not included in this review.

This section is focused on structural design strategies using molecular-/nano-scale building blocks including i) one-step hybrid strategy and ii) step-by-step hybrid strategy. The effects of functional groups on their mechanical properties are discussed comprehensively.

2.1. One-step hybrid strategy

2.1.1. One-step hybrid strategy based on the sol–gel reaction

Based on the sol–gel reaction, small molecules such as organosilanes, silicon alkoxides, and metal alkoxides can be directly used to form a hybrid network through a one-step hybrid strategy (Fig. 2a and b). The earliest use of sol–gel reaction can be traced to 1864 by Graham, and then the reaction was tailored by Warner Stöber [40,41]. The sol–gel reaction has been developed for more than 100 years and allows the efficient integration of asymmetric properties of organic and inorganic components into a material, which has been proven to be feasible in creating various shapes, such as coatings, uniform particles, aerogels, monoliths, and fibers [42,43]. This processing is simple and does not rely on any special instruments or harsh experimental conditions, which is favourable for the introduction of organic components because they are typically thermally unstable. Generally, alkoxysilanes, silicon alkoxides, or metal alkoxides can utilize the hydrolytic sol–gel reaction to hydrolyze under acidic-catalyzed or base-catalyzed conditions and subsequently condense with other hydroxyl and alkoxy groups. Chlorosilanes, hydroxyl silanes, or metal chlorides can condense directly via nonhydrolytic sol–gel reaction [44]. Through the sol–gel reaction, the aforementioned molecules are first formed into sols with nano sizes. Then the sol is applied to the substrate. Through heat treatment, the sol further develops into a liquid-phase interconnected gel, while the liquid phase in the gel is removed, resulting in a smooth coating [45].

The chemical structure of organosilane is particularly critical in the mechanical properties of coatings. Typically, through the hydrolytic condensation reactions of alkoxy groups, alkoxysilanes can form a high degree of cross-linking siloxane network by themselves or be introduced into the inorganic network formed by metal alkoxides. Only alkoxysilanes with three or four alkoxy groups can react with each other to form a 3D network, while alkoxysilanes having one or two alkoxy groups cannot form a 3D network. On the other hand, utilizing the different R^1 groups, such as alkane groups and functional groups, can enable the hybrid coating to exhibit different mechanical properties and functionality [46]. Note that one step sol–gel method is hard to obtain “polymer–ceramic hybrid coatings”, they need to incorporate silane with long chains or soft micelles to endow the material with a certain deformability to allow the high stresses to dissipate locally. For example, pure tetramethoxysilane (TMOS) coating could be reacted to form a hard inorganic silica network, and the corresponding film was easily cracked and poorly adherent (Fig. 3a and b). The introduction of *n*-propyltrimethoxysilane (C3-TMOS) or octyltriethoxysilane (C8-TEOS) with a flexible octyl group could apparently improve its flexibility and provide low-surface-energy [47,48]. In particular, the facile sol–gel reaction between C8-TEOS and TMOS could be used to fabricate a hybrid coating with low critical surface energy for biofouling release. The settlement of biofouling organisms could be released by the water shear force coming from the ship's navigation or mechanical cleaning (Fig. 3c). Unfortunately, due to the lack of biofouling resistant moieties, such coatings cannot prevent the adsorption of biofouling organisms. Meanwhile, these coatings exhibited limited flexibility. To overcome these shortcomings, our group introduced silane-bearing amphiphilic fluorocarbon acrylate-poly(ethylene glycol) (PEG) telomer

(FP) to endow the hybrid coating with biofouling resistance and improve its toughness (Fig. 3d) [49]. The effect of the FP on the mechanical properties of coatings was investigated in detail, and it was found that the introduction of FP remarkably increased its hardness and elastic modulus instead of degenerating these properties because of the self-enrichment of the rigid fluorocarbon moieties of FP. At the same time, the soft PEG portion of FP considerably improved the flexibility of the coating through plasticization. After 100 outward bending cycles, the surface of HC-FP-15 (50- μ m thick) remained intact, whereas numerous cracks appeared on the surface of HC-FP-0 (50- μ m thick) (Fig. 3e).

In addition to using simple organosilane molecules containing trifunctional silane units to participate in hydrolytic polycondensation, organosilanes containing difunctional silane units can form a linear chain with each other and combine with other organosilanes or metal alkoxides through the sol–gel process and provide the desired functions. For example, dimethoxydimethylsilane (DMDEOS) has been widely used to form a surface monolayer with omniphobicity, but this surface is vulnerable to external impacts [50]. Meanwhile, DMDEOS also can form a soft PDMS chain to improve the flexibility of coatings. Thus, a hard coating (8H pencil hardness) with anti-liquid adhesion ability was fabricated through the sol–gel reaction between TEOS (tetraethyl orthosilicate) and DMDEOS under acid catalysis [51]. Besides, some organosilanes with rigid groups, such as phenyl groups, can enhance hardness. By carefully tailoring the molar ratios of phenyltrimethoxysilane (PTMS) and DMDEOS based on sol–gel reaction, the *H* of resulting coatings increased from $\sim$ 12 MPa (PTMS/DMDMS = 1:9) to $\sim$ 218 MPa (PTMS/DMDMS = 6:4) [52].

It should be noted that the above coatings all exhibited good substrate adhesion characterized by the pull-off adhesion test or tape adhesion test. This is mainly attributed to the hydroxyl groups of these precursors can interlink the hybrid network to the substrate by reacting with the hydroxyl groups on substrate surfaces or forming strong physical hydrogen bonding interaction.

Alumina and zirconia networks exhibit different properties compared to silicon oxide networks. For example, alumina networks have better thermal stability and adsorption properties, while zirconia networks are more robust and have a high refractive index. The introduction of alumina or zirconia networks into a polymer–ceramic hybrid coating can endow the coating with some desired properties. Utilizing the one-step hybrid strategy based on the sol–gel reaction, various metallic oxides (e.g., aluminum oxide and zirconium oxide) or multicomponent oxide systems (e.g., Al–O–Si and Zr–O–Si) have been introduced to the polymer–ceramic hybrid coating. Notably, because of the significant difference in hydrolysis and condensation rates between silane coupling agents and transition metal alkoxides, using chelating ligands is necessary for controlling the hydrolysis and condensation kinetics of metallic precursors and preparing homogeneous mixtures [53,54]. For example, utilizing the sol–gel chemistry between methyltrimethoxysilane (MTMS), 3-glycidyloxypropyltrimethoxysilane (GPTMS), and aluminum tri-*sec*-butoxide (ATSB), a dense Al–O–Si hybrid network was obtained to enhance hardness and flexibility (Fig. 4a) [55]. In this system, acetylacetone was used to chelate with aluminum to control the hydrolysis and condensation reactions of ATSB. Particularly, an appropriate Al concentration is important for forming a uniform transparent coating at room temperature. An Al concentration of 5% could be induced to form uniform alumina as well as siloxane networks, but with up to 10% of the Al concentration, the network crystallized, and produced particles, which resulted in a heterogeneous coating. This phenomenon could be attributed to the strong Al oxidation behaviour and low chelation ratio [56]. The –OH group of the network formed strong hydrogen bonds with the N in the crystal violet (CV, photosensitizer) to uniformly

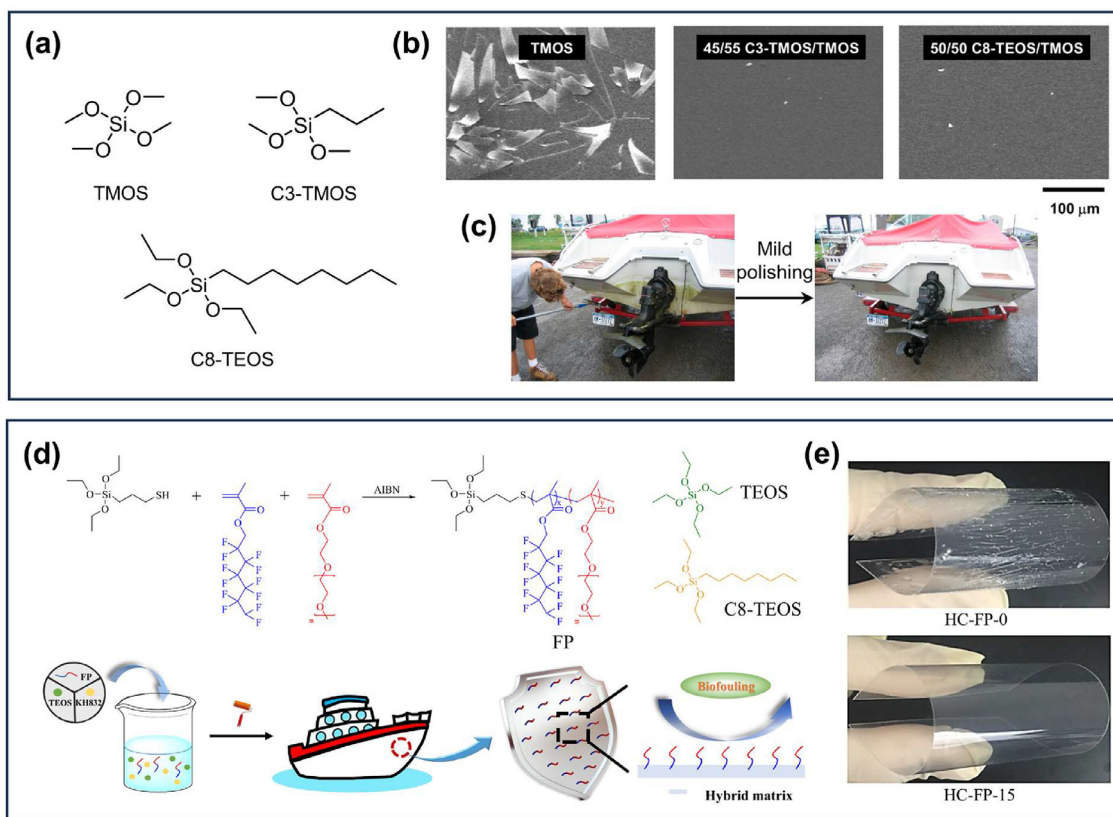


Fig. 3. (a) Chemical structures of TMOS, C3-TMOS, and C8-TEOS. (b) Scanning electron microscopy (SEM) images of pure TMOS, 45/55 C3-TMOS/TMOS, and 50/50 C8-TEOS/TMOS coatings. [47], Copyright 2015. Adapted with permission from John Wiley & Sons Inc. (c) The coated boat pulling from the Irondequoit Bay was subjected to mild polishing. [48], Copyright 2014. Adapted with permission from American Chemical Society. (d) Synthesis of telomer FP and hybrid coating. (e) Photographs of HC-FP-0 and HC-FP-15 coated on PET in the flexibility test. Adapted with permission. [49], Copyright 2020. Adapted with permission from Royal Society of Chemistry.

immobilize CV in the coating, achieving excellent antibacterial performance against *S. epidermidis* by generating reactive oxygen species under illumination (Fig. 4b). The resulting hybrid coating (7.1-μm thick) exhibited an outward bending radius of 0.5 mm and a pencil hardness of 5H (Fig. 4c).

ZrO₂ is a well-known inorganic material for its excellent mechanical robustness and chemical stability and can also be utilized to fabricate polymer–ceramic hybrid coatings. For example, transparent hard zirconia-based hybrid films were prepared utilizing the one-step sol–gel hybrid strategy of zirconium tetrapropoxide and branched-stearic acid [57]. Due to the thermal stability of branched-stearic acid (maximum of 200 °C), the hybrid coating (~340 nm thick) could maintain dynamic oleophobicity toward n-hexadecane even increasing thermal curing temperature from 100 °C to 200 °C, while the *H* markedly increased from 1.95 GPa to 3.03 GPa.

Multicomponent Zr–O–Si systems can be fabricated through a one-step sol–gel hybrid approach. Through a one-step sol–gel hybrid approach between zirconium alkoxide (TPOZ) and GPTMS, a glassy matrix with an average *H* of approximately 0.4 GPa was prepared [58]. To prevent precipitation, glacial acetic acid (GAA) was used as a metal-centric chelator of zirconium precursor. Meanwhile, nanocellulose fibrils were introduced into the glassy matrix as a stiffening and toughening phase (Fig. 4d). Due to the hydrophilicity of colloidal nanocellulose fibrils, nanocellulose could disperse homogeneously into the aqueous sol–gel precursors from 5 to 80 wt%. After heat curing, cellulose nanofibrils formed strong molecular-scale chemical interactions between polar groups of the glassy matrix, resulting in an apparent improvement in coating hardness and fracture energy (Fig. 4e). The effect of nanocellulose on the hardness of coatings was investigated and it was found that

the hard coating (130-μm thick) loading with 20 wt% nanocellulose also showed the best *H* (~1.1 GPa), flexibility, and no crack was observed on the coated poly(ether imide) film after subjecting to 20,000 bending cycles with a radius of 10 mm (Fig. 4f).

Although the organic groups of the organosilane precursor can be tuned in a broad range to obtain the target function [9], the above functional moieties should be end-capped with silanes or silanol groups that can participate in the sol–gel reaction; otherwise, the functional moieties can only be introduced through physical blending. Besides, the one-step hybrid strategy based on the sol–gel reaction is typically required to have an additional thermal process to increase the degree of condensation and thus form a dense inorganic SiO_x network, which limits its application on heat-sensitive substrates. The coatings that do not undergo a thermal process remain unreacted polar hydroxyl groups on the surface, which affects their application in flexible electronic devices, such as semiconductor devices [59,60].

2.1.2. One-step hybrid strategy based on cyclosiloxanes

In addition to using organosilane as a precursor, cyclosiloxane hybrid polymer, a ceramic precursor, can be utilized to fabricate polymer–ceramic hybrid coatings. The cyclosiloxane hybrid polymer is synthesized through the polymerization (e.g., hydrosilylation and Piers–Rubinsztajn reaction) of cyclosiloxane monomers [61,62]. For example, through the platinum-catalyzed hydrosilylation reaction between 1,3,5,7-tetramethylcyclotetrasiloxane (D₄H) and 1,3,5,7-tetravinyl-1,3,5,7-tetramethylcyclotetrasiloxane (D₄V), a highly crosslinked cyclosiloxane hybrid polymer with a *H* of approximately 0.34 GPa and a *E** of approximately 1.69 GPa was obtained [63]. This reaction used parts per million platinum atoms as the catalyst and had no side products. Cyclosiloxane hybrid

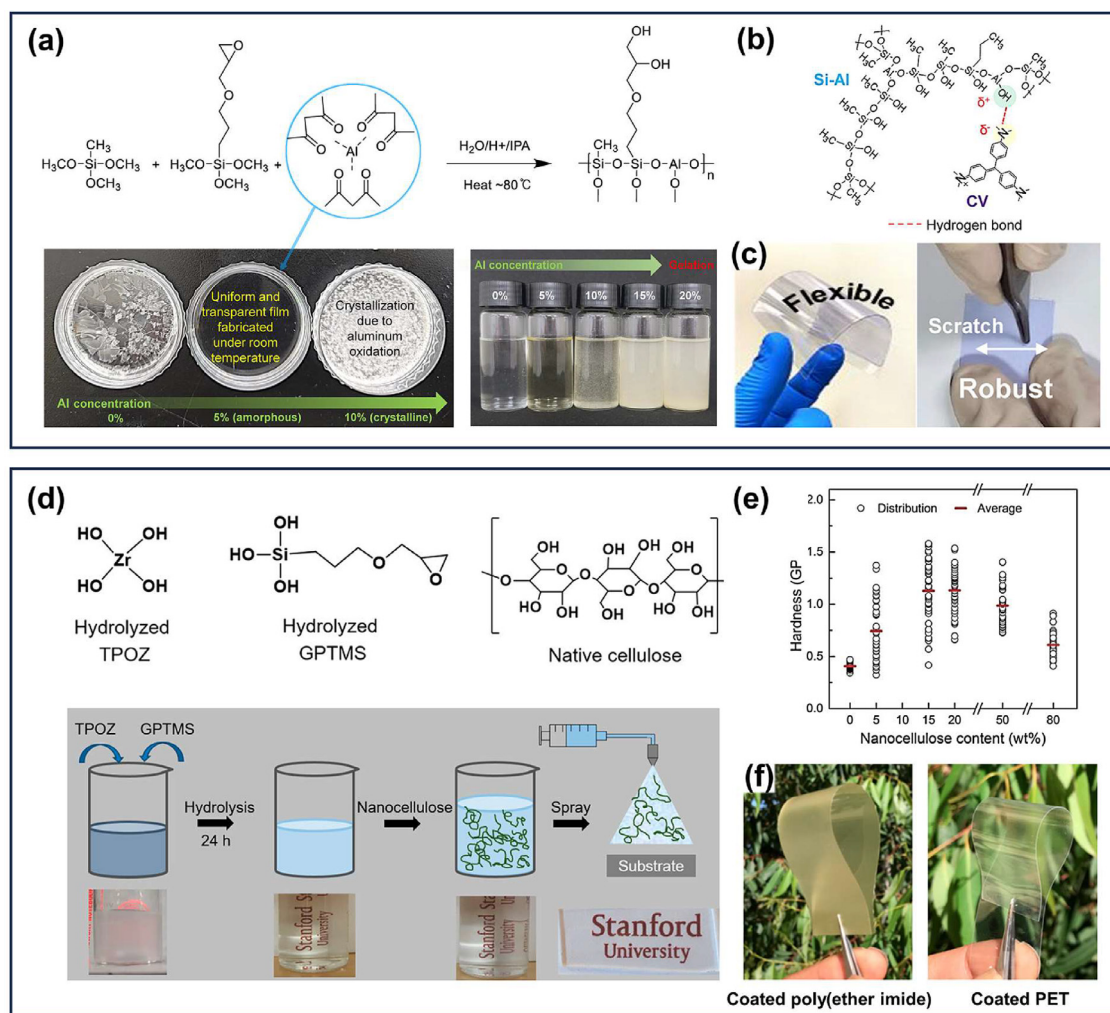


Fig. 4. (a) Synthesis of Si-Al films. (b) Schematic illustration of the proposed mechanism of CV immobilization by hydrogen bonding. (c) Photographs of Si-Al@CV films in the bending test and pencil hardness test. [55], Copyright 2022. Adapted with permission from American Chemical Society. (d) Chemical structures of hydrolyzed TPOZ, hydrolyzed GPTMS, and native cellulose. (e) The relationship between coating hardness and nanocellulose content. (f) Photographs of coated poly(ether imide) and PET films. [58], Copyright 2018. Adapted with permission from American Chemical Society.

polymer also has better thermal stability than linear polysiloxanes due to the reduction in “back-biting” of linear siloxane chains [64]. Meanwhile, it has a hard yet brittle hierarchical structure composed of assembled layers with varying orientations and the thickness of each layer is tens of nanometers, which indicates that each layer consists of several crosslinked layers of cyclosiloxane atomic layers [65]. Inspired by nacre structures with a “brick-and-mortar” hybrid structure, vinyl-terminated PDMS was incorporated to toughen materials through hydrosilylation with dangling Si-H in the layered structures of the cyclosiloxane hybrid polymer (Fig. 5a) [66]. With the increase in the PDMS content, the hardness and toughness increased because the homogeneously dispersed soft PDMS chain chemically bonded with the cyclosiloxane hybrid polymer. However, a subsequent increase in the PDMS content (0.8 wt%) resulted in the formation of phase separated PDMS particles, which reduced hardness and Young’s modulus. Besides, functional hyperbranched carboranes homogeneously anchored in cyclosiloxane networks could generate uniform B_2O_3 protective films upon reacting with oxygen in the high-temperature environment (1000 °C, air atmosphere), improving thermal-oxidation stability and toughness (Fig. 5b) [67].

Note that the cyclosiloxane hybrid polymer, as a novel emerging building block of polymer-ceramic hybrid coatings, is thermally stable but brittle. Functional applications are limited, and further

exploration is necessary, especially for high-temperature resistant coating.

2.1.3. One-step hybrid strategy based on polysilazanes

Polysilazane, as another ceramic precursor, can be used to fabricate polymer-ceramic hybrid coatings by a heat-assisted hydrolysis and condensation reaction or a photocleave and oxidation reaction induced through vacuum ultraviolet (VUV) irradiation. Polysilazanes consist of alternating Si and N atoms, such that the sites of Si and N atoms are substituted by various reactive groups (hydrogen, vinyl, trialkoxysilyl, etc.), functional groups, or inert methyl groups [68]. Polysilazane coatings can be converted into inorganic ceramic coatings at high temperatures, which is only suitable for certain applications. By contrast, VUV irradiation can be utilized to convert perhydropolysilazane (PHPS) into inorganic SiO_x with high thermal stability at room temperature within minutes [69]. Furthermore, functional organic polymers with the -OH group, such as PEG, polydimethylsiloxane (PDMS), perfluoropolyether, and trifluoroethanol can react with Si-N/Si-H bonds in polysilazanes [70]. For example, dihydroxyalkyl-terminated PDMS was grafted into PHPS through the reaction of -OH groups and Si-N/Si-H bonds for improving toughness and thermal stability (Fig. 6a) [71]. With the increase in the ratio of PDMS in SiO_x , the phase of the hybrid coating transitioned from a sea-land phase to a PDMS-dominated

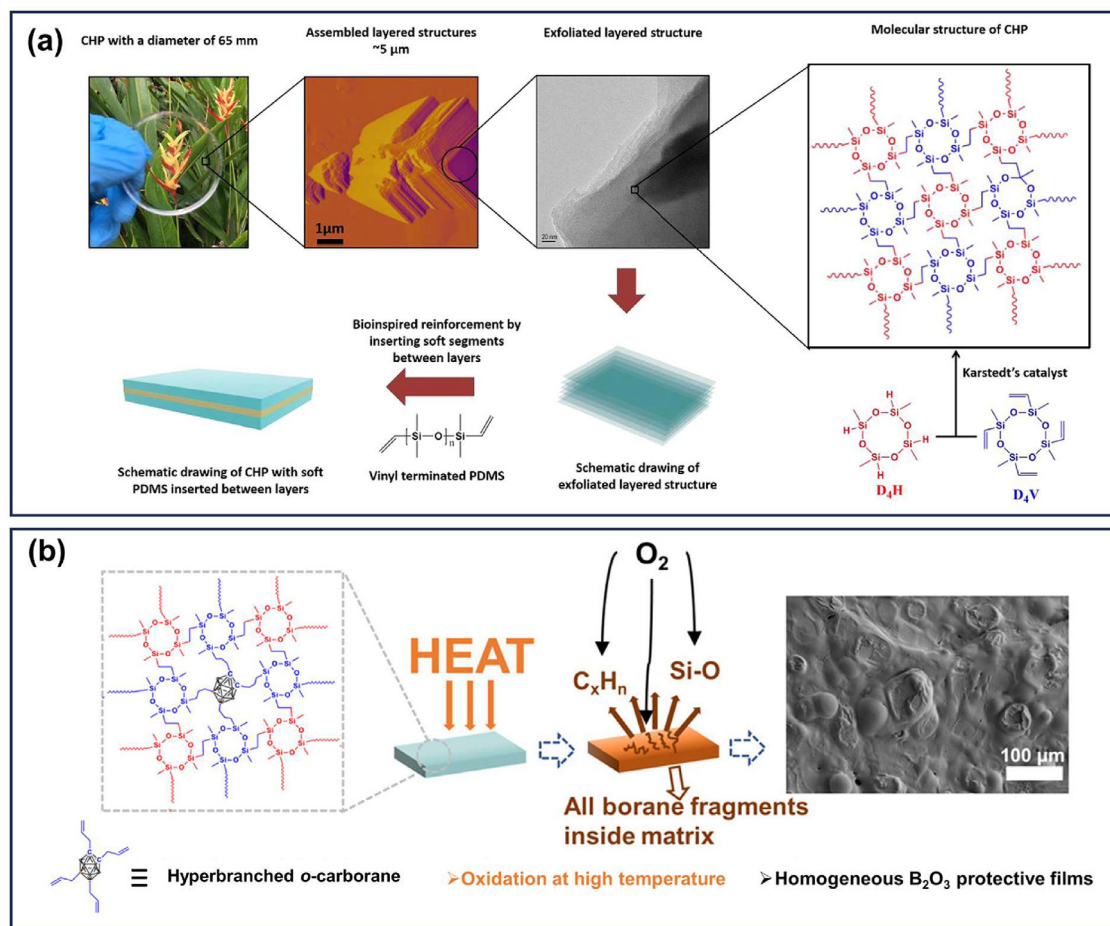


Fig. 5. (a) Hierarchical structure of the cyclosiloxane hybrid polymer and schematic illustration of the cyclosiloxane hybrid polymer with PMDS inserted between layers. [66], Copyright 2019. Adapted with permission from Elsevier Science Ltd. (b) Mechanism of surface morphogenesis of the cyclosiloxane network containing hyperbranched o-carborane. [67], Copyright 2023. Adapted with permission from American Chemical Society.

phase structure (Fig. 6b). Correspondingly, the hardness and elastic modulus decreased, while the H/E^* and W_e increased, which indicated improved toughness (Fig. 6c). The optimized nanocomposite coating (~1.2 μm thick) exhibited a H of approximately 3.07 GPa and a E^* of approximately 39.40 GPa, which was considerably higher than that of polymeric materials (Fig. 6d). The coating exhibited polymer-like flexibility (Fig. 6e), which could be bent with a radius of 2 mm for 10,000 times without noticeable cracks. Furthermore, the PDMS could self-enrich on the surface to obtain an antiadhesive coating toward water, ice, oil, as well as bacteria (Fig. 6f). Besides, Si–N/Si–H bonds in polysilazanes could react with the –OH group of the polar substrates to form strong chemical bonding and thus realize excellent substrate adhesion. Overall, polymer–ceramic hybrid coatings derived from polysilazane exhibit higher hardness and modulus than the coating derived from organosilane. This is attributed to the continuous inorganic phase formed by the highly efficient photocleave and oxidation reaction under VUV irradiation. This work provides an interesting way to construct highly hard yet flexible coatings by combining organic polymer with polysilazane.

In addition, Si–N and Si–H are moisture sensitive groups, so polysilazane can be cross-linked under humid atmosphere via hydrolysis and condensation. By using catalysts such as ammonia, fluoride, and tetrabutylammonium chloride, to accelerate the reaction rate, the curing time can be further reduced [72]. Some polysilazane-based polymer–ceramic hybrid coatings, such as polymethyl(hydro)/polydimethylsilazane resin have been commercialized.

2.2. Step-by-step hybrid strategy

The step-by-step hybrid strategy is to first prepare functionalized nano-scale building blocks and then cure them by polymerization to obtain coatings. Thus, compared with the one-step hybrid strategy, polymer–ceramic hybrid coatings prepared by the step-by-step hybrid strategy exhibit multiple functionalization and cross-linking methods (Fig. 2c). Polysiloxanes with different three-dimensional structures, i.e., stereoscopic polysiloxanes, can be prepared by combining different organosilanes or by regulating the reaction rate and the reaction degree between the hydrolysis and condensation reactions of organosilanes. Depending on the resulting inorganic SiO_x structure of stereoscopic polysiloxanes, they can be categorized as branched, cage-like, ladder-like, or spherical [73,74]. These 3D-structured polysiloxanes have a rigid inorganic core and organic periphery, in which the organic moieties of the organic periphery can be used for subsequent functionalization or curing by polymerization [75]. Controlling the final structure of polysiloxanes requires careful tailoring to avoid possible side reactions such as cyclization, gelation, and excessive cross-linking. The adjustable 3D-structured polysiloxanes and cross-linking methods enable a high degree of molecular-scale bonding of inorganic and organic phases, resulting in superior flexibility while maintaining high hardness [76–82]. Table 1 lists the specific polymerization methods, which will be reviewed in the following sections. In these systems, thermal or photopolymerization using cross-linking groups are two primary methods. In addition, for use as a coating in practice, reliable adhesion strength toward various

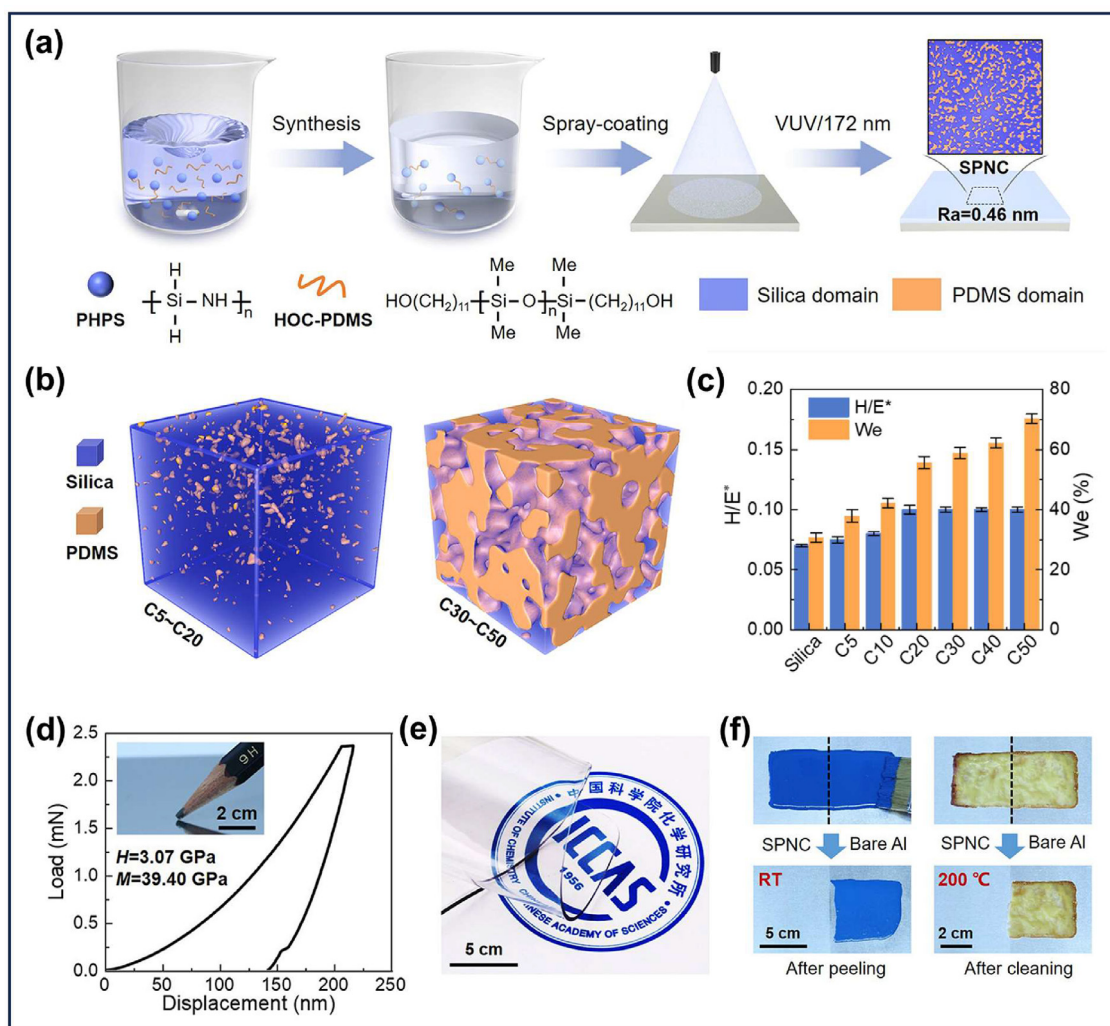


Fig. 6. (a) Synthesis of nanocomposite coatings and (b) their transition form. (c) The H/E^* and We of coatings at different PDMS contents. (d) Nanoindentation curve of the nanocomposite coating. Photographs of (e) transparency and (f) antiadhesion abilities of the coating. [71], Copyright 2023. Adapted with permission from American Chemical Society.

Table 1

Summary of polymerization methods for fabricating polymer–ceramic hybrid coatings.

Organic groups	Curing condition	Detailed polymerization	Ref.
Epoxy	Photo	Cationic	[83]
	Thermal (25 °C)	Epoxy-amine	[84]
	Thermal (120 °C)	Cationic	[85]
	Thermal (120 °C)	Epoxy-anhydride	[86]
Methacrylate	Photo	Free-radical	[87]
	Photo	Thiol-ene	[88]
Vinyl	Photo	Thiol-ene	[89]
	Thermal (200 °C)	Free-radical	[78]
Hydrogen	Thermal (150 °C)	Hydrosilylation	[90]
Thiol	Photo	Thiol-ene	[91]
Cyclic carbonates	Thermal (80/100 °C)	Cyclic carbonates-amine	[92]
Alkoxy-silyl	Thermal (80/120 °C)	Polycondensation	[93]
Amine	Thermal (360 °C)	Amine-anhydride	[94]
Dienophile and diene	Thermal (90 °C)	Diels-Alder	[95]

substrates, especially flexible substrates (e.g., PET and PI), should be considered. Based on the step-by-step hybrid strategy, excellent substrate adhesion can be obtained by tuning organic components. 3D-structured polysiloxanes exhibit various other advantages such as nano-scale size, good dispersion, and good thermal resistance. Particularly, in combination with a polymer matrix or structure at the nanometric dimensions, the final hybrid material can exhibit synergistic properties with each component. There-

fore, polymer–ceramic hybrid coatings prepared through various 3D-structured polysiloxanes or cross-linking methods have been reported [75,78,83–125].

2.2.1. Step-by-step hybrid strategy based on branched oligosiloxanes

Oligosiloxanes can be derived from sol-gel reactions between single organosilanes or multiple types of organosilanes. As mentioned above, organosilanes can form small hybrid units in the

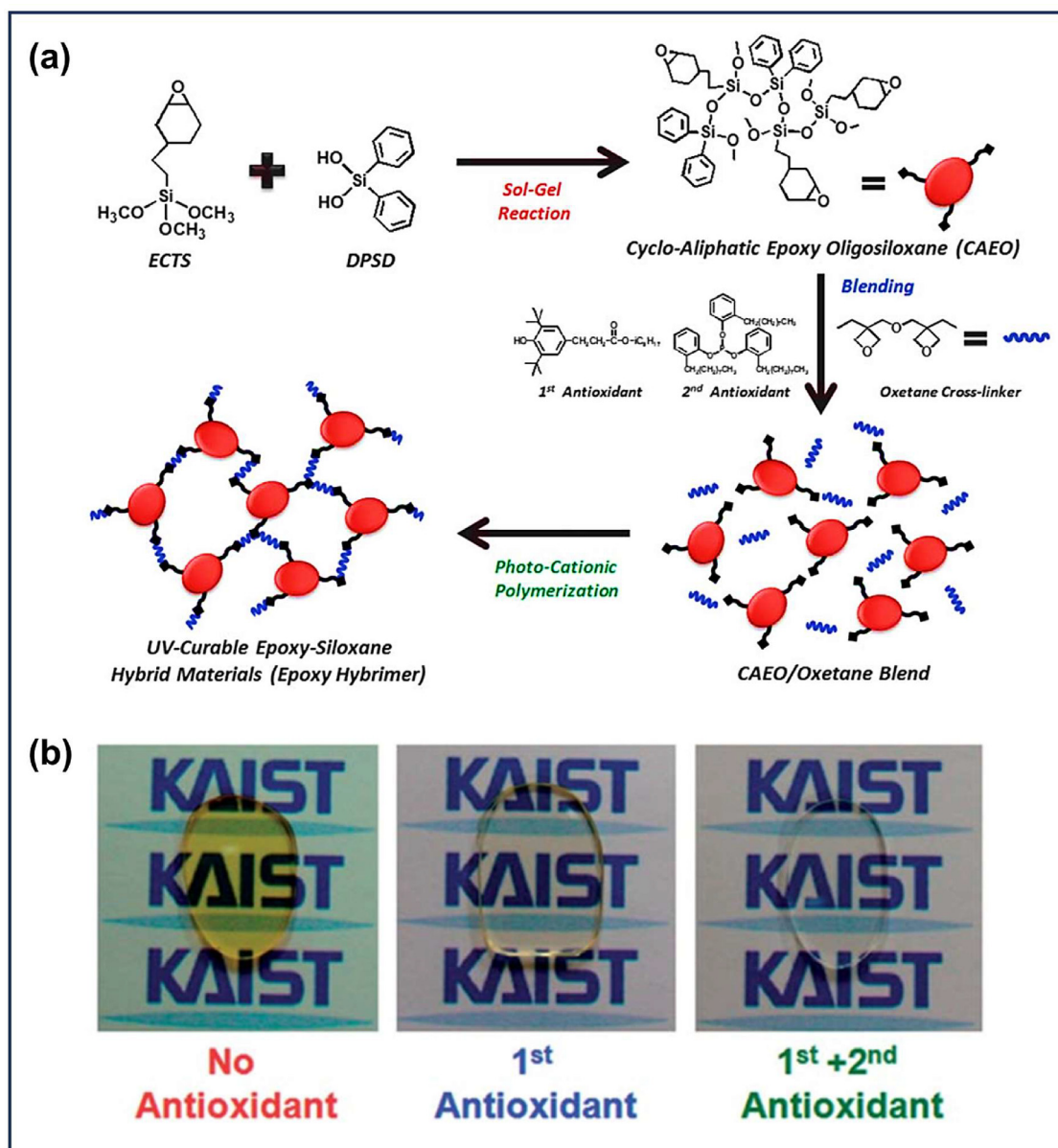


Fig. 7. (a) Synthesis of UV-curable epoxy-siloxane hybrid materials. (b) Photographs of the hybrid materials with/without antioxidants. [99], Copyright 2012. Adapted with permission from Royal Society of Chemistry.

liquid phase through the sol-gel reaction. Changing some variables including ratios of raw materials, catalysts (acidic or basic), and solvent can have a significant impact on the kinetics of the formation of individual reaction products, and thus on the final product.

Methacrylate-functionalized branched oligosiloxanes with randomly oriented structures were developed using hydrolytic sol-gel reactions under an acidic catalyst [96]. In this system, the hydrolysis reaction was preferentially promoted rather than the condensation reaction using an acidic catalyst, and the condensation reaction proceeded slowly. Therefore, branched oligosiloxanes remained a large amount of silanol groups, which exhibited better intermolecular compatibility with polar polymer compared to ladder-like polysiloxanes with small amount of silanol groups. On the other hand, cycloaliphatic epoxy-oligosiloxanes with the random branched structure were synthesized through the non-hydrolytic sol-gel condensation of 2-(3,4-epoxycyclohexyl)ethyltrimethoxysilane (ECTS) and diphenylsilanediol (DPSD) [97]. The

highly efficient non-hydrolytic sol-gel reaction left only a small amount of Si-OH and unreacted precursors, which avoided the self-crosslinking of the resin and improved the shelf stability. Then the cycloaliphatic epoxy groups underwent UV-induced ring-opening polymerization to form a hybrid coating for the encapsulation of organic light-emitting diodes (OLEDs). However, the high viscosity of epoxy-oligosiloxanes resin restricted the mobility of the cycloaliphatic epoxy groups and thus affected the efficiency of UV-initiated cationic ring-opening polymerization. The introduction of di-functional oxetane molecules into epoxy-oligosiloxanes reduced the viscosity of the resin and accelerated the curing of epoxides because of their greater basicity. As a result, the epoxy-oligosiloxanes/di-functional oxetane resin exhibited a faster photocationic polymerization rate and a higher cross-linking density than those of pure epoxy-oligosiloxanes (Fig. 7a) [98,99]. Blending them with antioxidants considerably increased thermal resistance against yellowing (Fig. 7b). Such hybrid materials could be used as light emitting diode (LED) encapsulants without

affecting the electroluminescent intensity. Notably, epoxy groups that underwent UV-induced ring-opening polymerization generated hydroxyl groups, which could impart the coating with good substrate adhesion.

Hyperbranched polysiloxanes have been widely used in the fabrication of high-performance materials because of their bulky hyperbranched architecture, high functionality, low viscosity, and good solubility. For example, hyperbranched polysiloxanes were prepared through the sol-gel reaction of γ -aminopropylmethyldiethoxysilane (APMS), PTMS, and TEOS [94]. Hyperbranched polysiloxanes containing numerous amino groups could be introduced into a polyimide network through polymerization. The introduction of hyperbranched polysiloxanes enhanced internal free volume and the dielectric confinement effect, which resulted in ultralow dielectric permittivity and simultaneously maintained the excellent mechanical properties of PI films. Amine-functionalized hyperbranched polysiloxanes could endow the material with self-healing properties and maintain high mechanical properties by forming dynamic vinylogous urethane groups [100]. In addition, hyperbranched polysiloxanes containing the Si-O-C network could be prepared by the transesterification polycondensation reaction [101]. This method provides a platform for the combination of silane coupling agents and polyols. Some resins have incorporated these Si-O-C hyperbranched polysiloxanes for improving toughness and decreasing the dielectric constant [102]. Thus, the use of hyperbranched polysiloxane as a building block is expected to fabricate high-performance polymer-ceramic hybrid coatings.

2.2.2. Step-by-step hybrid strategy based on ladder-like polysilsesquioxanes

In contrast to the randomly structured oligosiloxanes, ladder-like polysilsesquioxanes exhibit notable distinctions. These include elevated levels of condensation, substantial molecular weight, minimal unstable Si-OH groups, and consistent formations. These characteristics render them as compelling candidates among nanomaterials for the fabrication of polymer-ceramic hybrid coatings [103].

Ladder-like polysilsesquioxanes are mainly synthesized under base-catalyzed conditions, which can promote the deprotonation of silanol before the condensation step for the nucleophilic attack of partially positive Si atoms [104]. It is reported that ladder-like polysilsesquioxanes have been developed via hydrolytic polycondensation of ECTS under the conditions of solvent-free and strongly alkaline aqueous solution catalysis [83]. Numerous epoxy-cyclohexyl groups of ladder-like polysilsesquioxanes were used to form a highly crosslinking hybrid network via UV-initiated cationic ring-opening polymerization (Fig. 8a). Nevertheless, the growth of cycloaliphatic epoxy chains was readily isolated in the polymeric matrix due to tertiary oxonium ions (active center) becoming trapped and obstructing the formation of a highly cross-linking network [105,106]. To increase the epoxy conversion, the coating was exposed to 85% relative humidity at 85 °C for 2 h after photocuring (Fig. 8b). In this process, water molecules, as chain-transfer agents, could penetrate the whole reaction-limited matrix and react with tertiary oxonium ion to produce a hydroxyl group and an acidic proton, which the latter could reinitiate a neutral epoxide to produce a new secondary oxonium ion. These new secondary oxonium ions could participate in the chain propagation via the active monomer mechanism (Fig. 8c) [107,108]. Meanwhile, the diffusion of water molecules avoided the volume shrinkage of the coating. Thereby, the hard inorganic siloxane was chemically bonded at the maximum high level using flexible organic epoxy tethers as a bridge. The ladder-like polysilsesquioxane had a high degree condensation of approximately 97%, and the resulting hybrid coatings (50- μ m thick) displayed a H of approximately

0.78 GPa, a E^* of approximately 5.31 GPa, and a W_e of approximately 88.5% (Fig. 8d and e). The coating exhibited $H/E^* \geq 0.1$ and $W_e \geq 60\%$, which are satisfied with the key parameters of flexible hard nanocomposite coatings [33]. Even with inward bending (a bending radius of 1 mm) for more than 10,000 cycles, no cracks or permanent deformation were observed on the surface, and the coating did not separate from the substrate. The coated PET film could undergo the wear of steel-wool without visual scratches, revealing excellent wear resistance (Fig. 8f-h). The effect of ladder-like polysilsesquioxane molecular weight on the hardness of UV-cured coatings has been investigated based on the UV-curable epoxy [109]. The results revealed that the molecular weight of the product could be further increased by removing byproducts (e.g., alcohols) during the reaction and the final hardness and modulus of the coating increased with the increase in the molecular weight.

Another controlled and facile method for synthesizing ladder-like polysilsesquioxanes has been reported [110]. By the hydrolytic polycondensation of methacryloxypropyltrimethoxysilane (MOPTMS) and PTMS at room temperature using water/tetrahydrofuran mixture as the solvent and K_2CO_3 as the weak base catalyst, UV-curable ladder-like poly(phenyl-comethacryl)silsesquioxanes were synthesized [111]. In this system, K_2CO_3 was chosen because it has a pH comparable to the strong base KOH, with more potassium ions promoting the condensations of silanol groups than KOH. Besides, tetrahydrofuran, as a polar aprotic solvent, has a high polarity to accelerate hydrolysis rates, and its aprotic properties render it unable to promote the hydrolysis reaction by solvolysis and thus depreciate the condensation. Therefore, the ladder-like polysilsesquioxane had a superhigh degree condensation of approximately 99%. After UV polymerization, the optimal sample with a PTMS:MOPTMS mole ratio of 6:4 exhibited a high H of approximately 0.5 GPa and a E^* of approximately 6 GPa.

By changing the organic periphery groups in ladder-like polysilsesquioxane, the mechanical properties of coatings could be tuned over a broad range. Based on the same synthetic method, ladder-like polysilsesquioxane with distinct molar ratios of epoxy and phenyl groups was synthesized [112]. Here, epoxy groups were used for photocationic polymerization, and phenyl groups were used to enhance robustness. The mechanical performance of the coating can be further optimized by meticulously adjusting the molar ratio between phenyl and epoxy groups. The hardness of the coating increased with the increase in the phenyl content, and the molar ratio of optimal phenyl and epoxy group was 6:4. However, on increasing the molar ratio from 6:4 to 9:1, the hardness of the coating degraded apparently (from 9H to 4H pencil hardness), which could be attributed to the reduction of the molecular weight of ladder-like polysilsesquioxane and insufficient epoxy groups for cross-linking the network. Other types of ladder-like polysiloxanes, such as ladder-like poly(phenyl-comercaptopropyl)silsesquioxanes and poly(propyl carbazole)silsesquioxanes were synthesized, and the mechanical properties could be tailored by simply varying the ratio between various organic periphery groups [113,114].

In brief, ladder-like polysilsesquioxane, as a promising nanomaterial for fabricating polymer-ceramic hybrid coatings, has stable and regular structures, excellent functional tunability, and good solubility. It is also highly anticipated that ladder-like polysilsesquioxanes can be used as building blocks to fabricate other forms of functional materials.

2.2.3. Step-by-step hybrid strategy based on cage-like polysilsesquioxanes

The cage-like polysilsesquioxane (POSS) has a specific 3D network with a rigid silicon core surrounded by multiple organic functional groups, which can be synthesized by the direct and complete condensation of organosilane precursors [75,115]. It is

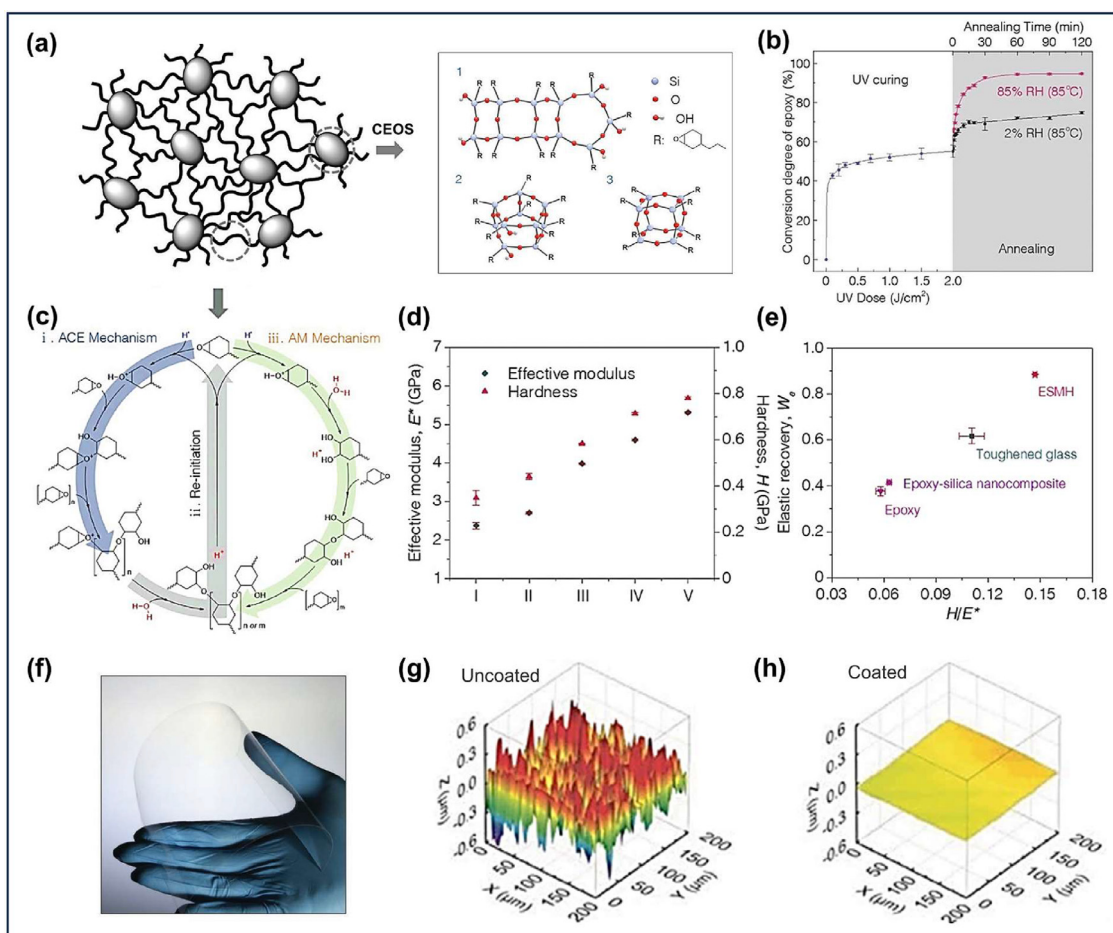


Fig. 8. (a) Chemical structures of ladder-like polysilsesquioxanes and the cross-linking network after UV polymerization. (b) Epoxy conversion of ladder-like polysilsesquioxanes. (c) Reaction mechanisms of UV-initiated cationic ring-opening polymerization in coating formation. (d) E^* and H of the coating increased with the increase in cross-linking density. (e) Graph of W_e versus H/E^* . (f) Photographs of the coated PET film. The white-light scanning interferometric images of the (g) uncoated and (h) coated PET film after steel-wool wear test. [83], Copyright 2017. Adapted with permission from John Wiley & Sons Inc.

considered to be the smallest stable silica particle in existence. If the POSS contains multiple polymerizable or reactive groups, they can be cross-linked to form a POSS-based matrix or polymerized with functional polymers [116].

Many POSS bearing various functional groups are commercially available (e.g., vinyl, methacrylate, and glycidyl) and have been used to fabricate polymer-ceramic hybrid coatings. Methacrylate (M) and glycidyl (G) are two classic UV-polymerizable groups. POSS bearing octakis(3-methacryloxypropyl) (MPOSS) could form a hard but brittle coating (50- μ m thick) via the free-radical polymerization. The coating exhibited high hardness (~ 0.99 GPa) but poor flexibility. By contrast, POSS bearing octakis(3-glycidyloxypropyl) (GPOSS) could form a more flexible coating compared with the MPOSS coating due to the ether and methylene groups produced by the cationic ring-opening polymerization of glycidyloxypropyl groups (Fig. 9a). Thus, the balance between hardness and flexibility could be adjusted by combining methacrylate and glycidyl groups in one POSS (G_mM_n POSS) [87]. It's also worth noting that the UV-curing behaviours of methacrylate and glycidyl groups are different. Methacrylate groups cross-link more effectively under the N_2 atmosphere than the air atmosphere, because O_2 in the air can react with carbon radicals to produce weakly reactive peroxy radicals, which quench the reaction with methacrylate. By contrast, glycidyl groups cross-link faster under air atmosphere via the active-chain end and the active monomer mechanism. Therefore, G_2M_6 POSS coatings exhibited the best balance between hard-

ness and flexibility by first curing in N_2 followed by curing in air (Fig. 9b). Based on the same strategy, the glycidyloxypropyl groups of GPOSS could be chemically fixated with CO_2 via the carbonylation reaction to synthesize POSS bearing cyclic carbonates (Fig. 9c) [92]. The cyclic carbonates of POSS were cured by diamines, including hexamethylenediamine and isophoronediamine to prepare POSS/non-isocyanate polyurethane (NIPU) hybrid coatings. Compared with the NIPU coating without POSS, its scratch resistance was markedly improved (Fig. 9d). Besides, a novel fluorinated and thermal-curable POSS was synthesized [117]. Such a POSS contains aryl-trifluoro-vinyl-ether groups, which could connect other POSS by transforming into a perfluorocyclobutane under heating, and the resulting perfluorocyclobutane could improve the dielectric properties and hydrophobicity of the films.

The mechanical properties of POSS-based hybrid coatings are affected not only by the cross-linking methods but also by the cage size. By conventional flash column chromatography, three vinyl POSS species, including octameric (T8), decameric (T10), and dodecameric (T12) were separated from a commercial POSS mixture [118]. Next, they were modified with benzocyclobutene through a hydrosilylation reaction to prepare a thermal-curable POSS. After the thermal cross-linking process, hybrid materials based on different cage sizes of POSS were obtained. The hardness and modulus increased with the increase in the cage size. The dielectric constant decreased with the increase in the cage size, which is favorable for application in microelectronic integrated circuits.

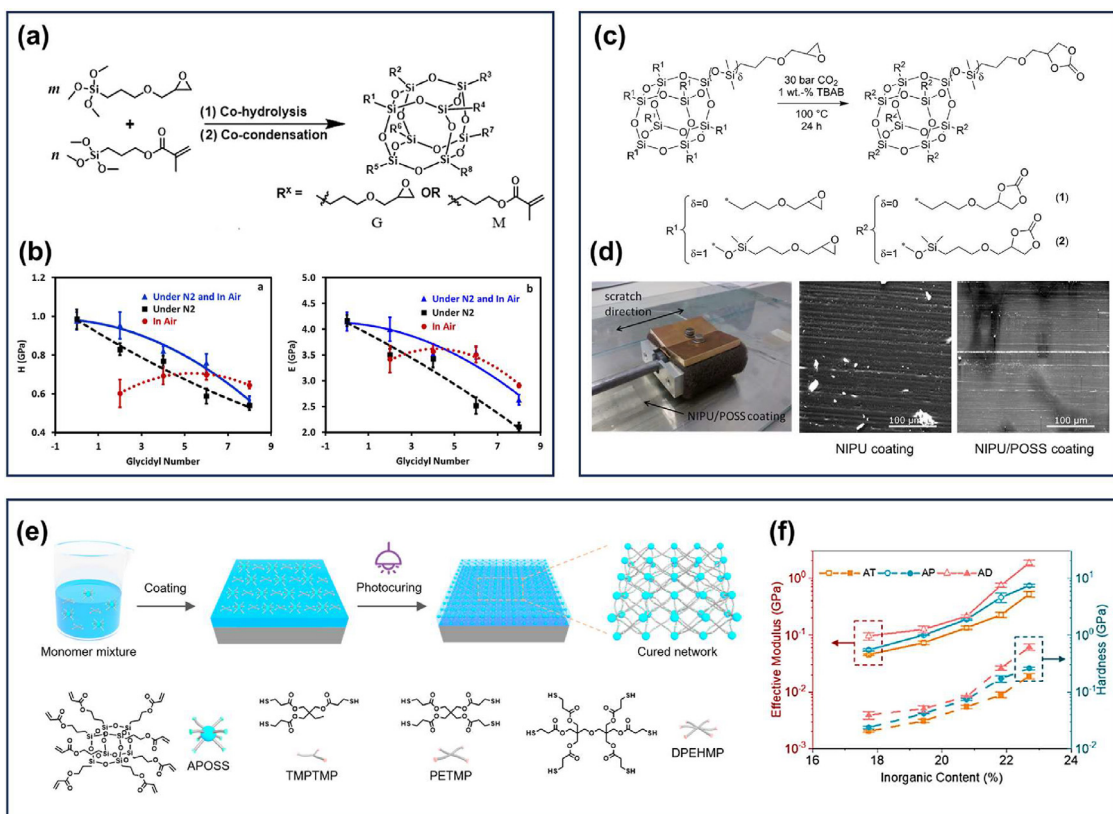


Fig. 9. (a) Chemical structures of the POSS bearing methacryloxypropyl or glycidyoxypropyl. (b) H and E^* values of the coatings cured under different atmospheric conditions as a function of the amount of glycidyl in each POSS cage. [87], Copyright 2022. Adapted with permission from American Chemical Society. (c) Synthesis of POSS cyclic carbonates. (d) SEM images of NIPU coatings and NIPU/POSS hybrid coatings after scratching by an abrasive nonwoven. [92], Copyright 2016. Adapted with permission from American Chemical Society. (e) Schematic illustration of the fabrication procedure of the hybrid coating via thiol-ene click reaction between APOSS and thiols. (f) E^* and H of coatings with different inorganic content. [88], Copyright 2022. Adapted with permission from American Chemical Society.

Another typical approach to balance hardness and flexibility is the incorporation of flexible organic monomers. Utilizing the highly efficient thiol-acrylate photocatalytic cross-linking reaction between commercial acrylo POSS and multifunctional thiols under the N₂ atmosphere, polymer-ceramic hybrid coatings were achieved within 5 s (Fig. 9e) [88]. In this system, the cross-linking reaction of thiol-acrylate was more effective under the N₂ atmosphere because only highly reactive carbon radicals involve in the reaction without weakly reactive peroxy radicals that form by O₂. Furthermore, by varying the thiol functional degree and acrylate/thiol ratio, the optimal coating (50-μm thick) exhibited high H (0.609 GPa) and low inward bending diameter (0.33 mm) (Fig. 9f). Interestingly, the incorporation of organic hexa-functionality thiols effectively reduced the E^* of the coating, while hardness was not lost considerably because of the efficient cross-link of hexa-functionality thiols. Such a coating had a high value of H/E^* (~ 0.33) and an excellent W_e of over 80%.

POSS exhibits considerable potential in the fabrication of polymer-ceramic hybrid coatings. However, the preparation methods of POSS are complex and time-consuming, and require a large usage of organic solvent, which limits its large-scale application. Therefore, more effort should be made to develop relevant low-cost POSS.

2.2.4. Step-by-step hybrid strategy based on in situ generation of siloxane nanoclusters

Recently, molecular necklace-like cross-linkers were successfully developed by substituting a part of the primary hydroxyl groups in the polyrotaxane with alkoxypropyl groups [93]. The polyrotaxane consisted of a ring-shaped host molecule (α -cyclodextrin)

threaded onto a linear guest molecule (PEG10000). During coating formation, these cross-linkers could cross-link with alkoxypropyl-functionalized linear polymer to form siloxane nanoclusters via the hydrolytic polycondensation reaction of alkoxypropyl groups, and melamine agents were introduced for maximizing the cross-linking points (Fig. 10a). The organic-inorganic hybrid cross-linking points with a size of 5–10 nm homogeneously formed along the polyrotaxane backbone. The cross-linked α -cyclodextrin rings could slide along the PEG axis in the highly cross-linking hybrid network during physical stretch (Fig. 10b). Due to the dynamic movement of modified α -cyclodextrin, the coated indium tin oxide (ITO) films exhibited excellent flexibility ($\sim 110\%$ strain). Because the dynamic movement could effectively buffer external stress, the resulting coating exhibited excellent hardness (9H pencil hardness) and scratch resistance (Fig. 10c-e). Note that the polymer usually requires alkoxypropyl-functionalized to realize a highly cross-linking network between the hybrid cross-linker and the polymer, and thus the polymer needs to be judiciously designed. Overall, the introduction of organic-inorganic hybrid cross-linkers into polymers to generate siloxane nanoclusters *in situ* provides important insights for circumventing the limitation of simultaneously achieving superior flexibility and hardness in a coating.

2.2.5. Step-by-step hybrid strategy based on spherical nanoparticles

Spherical nanoparticles (SiO₂, ZrO₂, etc.) have been widely used as fillers to improve the mechanical properties of resins and elastomers [126]. These nanoparticles exhibit high hardness, excellent chemical and thermal stability [120,127]. Unlike using spherical nanoparticles as fillers with low addition content, they can be used as building blocks in polymer-ceramic hybrid coatings.

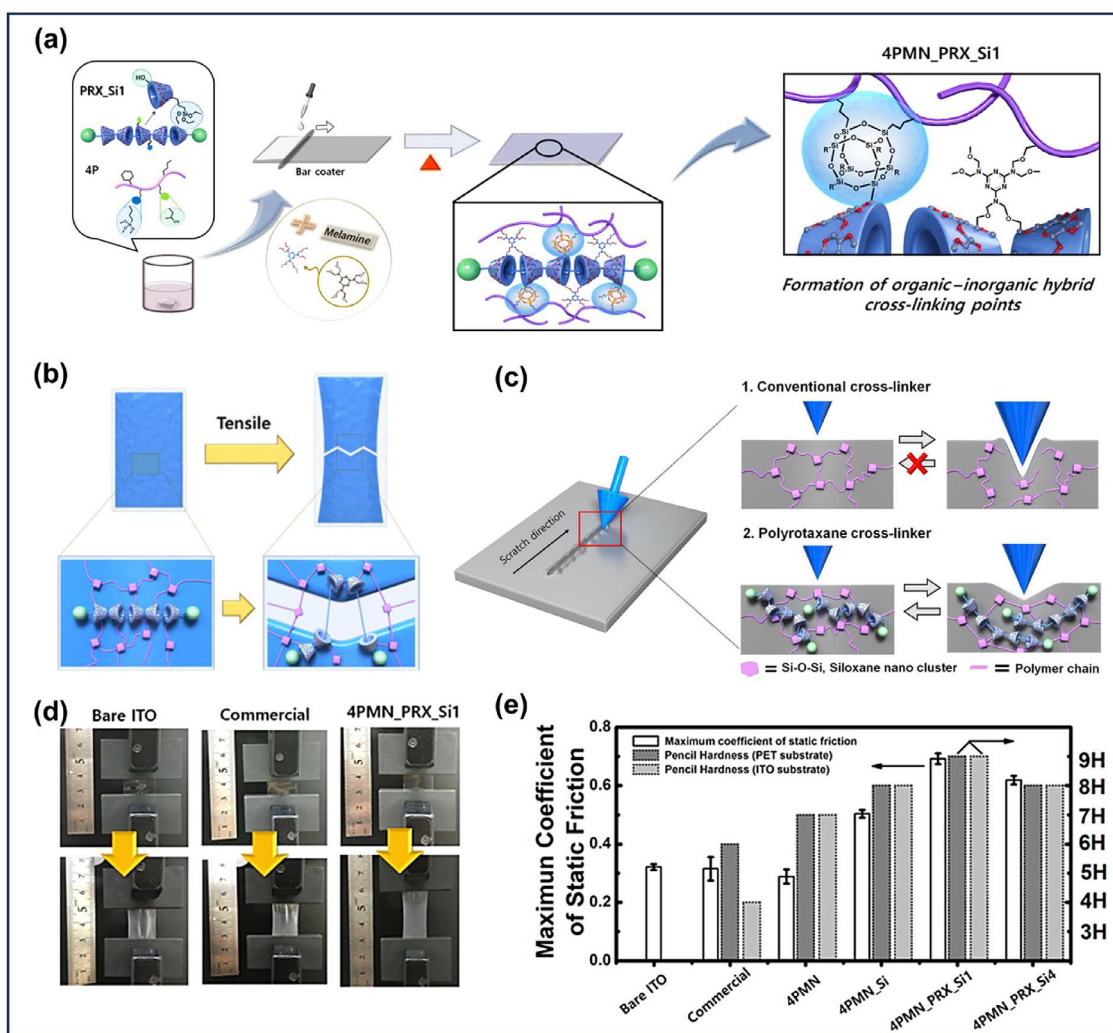


Fig. 10. (a) Schematic illustration of siloxane nanoclusters formation. Schematic illustration of the proposed dynamic movement of necklace-like cross-linkers under (b) tensile and (c) scratch. (d) Photographs of various coated ITO films in a tensile test. (e) Pencil hardness value of each sample. [93], Copyright 2019. Adapted with permission from American Chemical Society.

However, the incorporation of nanoparticles into a coating in a large proportion without precipitation, phase-separation, and affecting surface topography is challenging for a hybrid coating [128]. To overcome this shortcoming, the development of reactive inorganic nanoparticles with polymerizable groups has attracted considerable attention. For example, by modifying silica nanoparticles with methyl and methacryl groups to improve the compatibility of components, modified nanoparticles were uniformly dispersed in branched methyl-methacryl oligosiloxanes [119]. These modified organic groups impeded hydrogen bonds formed by silanol groups, which avoided the creation of aggregates. Next, modified nanoparticles were chemically cross-linked with the hybrid matrix via the UV-induced free-radical polymerization, which maximized the synergistic effect of the hybridization (Fig. 11a). The modified nanoparticles are homogeneously dispersed in the matrix without aggregation (Fig. 11b). With the increase in nanoparticle contents, the hardness of the coating (3- μm thick) was increased from 4H pencil hardness to 9H pencil hardness (Fig. 11c).

This strategy could also be applied to realize the combination of polymerizable inorganic nanoparticles with oligomers [120]. The methyl-/methacryl-modified nanoparticles (15 nm diameter) could be chemically embedded in urethane acrylate to prepare a coating that exhibited excellent optical transparency and reliable scratch resistance (Fig. 11d-f). Notably, the high conversion of

UV-curable acryl and methacryl groups enabled the successful photo-fabrication of micropatterns with various shapes on hybrid coatings.

ZrO₂ nanoparticles exhibit high hardness, low thermal conductivity, and excellent chemical stability. Therefore, ZrO₂ nanoparticles adorned with polymerizable groups are attractive as reactive inorganic nanoparticles for hybrid coatings. For example, ZrO₂ nanoparticles modified with butyric acid and 3-methacryloxypropyl trimethoxysilane were well-compatible with the methacryloxy-oligosiloxane matrix so that the resin could be loaded with 51.7 wt% of the organically-modified ZrO₂ nanoparticles while exhibiting high transparency [121]. Besides, utilizing the sol-gel reaction, an inorganic Zr-O-Si network with higher hardness than soda lime silicate glass could be prepared [122]. Zirconium *n*-propoxide also can first react with functional organosilanes to form zirconium-oligosiloxanes and form a hybrid network by subsequent polymerization curing. For example, vinyl- and hydride-functionalized zirconium-phenyl oligosiloxanes were prepared via the sol-gel reaction. Next, the robust hybrid network was formed via the hydrosilylation reaction (Fig. 11g) [90]. In this system, zirconium exerted an additional catalytic effect on the hydrosilylation reaction, and the Zr-O-Si heterometallic phase improved the refractive index. After thermal aging, the hybrid encapsulant maintained its high transmittance and small yellowness

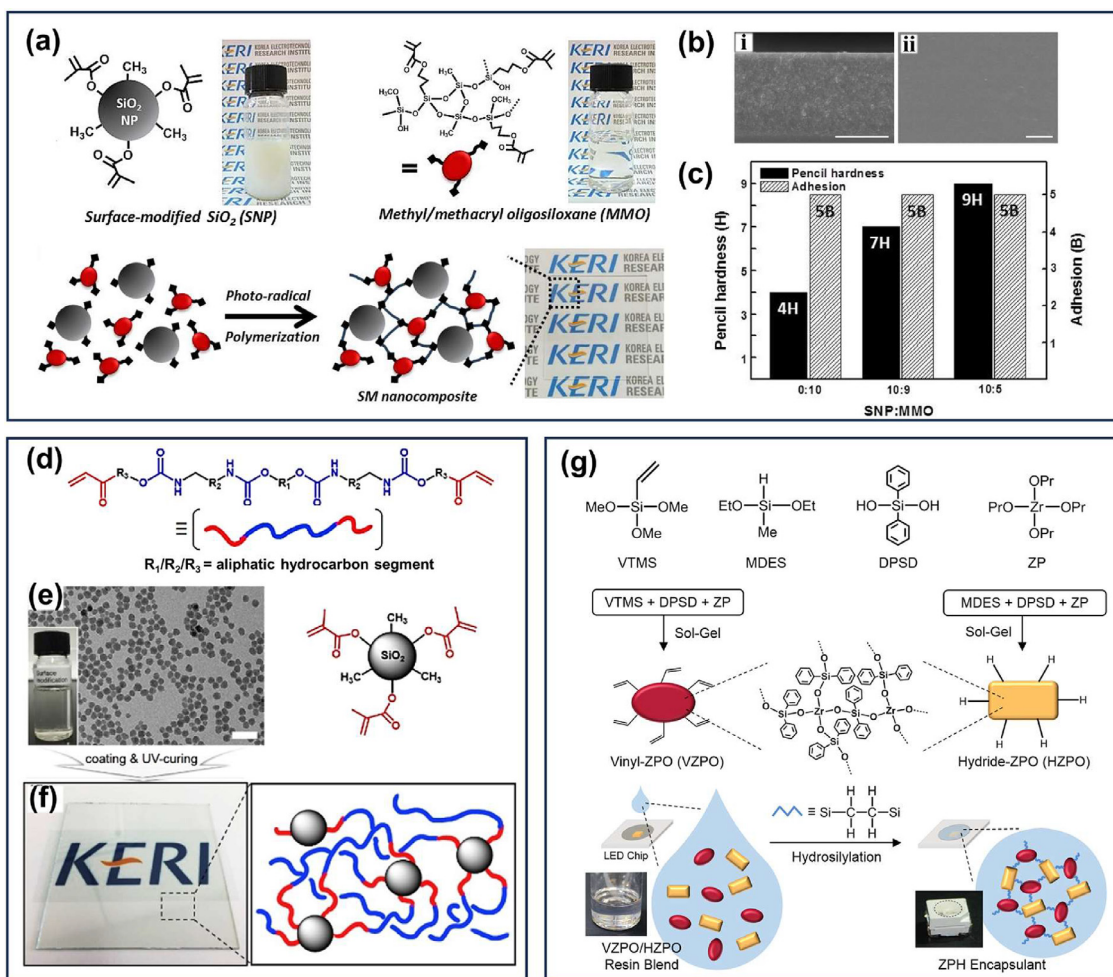


Fig. 11. (a) Schematic fabricating illustration of the fabrication of silica-oligosiloxane hybrid coatings. (b) The cross-sectional (i) and perspective-view (ii) SEM images of the silica-oligosiloxane hybrid coating. (c) Pencil hardness and adhesion strength of the coating with different silica / oligosiloxane ratios. [119], Copyright 2017. Adapted with permission from Elsevier Science Ltd. (d) Chemical structures of urethane acrylate and methyl-/methacryl-modified SiO_2 . (e) TEM and photographic image of the resin incorporating organically-modified SiO_2 . (f) UV-cured coating and its cross-linking network. [120], Copyright 2016. Adapted with permission from Elsevier Science Ltd. (g) Chemical structure of precursors and schematic illustration of the synthesis of zirconium-oligosiloxanes and hybrid materials derived from zirconium-oligosiloxanes by the hydrosilylation reaction. [90], Copyright 2014. Adapted with permission from American Chemical Society.

index, whereas the commercial LED encapsulant degraded considerably. Besides, the addition of an organic polymer can readily endow the coating with desired functions. For example, highly weather-resistant fluoropolymers with silane pendant groups were introduced to $\text{Zr}-\text{O}-\text{Si}$ via the hydrolytic polycondensation reaction among silane pendant groups, silicon sol, and zirconium sol [123]. After exposure to UV-vis light for 1500 h, the hardness of coated polycarbonate (PC) films changed negligibly, whereas the hardness of pure PC films decreased by 35%. Through modifying reactive functional groups, spherical nanoparticles can homogeneously disperse in the matrix and exhibit good compatibility with other components. These spherical nanoparticles can be used as building blocks and endow the coating with desired physicochemical properties. At the same time, it is crucial to explore polymer-ceramic hybrid coating based on other shapes of nanoparticles, such as nanotubes and nanoplatelets.

2.2.6. Step-by-step hybrid strategy based on Si-doped carbonized polymer dots

Si-doped carbonized polymer dots (Si-CPDs) with a carbon core and polymer chains shell structure exhibit considerable potential for application in photoelectricity materials, photonic crystals, as well as polymer-ceramic hybrid coatings [129–131]. For example,

using the cross-linking reaction and carbonization between amino-oligosiloxane and citric acid, Si-CPDs were synthesized and served as building blocks (Fig. 12a) [124]. The residue $\text{Si}-\text{OH}$ of Si-CPDs was dehydrated and condensed to form a polymer-ceramic hybrid coating after the heating process. In this system, the Si-CPDs exhibit dot shapes with an average diameter of 4.5 nm, which can achieve nano-scale hybridization, thus exhibiting high transmittance of the resulting coating (Fig. 12b). Notably, citric acid (CA) was used to bind SiO_x segments tightly through a cross-linking reaction and carbonization to form Si-CPDs with lattice. Therefore, the hardness and elastic modulus increased with the increase in the CA content (Fig. 12c). The optimal coating (50- μm thick) exhibited 9H pencil hardness and 5-mm outer bending diameters. However, numerous amino groups of the as-prepared coating rendered the coating hygroscopic, which may affect its mechanical performance. By using the epoxy group of GPTMS to react with the amino groups of Si-CPDs derived from tetrahydric acid and amino-oligosiloxane, the hybrid coating exhibited good water and solvent resistance (Fig. 12d) [125]. The hardness and elastic recovery of the hybrid coating were much higher than that of some engineering polymers, showing excellent wear resistance (Fig. 12e). Furthermore, by introducing the low-surface-energy PDMS, the solvent resistance and hydrophobicity of hybrid

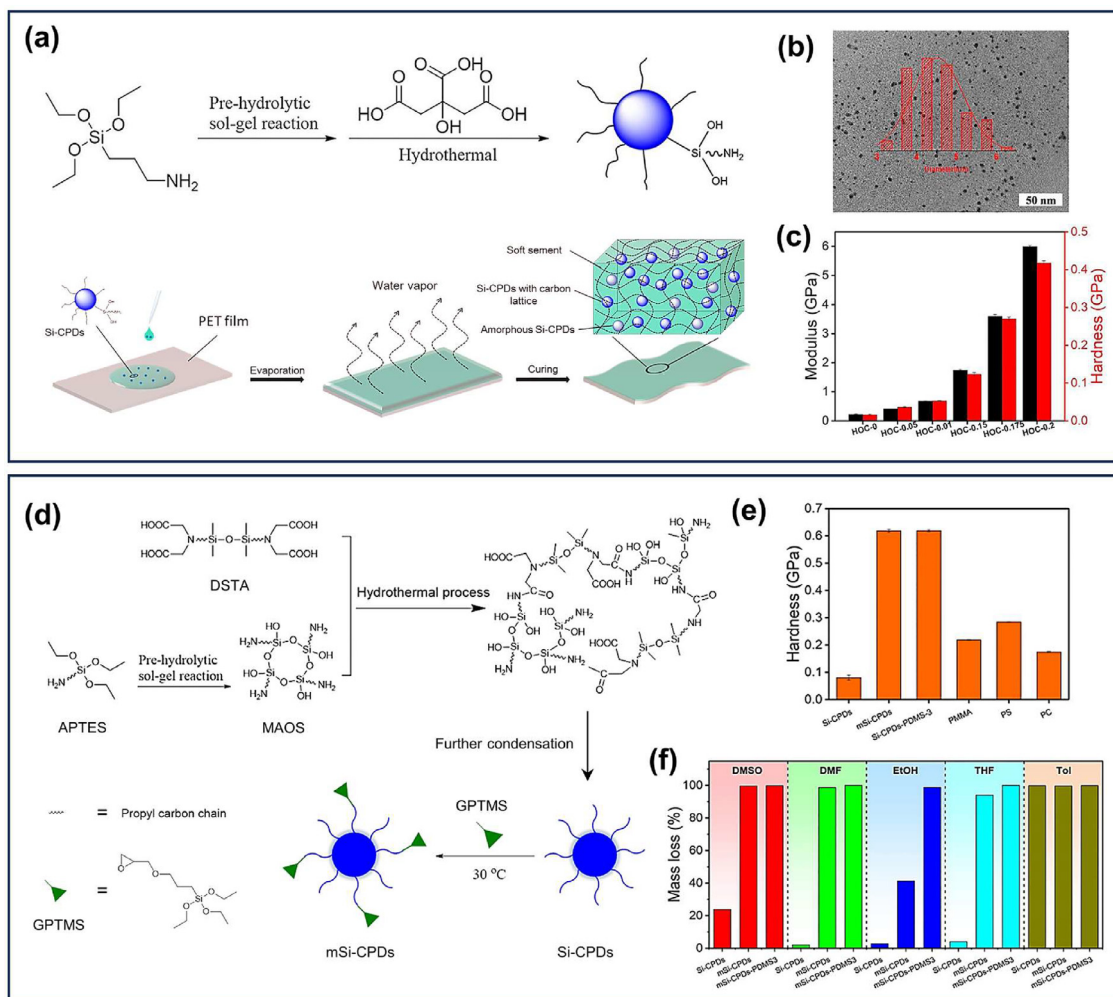


Fig. 12. (a) Synthesis of Si-CPDs and schematic illustration of the fabrication of the hybrid coating using Si-CPDs as building blocks. (b) TEM image of Si-CPDs. (c) E^* and H of hybrid coating with different CA contents. [124], Copyright 2022. Adapted with permission from American Chemical Society. (d) Schematic illustration of the fabrication of the epoxy modified Si-CPDs. (e) H of the coatings and counterpart materials. (f) Resistance test of solvent erosion. [125], Copyright 2023. Adapted with permission from American Chemical Society.

coating improved considerably (Fig. 12f). Si-doped carbonized polymer dots have exhibited great potential as hard building blocks, but relevant research on whether their photophysical properties can be applied in polymer–ceramic hybrid coatings is yet to be explored.

Table 2 summarizes the key points (building blocks, curing method, mechanical properties, and functions) of the recently developed polymer–ceramic hybrid coatings. Overall, the step-by-step hybrid strategy exhibits superior structural/functional tunability, milder reaction conditions, and better mechanical properties than the one-step sol-gel hybrid strategy. Polymer–ceramic hybrid coatings derived from the one-step hybrid strategy of polysilazanes exhibit superior mechanical properties but require tight control of coating thicknesses. Based on the step-by-step hybrid strategy, 3D-structured polysiloxanes (branched, cage-like, ladder-like, and spherical) can be used as building blocks. The branched oligosiloxane has lower viscosity and condensation degree compared with other building blocks, which makes it have good compatibility with the introduced organic components. The bulky functional branched architecture can improve the toughness of the coating while maintaining a high cross-linking density. Conversely, ladder-like polysilsesquioxanes and cage-like polysilsesquioxanes have stable and regular structures and high condensation degrees. The coatings prepared using these two polysiloxanes as building

blocks have superior hardness and flexibility. These building blocks have tailor-designed active sites such as epoxy, amine, vinyl, acrylate, thiol, etc. for obtaining highly cross-linking hybrid networks. Note that the cage-like polysilsesquioxane (POSS) has a more specific 3D network than ladder-like polysilsesquioxanes, which is advantageous for some reactions that require tight stoichiometric control. However, the preparation methods of POSS are complex and time-consuming, which limits its large-scale application. In comparison, the fabrication process of ladder-like polysilsesquioxanes is simple and cost-effective. Polymer–ceramic hybrid coatings based on *in situ* generation of siloxane nanoclusters also exhibit superior mechanical properties, but require judicious design of silane-containing organic components for efficient formation of siloxane nanoclusters. Si-doped carbonized polymer dots offer a novel building block for fabricating polymer–ceramic hybrid coatings. The prepared coatings exhibit similar hardness and flexibility to those prepared based on 3D-structured polysiloxanes. In addition, by modifying reactive functional groups on the surface of spherical nanoparticles, they can be used as building blocks to fabricate polymer–ceramic hybrid coatings. Spherical nanoparticles have the largest size compared to the building blocks described above, which makes them difficult to use alone and need to be combined with other organic polymers or smaller building blocks to form a coating.

Table 2
Selected examples of recent/innovative polymer–ceramic hybrid coatings.

Materials	Curing method	Hardness	Flexibility	Function	Ref.
One-step hybrid strategy					
TEOS + C8 + silane-bearing amphiphilic fluorocarbon acrylate-PEG telomer	Thermal	$H = \sim 57$ MPa $E^* = \sim 420$ MPa (~ 50 μm thick)	Bending 100 times with a radius of 2.5 mm (~ 50 μm thick)	Anti-biofouling	[49]
MTMS + GPTMS + ATSB	Thermal	5H pencil hardness (2.3 to 7.1 μm thick)	Bending with a radius of 0.5 mm	Anti-biofouling	[55]
Aminopropyl-functionalized poly(silsesquioxane) + triethylamine + PTF-PMS micelle	Thermal	9H pencil hardness $H = \sim 0.83$ GPa $E^* = \sim 5.89$ GPa $H/E^* = 0.141$ (~ 200 μm thick)	Bending 10,000 times into a tube $W_e = 0.863$ (~ 200 μm thick)	Self-healing	[132]
PHPS + dihydroxyalkyl-terminated PDMS	Vacuum ultraviolet	9H pencil hardness $H = \sim 3.07$ GPa $E^* = \sim 39.40$ GPa (~ 1.2 μm thick)	Bending 10,000 times with a radius of 2 mm (~ 1.2 μm thick)	Anti-adhesion	[71]
PHPS	AO and ultraviolet	3H pencil hardness $H = \sim 1$ GPa $E^* = \sim 5$ GPa (~ 572 nm thick)	Bending 10,000 times with a radius of 1.5 mm $W_e = 0.9$ (10 μm thick)	Atomic oxygen resistance	[133]
Polysilazane + hydroxy silicone oil + PARALOID™ B72	Moisture	2H pencil hardness $H = \sim 1.22$ GPa $E^* = \sim 22.98$ GPa	Bending the coated sample to a 60°	Anti-icing	[72]
Step-by-step hybrid strategy					
Methacrylate- functionalized oligosiloxane + urethane prepolymer	Photo	$H = \sim 0.168$ GPa $E^* = \sim 1.72$ GPa	Outward bending with a radius of 2.5 mm	Transparency	[96]
Epoxy-oligosiloxane nanocluster + amine-terminated curing agents	Thermal	6H pencil hardness $H = \sim 0.121$ GPa $E^* = \sim 0.437$ GPa (~ 100 μm thick)	Bending with a radius of 5 mm (30 μm thick)	Anti-biofouling / anti-liquid adhesion	[84]
Epoxy-oligosiloxane nanocluster + diglycerol decarbonate + 4,4-diaminodicyclohexylmethane	Thermal	$H = \sim 0.21$ GPa $E^* = \sim 2.23$ GPa (~ 200 μm thick)	Bending with a radius of 1 mm (200 μm thick)	Anti-liquid adhesion / anti-icing	[134]
Epoxy-functionalized siloxane hybrid matrix + glass-fabric reinforced siloxane hybrid composite film	Photo	6H pencil hardness (~ 15 μm thick)	Inward bending with a radius of 3 mm Outward bending with a radius of 5 mm (~ 15 μm thick)	Transparency	[135]
Epoxy-functionalized siloxane + glass fabric (hierarchically surface-textured)	Thermal	$H = \sim 0.56$ GPa $E^* = \sim 11.9$ GPa (~ 50 μm thick)	–	Triboelectric surface	[136]
Cycloaliphatic epoxy-functionalized oligosiloxane + PEMA-g-PDMS	Thermal	9H pencil hardness $H = \sim 0.52$ GPa $E^* = \sim 8.8$ GPa (~ 100 μm thick)	Bending 500 times with a side-to-side distance of ~ 0.8 cm (~ 40 μm thick)	Anti-liquid adhesion	[86]
Ladder-like structured cycloaliphatic epoxy-functionalized oligosiloxane	Photo	9H pencil hardness $H = \sim 0.78$ GPa $E^* = \sim 5.31$ GPa $H/E^* = 0.147$ (~ 50 μm thick)	Bending 10,000 times with a radius of 1 mm $W_e = 0.885$ (50 μm thick)	Transparency	[83]
Ladder-like structured methacrylate siloxane	Thermal	$H = \sim 0.6$ GPa $E^* = \sim 6$ GPa $H/E^* = 0.1$	$W_e = 0.861$	Transparency	[104]
Ladder-like structured methacrylate oligosiloxane + butylene–urethane dimethacrylate	Photo	8H pencil hardness (60 μm thick)	Inward/outward bending with a radius of 3.2 mm $W_e = 0.68$ (10 μm thick)	Transparency	[137]
Ladder-like structured poly(phenyl-co-methacryl silsesquioxane)	Photo	$H = \sim 0.4$ GPa $E^* = \sim 8$ GPa	–	Transparency	[111]
Ladder-like structured poly(phenyl-co-mercaptopropyl)silsesquioxane + ladder-like structured poly(phenyl-co-methacryloxypropyl)silsesquioxane	Photo	9H pencil hardness $H = 0.3\text{--}0.4$ GPa $E^* = 4\text{--}5$ GPa $H/E^* = 0.068$ (~ 40 μm thick)	Bending without macroscopic cracks (~ 40 μm thick)	Transparency	[113]
Ladder-like poly(n-propylpyrrole-co-cyclohexenylethyl)silsesquioxane	Thermal	$H = \sim 0.1$ GPa $E^* = \sim 6$ GPa $H/E^* = 0.015$ (~ 1.5 μm thick)	–	Self-healing	[95]
Ladder-like epoxycyclohexyl polysilsesquioxane	Thermal	8H pencil hardness $H = \sim 0.49$ GPa $E^* = \sim 5.15$ GPa $H/E^* = 0.094$ (~ 25 μm thick)	Inward bending with a radius of 0.5 mm Outward bending with a radius of 5 mm $W_e = 0.801$ (~ 25 μm thick)	Transparency	[138]

(continued on next page)

Table 2 (continued)

Materials	Curing method	Hardness	Flexibility	Function	Ref.
Ladder-like poly(acryloxypropyl-co-n-propylpyrrole-co-cyclohexenylethyl)silsesquioxane	Photo	6H pencil hardness $H = \sim 0.2$ GPa $E^* = \sim 9$ GPa $H/E^* = 0.025$ (1.5 μm thick)	–	Self-healing	[139]
Ladder-like epoxycyclohexyl polysilsesquioxane + perfluorinated poly(propylene oxide)	Photo	9H pencil hardness $H = \sim 1.39$ GPa $E^* = \sim 6.4$ GPa (~ 40 μm thick)	Inward bending with a radius of 1 mm Outward bending with a radius of 2 mm $W_e = 0.81$ (~ 10 μm thick)	Anti-liquid adhesion	[140]
Acrylo polyhedral oligomeric silsesquioxane + multifunctional thiols	Photo	9H pencil hardness $H = \sim 0.609$ GPa $E^* = \sim 1.85$ GPa $H/E^* = 0.33$ (~ 200 μm thick)	Bending 10,000 times with a radius of 0.8 mm (100 μm thick)	Transparency	[88]
GPOSS + PDMS	Photo	9H pencil hardness $H = \sim 0.7$ GPa $E^* = \sim 3.3$ GPa $H/E^* = 0.21$ (~ 100 μm thick)	Bending 500 times into a U $W_e = 0.79$ (~ 100 μm thick)	Anti-liquid adhesion	[141]
GPOSS + FC ₆ -COOH	Photo	7H pencil hardness $H = \sim 0.59$ GPa $E^* = \sim 3.32$ GPa (~ 100 μm thick)	Bending 100,000 times with a radius of 2 mm $W_e = 0.705$ (~ 50 μm thick)	Anti-liquid adhesion	[142]
G ₂ M ₆ POSS	Photo	9H pencil hardness $H = \sim 0.95$ GPa $E^* = \sim 4$ GPa $H/E^* = 0.24$ (~ 50 μm thick)	Inward bending with a radius of 1 mm Outward bending with a radius of 5 mm $W_e = 0.759$ (~ 50 μm thick)	Transparency	[87]
GPOSS + amine-terminated hyperbranched polysiloxane + epoxy groups modified poly(3,3,3-trifluoropropylmethylsiloxane)	Thermal	9H pencil hardness $H = \sim 0.94$ GPa $E^* = \sim 11$ GPa (~ 50 μm thick)	–	Anti-icing / anti-corrosion	[143]
Decameric UPy-POSS	Thermal	5H pencil hardness $E^* = \sim 5.0$ GPa (~ 50 μm thick)	Bending with a radius of 1 mm (~ 50 μm thick)	Self-healing	[144]
Quaternary random copolymer + hexa(methoxymethyl)-melamine + Paraloid B-72 + alkoxysilyl-functionalized polyrotaxane cross-linker	Thermal	9H pencil hardness (~ 10 μm thick)	Completely folding 5000 times without fracture	Transparency	[93]
Glycidylxypropyltrimethoxysilane-modified Si-doped carbonized polymer dots + PDMS	Thermal	$H = \sim 0.6$ GPa $E^* = \sim 5.5$ GPa (~ 200 μm thick)	Inward/outward bending 1000 times with a radius of 1 mm $W_e = 0.8594$ (~ 10 μm thick)	Anti-liquid adhesion	[125]
Si-doped carbonized polymer dots	Thermal	9H pencil hardness $H = \sim 0.40$ GPa $E^* = \sim 6$ GPa	Bending 1000 times with a radius of 2.5 mm (~ 50 μm thick)	Transparency	[124]
Epoxy-zirconium particle + amine-terminated hyperbranched polysiloxane	Thermal	7H pencil hardness $H = \sim 0.17$ GPa $E^* = \sim 1.5$ GPa $H/E^* > 0.1$ (~ 10 μm thick)	Bending with a radius of 1.5 mm (10 μm thick)	Anti-biofouling	[79]
Pentaerythritol tetrakis(3-mercaptopropionate) + (acryloxypropyl)methylsiloxane homopolymer + alkyl-fluorine and alkyl silane-based silica particles	Photo	8H pencil hardness (~ 50 μm thick)	–	Anti-liquid adhesion	[145]
Epoxy MTQ silicone resins + amine-terminated hyperbranched polysiloxane	Thermal	9H pencil hardness $H = \sim 0.46$ GPa $E^* = \sim 3.3$ GPa $H/E^* = 0.139$ (~ 10 μm thick)	Bending with a radius of 1 mm (~ 20 μm thick)	Anti-liquid adhesion	[146]
Surface-modified silica nanoparticles + methyl-methacryl oligosiloxane	Photo	9H pencil hardness (3 μm thick)	–	Transparent	[119]

2.3. Modification of polymer–ceramic hybrid coatings for reinforcement

The physicochemical properties of polymer–ceramic hybrid coatings can be further tuned by many other approaches, which

can be classified into two general categories, namely fillers (e.g., chitin nanofibers, silica nanoparticles, and glass-fabrics) and the structure reinforcement (e.g., 3D structures, layer-by-layer (LBL) assembly structure, and gradient structure). In this section, some examples from each of these categories are reviewed.

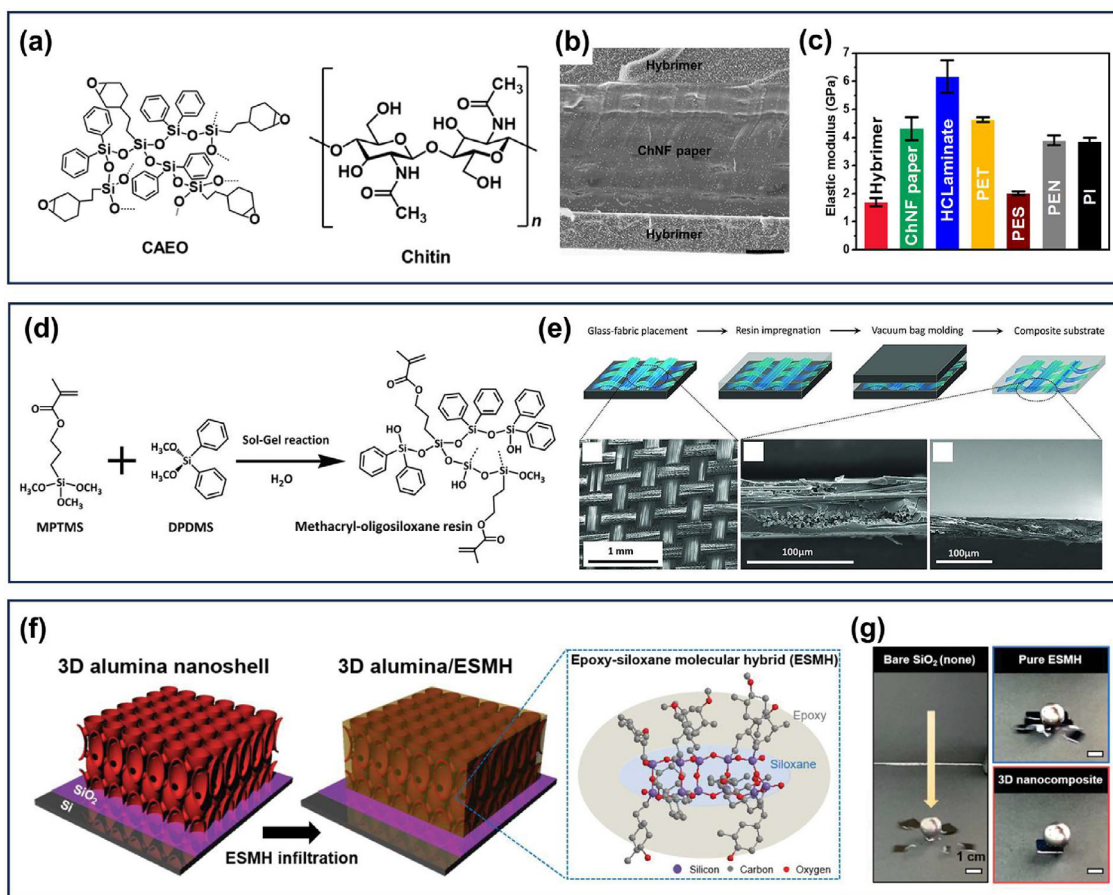


Fig. 13. (a) Chemical structures of oligosiloxane and chitin. (b) SEM image of the chitin nanofiber-laminated composite film. (c) Elastic modulus of the composite film and commercial plastics. [149], Copyright 2010. Adapted with permission from American Chemical Society. (d) Synthesis of the methacryl-phenyl-oligosiloxanes resin. (e) Schematic illustration of the fabrication of the composite film reinforced by glass-fabrics. [151], Copyright 2010. Adapted with permission from John Wiley & Sons Inc. (f) Schematic illustration of the hybrid nanocomposite reinforced by the 3D alumina nanoshell. (g) Photographs of the pure ESMH and 3D nanocomposite coated on the SiO_2 substrate in the ball drop test. [155], Copyright 2021. Adapted with permission from John Wiley & Sons Inc.

2.3.1. Fillers

Physically blending polymer–ceramic hybrid coatings with fillers can improve their mechanical properties [147,148]. Chitin is the second most abundant polysaccharide in nature and is expected to be a sustainable source of polymeric materials. Stiff chitin nanofibers could reinforce the oligosiloxane-based film without declining its optical transparency through a hierarchical biocomposite (Fig. 13a and b) [149]. To produce a hierarchical structure, the chitin nanofiber paper was first impregnated with UV-curable resin through vacuum bag compression. The hierarchical structure was obtained after UV curing of the composite resin. Because of the combination of chitin nanofibers with the hydrogen-bond-driven crystal structure, the elastic modulus of composite films was enhanced (Fig. 13c). As a result, the optimal films exhibited a transparency of approximately 91%, an elastic modulus of 6.2 GPa, an outward bending diameter of 3.2 mm, and a coefficient of thermal expansion of $22 \text{ ppm}\cdot\text{K}^{-1}$. Such a lightweight film has been used in flexible perovskite solar cells and touchscreen panels.

In addition to using chitin nanofibers to reinforce the matrix, silica nanoparticles, as common fillers used for elastomer reinforcement, can be utilized. For example, 3-aminopropyltriethoxysilane (KH550) was used as a curing agent in the hybrid network prepared by phenyltrimethoxysilane, 3-glycidypropylmethylmethoxysilane, and epoxy-modified silica nanoparticles. When 0.75 wt% epoxy-modified silica was incorporated into a hybrid coating, its hardness increased by 58.3% compared with the pure polysiloxane coating [150]. The resulting

coating was transparent (91% transmittance), hard (8H pencil hardness), and wear-resistant.

The coating reinforced by glass-fabrics exhibits a decrease in the coefficient of thermal expansion and an increase in elastic modulus and flexibility [76,148]. Transparent polymer–ceramic hybrid coatings reinforced by glass-fabrics were reported using UV-curable methacryl–phenyl–oligosiloxanes resin as the matrix (Fig. 13d and e) [151]. The refractive index of the hybrid coating could be precisely tailored by controlling the content of phenyl groups to match the refractive index of the glass-fabrics. Such a coating (55–60 μm thick) exhibited high transparency ($\sim 89\%$), a low coefficient of thermal expansion ($\sim 13 \text{ ppm}\cdot\text{K}^{-1}$), and could be repeatedly bent with a radius of 5 mm for 10,000 times without any cracking. The glass-fibers can also be used to reinforce UV-/thermal-curable epoxy-functionalized oligosiloxanes-based coating [152]. After glass-fibers reinforcement, the mechanical properties of thermal- and UV-curing samples improved considerably. Specifically, the thermal-curing method demonstrated reduced microporosity within the matrix, resulting in improved wetting of the matrix resin into the glass-fiber. This improvement translated to higher flexural stress (48.11 MPa) and flexural modulus (3.10 GPa). Introducing elongated glass fabric further refined the mechanical properties of the composites. The optimized composite exhibited a 3H pencil hardness, along with a flexural strength of 251 MPa and a flexural modulus of 9 GPa. Additional enhancements in mechanical properties of polymer–ceramic hybrid coatings have been explored through the incorporation of alternative fillers,

such as boron nitride nanoplatelets and graphene nanoplatelets [153,154].

2.3.2. Structure reinforcement

3D structures. In contrast to fillers, the synergy of organized and uninterrupted 3D nanostructures within a hybrid coating adeptly facilitates the transfer of loads between these nanostructures and the matrix. For instance, the voids within the 3D alumina nanoshell structure were effectively filled with cycloaliphatic epoxy-functionalized oligosiloxanes (EMSH). This was followed by UV-curing to produce a robust nanocomposite (Fig. 13f) [155]. By heat treatment at 700 °C, 3D alumina could enhance the mechanical properties of the nanocomposite. More interestingly, the 3D nanocomposite could absorb energy impacted by a falling steel ball, whereas the control sample was fractured, which indicated that the 3D alumina nanoshell could block crack propagation (Fig. 13g). The film (10- μ m thick) loading with 17.1 vol% alumina exhibited superior metal-like H (1.3 GPa) and excellent flexibility (20,000 outer bending cycles with 5-mm radius without failure).

By contrast, 3D continuous porous polymeric structures could enhance the flexibility of coatings [156]. First, flexible poly(vinylidene fluoride-co-hexafluoropropylene) (PVDFP) and its poor solvent dibutyl phthalate (DBP) were dissolved in acetone and subsequently cast onto the substrate. With the evaporation of acetone, PVDFP phase and DBP phase were separated. Next, after removing DBP, a continuous porous PVDFP scaffold was obtained. Finally, the UV-curable fluorinated POSS and photoinitiator were infiltrated into the scaffold and photocured to obtain the composite. The PVDFP scaffold could effectively transfer or spread loads between the PVDFP scaffold and the cross-linked POSS network. The outer bending diameter of the composite was reduced to 2 mm compared with the 6 mm bending diameter of the pure POSS coating with simultaneous maintenance of the same pencil hardness (8H).

LBL assembly. The LBL assembly technique is a versatile method for preparing films by alternately depositing materials with complementary noncovalent interactions onto substrates. The chemical composition and structure of the film can be precisely controlled at the nano-scale level by predesigning LBL [157,158]. For example, polycation polyethylenimine (PEI) and polyanion poly(acrylic acid) (PAA) with opposite charges were alternately assembled to form a polymeric film using electrostatic interaction and hydrogen bonding as the primary driving force [25]. The polymeric film exhibited a H of approximately 0.43 GPa and a Young's modulus of approximately 9.2 GPa. However, due to the weak interaction between polyelectrolytes with opposite charges, the scratch resistance of PEI/PAA films was poor. Based on this LBL assembly system, inorganic CaCO_3 nanoparticles (7–9 nm diameter) were introduced through *in situ* generation manner to enhance the scratch resistance while retaining the self-healing ability [159]. The resulting film exhibited a high H of approximately 0.95 GPa and a Young's modulus of approximately 17.6 GPa.

In addition to forming strong noncovalent interactions between inorganic components and organic components, inorganic components can be dispersed in the LBL system by forming strong covalent bonds. BN nanoplatelets with 2D plate-like structures have been widely used in the synthesis of polymer nanocomposites owing to their excellent mechanical, electrical, and thermal properties. Utilizing the B–OH groups of BN nanoplatelets to react with poly(vinyl alcohol) (PVA) could form ether bonds that not only helped to disperse BN nanoplatelets but also achieved the better synergetic effect of the hybridization [154]. In contrast to PVA/tannic acid LBL coatings, hybrid coatings (40- μ m thick) could effectively transfer loads between the soft and hard interfaces, simultaneously improving H (0.25–0.60 GPa) and Young's modulus (8.60–12.9 GPa). The LBL assembly technique can also be di-

rectly used to modify the surface of polymer–ceramic coatings to improve scratch resistance [160]. The UV-curable ladder-like polysilsesquioxane was used as a flexible hard matrix. Two types of LBL coatings were applied. One LBL was an alternately stacked polyacrylamide hydrochloride (PAH)/PAA LBL coating and the other LBL was an alternately stacked PEI/montmorillonite (MMT) LBL coating. Compared with PAH/PAA LBL coatings, PEI/MMT LBL coatings showed an enhancement in scratch resistance by > 12% because of the reduction of frictional force by the planar MMT layer. However, the LBL assembly technique is time-consuming compared with other approaches.

Gradient structures. Gradient structure reinforcement can simultaneously realize ceramic-like hardness and polymer-like flexibility in a hybrid coating. Many biological materials such as clam hinges and coconut crab pincers use organic proteins and inorganic calcium carbonate calcification to form a gradual transition structure with high mechanical performance [161,162]. Inspired by this, some hybrid materials with gradient structures were fabricated using surface free energy or magnetic field as the driving force.

For example, organic trimethylolpropane triacrylate (TMPTA) and inorganic poly[3-(trimethoxysilyl)propyl acrylate] (PTSA) dispersed uniformly in tetrachloromethane (CCl_4). As the solvent is slowly evaporated by heat, two phases with distinct surface free energy would gradually segregate to form an organic–inorganic gradient structure instead of a sea-island structure (Fig. 14a) [163]. At this stage, it was necessary to employ radical photopolymerization to prevent the hybrid network from undergoing full segregation to form a fully separated two-phase structure (Fig. 14b). In this system, surface free energy is the driving force, and the viscosity is a key parameter for controlling the movement to form a gradient structure. During solvent evaporation, low-viscosity resins separated swiftly, and its influence was greater than that of the surface free energy on resin movement. By contrast, high-viscosity resins separated slowly, and the surface segregation could be fully driven by surface free energy. However, high-viscosity resins with small surface free energy differences could only form a dispersed structure because of an inadequate driving force (Fig. 14c). The gradient-structured coating exhibited higher pencil hardness (3H) than dispersion-structured coating, and the coating hardness could be improved to 9H by adding coumaric- N,N' -dicyclohexyl-4-morpholinecarboxamide (Cou-DCMC) as a catalyst for photobase-catalyzed sol–gel reactions (Fig. 14d). Meanwhile, the bottommost organic polymers ensured excellent flexibility and substrate adhesion.

Another simple technique for constructing a gradient structure on the surface was using the magnetic field as the driving force [164]. By changing the ratio of the TEOS precursor to Fe_3O_4 nanoparticles, $\text{Fe}_3\text{O}_4@$ SiO_2 nanoparticles were synthesized with total diameters of approximately 20 and 50 nm [165]. These nanoparticles were modified by MOPTMS to realize chemical bonding with a UV-curable organic resin. The resulting $\text{Fe}_3\text{O}_4@$ SiO_2 nanoparticles could be driven by the magnetic field to achieve the rich domain of inorganic nanoparticles at the air interface of the organic resin matrix. Thus, at top surfaces, the gradient-structured hybrid coating (nanoparticle content of 20 vol%) exhibited remarkably higher H (~780 MPa vs. ~350 MPa) and E^* (~12.5 GPa vs. ~5.3 GPa) than the homogeneous hybrid coating did. On the contrary, at the bottom interfaces, the gradient-structured hybrid coating exhibited a lower hardness and elastic modulus than the homogeneous hybrid coating did, which showed superior flexibility. By using mixed particle sizes, the problem of limited increase in mechanical gradient when using small particles and step-changes in mechanical performance when using large particles were overcome. Although this method provided a hard surface and flexible bottom, the flexibility of the resulting film was not further investigated.

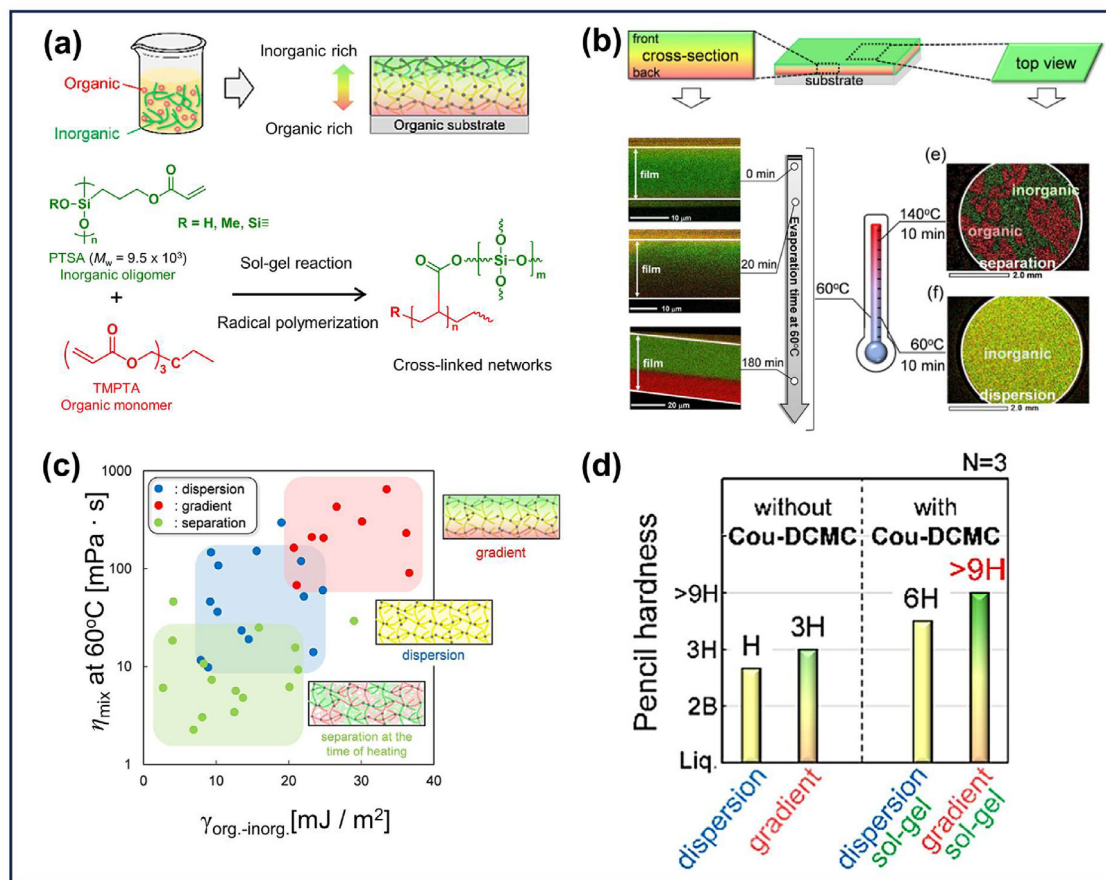


Fig. 14. (a) Schematic illustration of the fabrication of the hybrid coating with gradient structure. (b) EDX images of cross-sections and top view of hybrid films cured with curing with different time. (c) Graph of the relationship among surface free energy, mixed viscosity, and film states. (d) Pencil hardness of the coatings with and without Cou-DCMC. [163]. Copyright 2023. Adapted with permission from American Chemical Society.

3. Functional polymer–ceramic hybrid coatings

The fabrication of polymer–ceramic hybrid coatings having additional functional properties can enable their use in numerous applications, such as foldable displays, wearable devices, maritime industries, and biomedical engineering. In particular, the organic–inorganic hybrid approach can combine high hardness and wear resistance of inorganic components with flexibility and tunability of organic components, which can be tailored to prepare flexible hard coatings with the desired physicochemical properties for application in certain scenarios [166,167]. In this section, we provide an overview of their functions mainly in transparency, anti-liquid adhesion, anti-biofouling, self-healing, and multifunction. Other functions such as atomic oxygen resistance, anti-icing, anti-fogging, anti-corrosion, and flame-retardant are mentioned (Fig. 15). The structure–property relationship of the functional polymer–ceramic hybrid coatings is discussed.

3.1. Transparency

Conventional silicate glasses are transparent, hard, slick, and thermally stable, and have been widely applied in conventional electronic devices. However, applying these glasses to emerging flexible devices is difficult because of their brittle nature [168]. Conversely, organic polymer films exhibit superior flexibility and toughness but poor hardness and wear resistance. The transparent polymer-ceramic hybrid coating with high thermal stability, low permeability to gases, water vapor, and chemicals, and a low coef-

ficient of thermal expansion opens novel prospects in flexible devices, such as foldable displays and wearable electronics [169,170].

Polymer-ceramic hybrid coatings with high transmittance and low haze can be readily achieved by the organic-inorganic hybrid approach. First, the size of inorganic components is smaller than the visible light wavelength. By using the organic-inorganic hybrid approach, the inorganic phase and organic phase can bond at nano-scale or even at the molecular level, eliminating the macro-phase separation [88,151]. For example, epoxycyclohexyl polyhedral oligomeric silsesquioxane (EPOSS) and ladder-like epoxycyclohexyl polysilsesquioxane (EPSQ) were synthesized via the hydrolytic polycondensation reaction [138]. Next, highly transparent EPOSS coatings and EPSQ coatings were obtained after UV-induced ring-opening polymerization. The haze value of EPOSS and EPSQ coatings (25- μm thick) was 0.9 or less, whereas the haze value of pure bare PI films was 1.1. The total transmittance of EPOSS and EPSQ coatings was more than 90.0% owing to the low refractive index (1.55) of the silsesquioxane structure. Besides, because of the regular double-stranded structure and uniform surface morphology of EPSQ, the EPSQ coating exhibited higher hardness and elastic recovery than the EPOSS coating. Moreover, the transparent polymer-ceramic hybrid coating with a low refractive index has the potential to be used as a robust antireflection coating. UV-curable silsesquioxanes were introduced into acidic silica sols to prepare mesoporous silica antireflection coatings (MSCs) with a low refractive index (1.47) and high average transmittance (92.71% from 400 to 2000 nm) (Fig. 16a) [133]. The reflection of coated PET reduced by 4.33% from 400 to 2000 nm compared with the bare PET film (Fig. 16b). In addition to the cross-linking of

Functional polymer–ceramic hybrid coatings

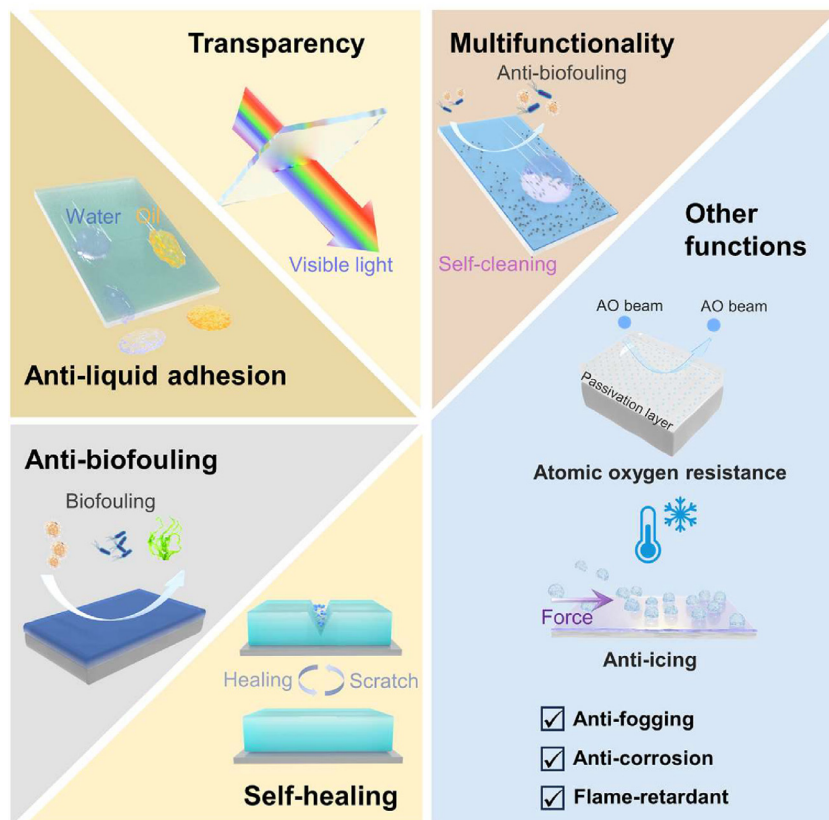


Fig. 15. Schematic illustration of different functional polymer–ceramic hybrid coatings.

3D-structured polysiloxanes themselves, transparent polymer–ceramic hybrid coatings could also be fabricated by combining hard 3D-structured polysiloxanes with flexible oligomers via a polymerization reaction (Fig. 16c) [131]. The incorporation of the urethane dimethacrylate oligomers into ladder-like structured methacrylate oligosiloxanes can improve the mechanical and optical properties of the coating. The urethane bonding of urethane dimethacrylate monomer reduced the accumulated stress and kept the hybrid material intact, whereas the sample without urethane bonding could not maintain its structural integrity and was cracked into pieces after curing (Fig. 16d). Such a film exhibited high hardness (8H pencil hardness) and transparency ($> 90.0\%$) comparable to conventional silicate glass, showing potential as a flexible window cover for foldable optoelectronic devices.

Differing from the above coatings with high transmittance and low haze, polymer–ceramic hybrid coatings with high transmittance and high haze are desirable in many applications. For example, these coatings can improve the efficiency of optical coupling in and out of OLED, solar cells, and backlight units [171]. These coatings can also efficiently reduce the glare effect of OLED displays [172]. The incorporation of nanoparticles into polymer–ceramic hybrid coatings can increase the transmission haze, but nanoparticles easily agglomerate and increase surface roughness. Alternatively, the use of fibrous materials such as cellulose and glass-fabrics can overcome the shortcomings of agglomeration and surface protrusion. Based on these concepts, a series of transparent and hazy coatings were developed by introducing glass-fabrics to ladder-like polysilsesquioxane functionalized with various phenyl and epoxy molar ratios [173]. By controlling the content of phenyl groups of the hybrid matrix, the refractive index of the hybrid matrix ($n_H = 1.50$) differed considerably from that of the glass-fabric

($n_{GF} = 1.56$), which increased the scattering angle and resulted in high total transmittance ($\sim 89\%$) and high haze ($\sim 89\%$) (Fig. 16e and f). Furthermore, OLEDs were successfully prepared on these films, and test results showed that their external quantum efficiency improved by 74%.

Besides, there are several other interesting examples combining transparent polymer–ceramic hybrid films with luminescent materials for flexible display applications. For example, quantum dots, as semiconductor nanoparticles, have considerable potential for application in LEDs. However, quantum dots are susceptible to heat, humidity, and chemicals. To overcome these shortcomings, quantum dots were embedded in silica and subsequently encapsulated by oligosiloxanes via the sol–gel reaction [174]. Luminescent materials can be chemically cross-linked with matrix. By covalently bonding red and green dyes to UV-curable oligosiloxanes, hybrid materials exhibited wide color expression [175]. Moreover, this hybrid method enabled the structure with long-term thermal stability and less temperature-dependent photoluminescence. Besides, perovskite nanoparticles could be chemically cross-linked in the matrix through free radical polymerization between perovskite nanoparticles colloids covered with unsaturated hydrocarbon ligands and oligosiloxanes functionalized with methacrylate, which could effectively inhibit the decomposition of perovskite crystals [176]. More interestingly, the surface defects of nanoparticles could be chemically passivated by water vapor when the ligands were unsaturated double bonds. The resulting films maintained the photoluminescence quantum for more than 1.5 years. Additionally, the ladder-like polysilsesquioxane features excellent thermal, mechanical, and chemical stability, which is suitable for application in organic field-effect transistors as dielectric materials. Capping the minor unstable Si–OH groups of ladder-like polysilsesquioxane

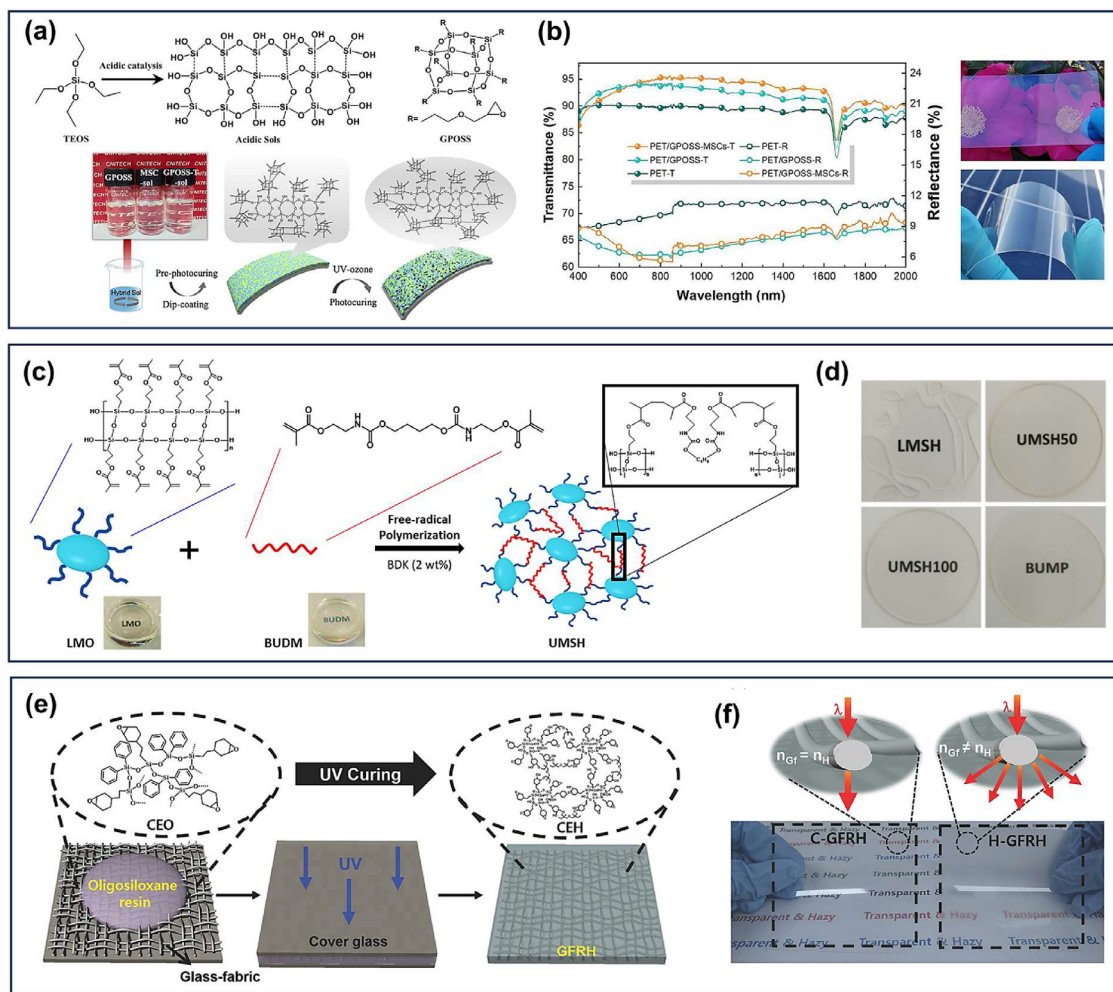


Fig. 16. (a) Chemical structures GPOSS and acidic silica sols derived from TEOS, and the fabrication of antireflection hybrid coatings. (b) Transmittance and reflectance spectra of GPOSS and GPOSS-MSCs coated PET films. [178], Copyright 2022. Adapted with permission from Elsevier Science Ltd. (c) Schematic illustration of the fabrication of urethane-siloxane hybrid coatings. (d) Photographs of the bulk materials with different butylene-urethane dimethacrylate content. [137], Copyright 2018. Adapted with permission from American Chemical Society. (e) Schematic illustration of the fabrication of the composite film reinforced by glass-fabrics. (f) The effect of the refractive index difference between the glass-fabrics (n_{Gr}) and the hybrid matrix (n_H). [173], Copyright 2018. Adapted with permission from John Wiley & Sons Inc.

with trimethylsilyl groups could improve its thermal stability [177].

3.2. Anti-liquid adhesion

Anti-liquid-adhesion surfaces have attracted considerable research attention in many fields, such as antifouling (water, oil, and corrosive liquids), self-cleaning, liquid collection, and drag reduction. Endowing polymer-ceramic hybrid coatings with anti-liquid adhesion ability can simultaneously improve their chemical corrosion resistance and self-cleaning properties, which increases the longevity of coated materials and devices [156]. To impart smooth coatings with the anti-liquid-adhesion ability, current strategies mainly focus on constructing high-mobility components on the coating surface to mimic the liquid-liquid interface, that is, preparing a "liquid-like" surface. These strategies do not rely on the fragile nano-/micro-scale hierarchical surface structures to ensure the liquid in the Cassie states [179,180]. Polymer-ceramic hybrid coatings with excellent anti-liquid adhesion ability were obtained by introducing suitable low-surface-energy components (long-chain alkyl, perfluoroalkyl, or PDMS) on the surface to reduce energy barriers of liquid sliding on the surface [181].

Long alkyl chains have been widely used for hydrophobic modification. In particular, alkylsilanes are used to toughen and en-

hance hydrophobicity in hybrid coatings fabricated by the one-step sol-gel strategy [4,182]. However, their oleophobicity is yet to be fully investigated. Urata et al. [183] first reported the "liquid-like" hybrid coating derived from alkylsilanes. The coating was prepared by a sol-gel reaction of alkylsilane and TMOS. It was found that the surface dynamic oleophobicity increased with the increase in the alkyl chain length and showed the best dynamic oleophobicity at $n = 12$. A further increase in the alkyl chain length would result in unsolidified opaque coatings. This dynamic oleophobicity can be attributed to the appropriate space between the alkyl chains for rotation/movement through the introduction of TMOS, as well as the high affinity between the alkane chain and the alkane liquid that enhances the mobility of the alkane chain.

A perfluorinated monomer is another commonly used low-surface-energy molecule [145]. For example, through the sol-gel reaction of fluorosilane and TEOS, a smooth coating with liquid-like surface properties could be produced [184]. Besides, the effect of perfluoroalkyl chain length on the dynamic oleophobicity of hybrid systems was investigated [185]. The authors theorized that the addition of TMOS effectively improved the motion of perfluoroalkyl chains and long perfluoroalkyl chains were not always the key to achieving dynamic oleophobic performance. Unfortunately, even though these studies have provided many methods for constructing anti-liquid adhesion hybrid coatings, their mechanical

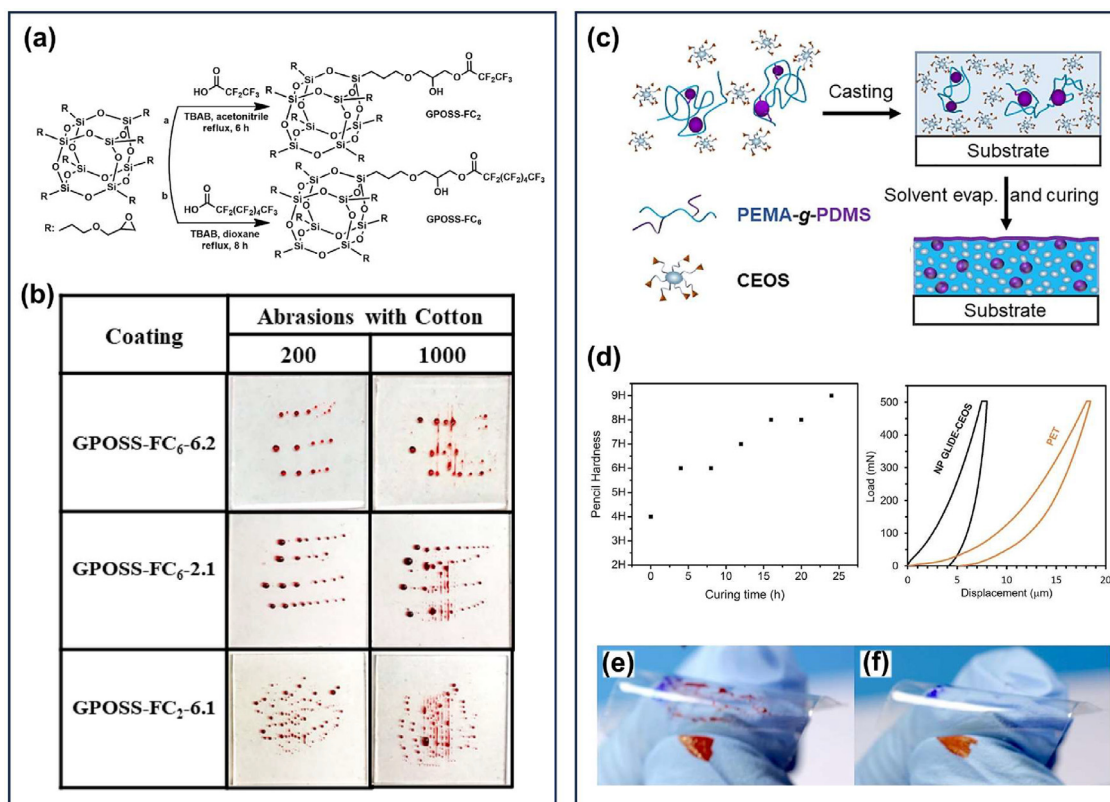


Fig. 17. (a) Synthesis of GPOSS-FC₂ and GPOSS-FC₆. (b) Photographs of the contraction patterns of ink traces varying with the number of wear times with cheesecloth on various coatings. [142], Copyright 2021. Adapted with permission from American Chemical Society. (c) Schematic illustration of the fabrication of the hybrid coating. (d) Pencil hardness of the hybrid coating with different curing time at 120 °C and nanoindentation curve of the hybrid coating and a PET reference film. (e) Photographs of the contraction patterns of ink traces on the coatings after bending into a U-shape for 500 times and (f) the coating surface after the ink has been erased. [86], Copyright 2020. Adapted with permission from Elsevier Science Ltd.

properties (e.g., hardness, flexibility, and adhesion) have been ignored.

Based on “liquid-like” surfaces, various polymer–ceramic hybrid coatings with anti-liquid adhesion ability have been successfully developed. Since low-surface-tension molecules can self-enrich at the coating surface during the formation of highly cross-linking networks, it is necessary to avoid macrophase separation that can affect the bulk performance. For example, utilizing the pentafluoropropionic acid (FC₂–COOH) or tridecafluoroheptanoic acid (FC₆–COOH) to react with excess GPOSS, the macro-phase separation was avoided and the corresponding coating exhibited excellent dynamic omniphobicity and transparency (Fig. 17a) [142]. Perfluorinated agents with higher molecular weight could impede surface reconstruction more effectively and exhibited superior abrasion resistance (Fig. 17b). In addition, low-surface-tension perfluorinated poly(propylene oxide) bearing one terminal carboxyl group could be used to react with the UV-curable ladder-like epoxycyclohexyl polysilsesquioxane for endowing hybrid coatings with dynamic omniphobicity [140]. During coating formation, low-surface-tension monomers were enriched on the surface and formed nanopools of a grafted lubricating ingredient, which endowed the coating (40-μm thick) with omniphobicity without compromising its *H* (1.4 GPa) and flexibility (<1 mm inward bending radius and <2 mm outward bending radius). The coating with 6.0 wt% fluorine retained its impressive ink contraction properties even after 500 abrasions by steel wool. In another study, an epoxy-amine reaction was utilized to cross-link fluorinated oligosiloxanes. First, the oligosiloxanes grafted with perfluorinated side chains were derived from the co-hydrolysis/condensation of functional silanes [186]. Next, the amino silane was introduced as a cur-

ing agent to react with the epoxy groups of oligosiloxanes and as an alkaline catalyst to accelerate the further condensation of oligosiloxanes. The resulting coating (156-nm thick) exhibited not only high hardness (8H pencil hardness) but also excellent self-cleaning properties.

PDMS, as a non-fluorinated alternative, is another typical surface treatment agent used for low-surface-energy modification [27,187,188]. Compared with the use of perfluorinated substances affecting the ecological environment, environmentally friendly and non-toxic PDMS is a promising candidate. Poly[ethylene-alternating-(maleic anhydride)] (PEMA) bearing PDMS side chains, as a flexible organic component, was introduced into cycloaliphatic epoxy–oligosiloxanes through heat-induced ring opening of the epoxide groups by maleic anhydride groups (Fig. 17c) [86]. The anti-liquid adhesion mechanism relied on the entrenchment of lubricating silicon segments with weak intermolecular forces and high molecular mobility on the coating surface [187]. The coating (40-μm thick) exhibited ceramic-like hardness (> 9H pencil hardness), polymer-like flexibility (U-shape with a side-to-side distance of ~0.8 cm), and repellency against various liquids (water, oil, and oily ink) (Fig. 17d–f).

The desirable function can be endowed not only by combining functional polymers but also by directly modifying 3D-structured polysiloxanes. PDMS was introduced into GPOSS by an epoxy-amine reaction. Then, the modified GPOSS was cross-linked via cationic ring-opening polymerization (Fig. 18a and b) [141]. During photocuring, low-surface-energy PDMS self-enriched on the coating (100-μm thick) surface to achieve dynamic omniphobicity without affecting flexibility (U-shape inward bending) and hardness (> 9H pencil hardness) (Fig. 18c and d). The

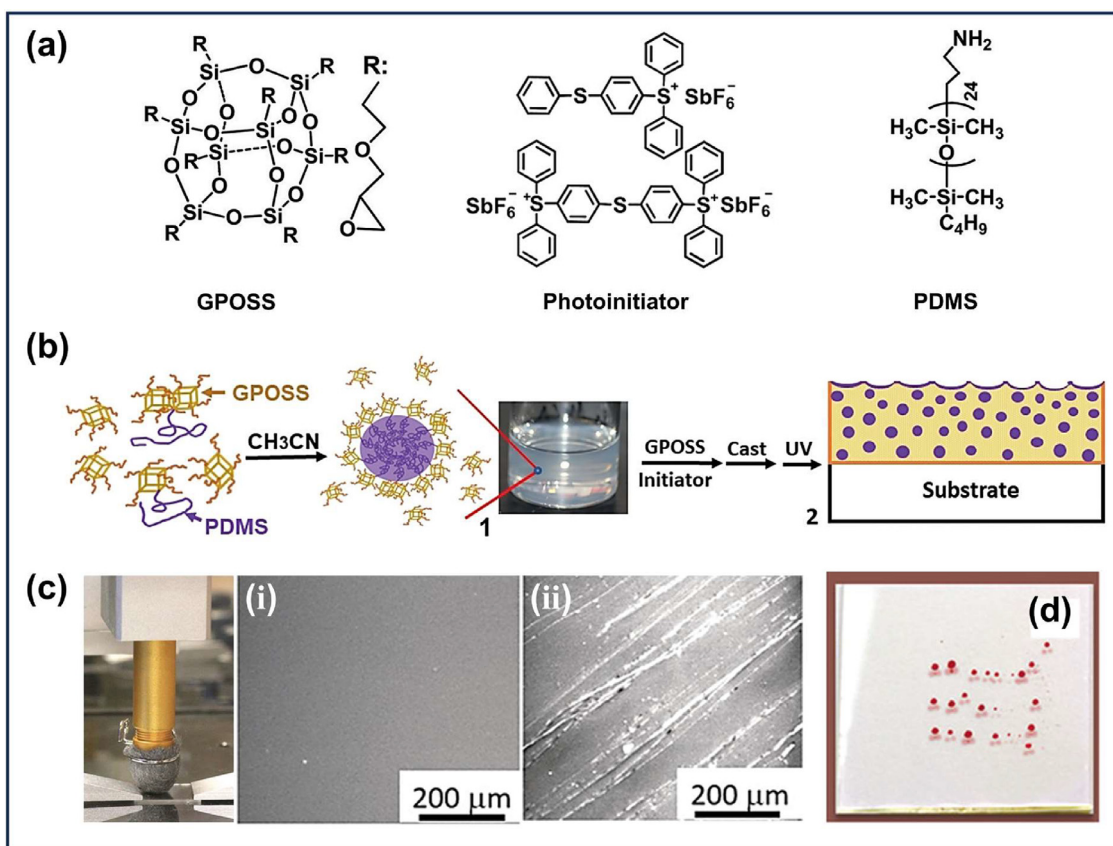


Fig. 18. (a) Chemical structures of GPOSS, photoinitiator, and PDMS. (b) Schematic illustration of the fabrication of GPOSS coatings through UV polymerization. (c) SEM images of the GPOSS coating (i) and PU coatings (ii) after the steel-wool wear test. (d) Photographs of the anti-smudge properties of the GPOSS coating. [141], Copyright 2019. Adapted with permission from John Wiley & Sons Inc.

omniphobicity of the resulting GPOSS coating changed negligibly even after 200 abrasions by steel wool or 500 times U-shape bending.

In contrast to “liquid-like” surfaces derived from covalent grafted hydrophobic alkyl, perfluoroalkyl, or PDMS, low-surface-energy monomers can also be physically blended with resin to form a semi-interpenetrating network. For example, utilizing the quick and efficient thiol–ene click reaction, sulfhydryl-functionalized POSS could react with poly(styrene-*b*-butadiene-*b*-styrene) (SBS) to form a hybrid network after 30 s UV irradiation in the air or underwater (Fig. 19a) [91]. The PDMS, which was originally physically dispersed in the resin, was immobilized in the hybrid network, thus forming a semi-interpenetrating network. The resulting coating surface exhibited excellent resistance to organic solvents as well as crude oil (Fig. 19b and c).

Chemisorption is another surface modification approach for endowing polymer–ceramic hybrid coatings with anti-liquid adhesion ability. Before the chemisorption of fluorosilane, the oxygen plasma-treated hybrid coating was coated through a hydrolyzed TEOS primer to increase hydroxyl groups on the surface for producing strong chemical bonds with fluorosilane [189]. The hydrophobicity and optical transparency of the fluoro-treatment coating (35-μm thick) changed negligibly even after 5000 abrasions by steel wool and 2000 abrasions by eraser. The combination of low-surface-energy chemistry and nano/micro-scale hierarchical roughness in a polymer–ceramic hybrid coating can prepare a superhydrophobic surface [179]. Hybrid coatings with moth-eye structures were fabricated using nanoimprint lithography [190]. The superhydrophobic hybrid coating was obtained from the self-assembly of a fluorosilane. Unlike soft polymeric materials, using flexible hard

hybrid materials as nanosized patterns provided excellent mechanical endurance.

3.3. Anti-biofouling

Biofouling adhesion and reproduction onto surfaces result in numerous detrimental consequences for electronic displays, biomedical engineering, maritime industry, and other fields [191–194]. Anti-biofouling coating is the most common and efficient method for anti-biofouling [195,196]. In particular, the demand for the simultaneous realization of excellent wear resistance and biofouling resistance in a coating has increased continuously since the breakout of the COVID-19 pandemic, which cannot be satisfied with conventional anti-biofouling coatings, such as silicone-based coating. Thus, many researchers have focused on endowing polymer–ceramic hybrid coatings with the anti-biofouling ability.

In 2005, Detty and coworkers successfully developed a series of hybrid coatings for marine biofouling release [48,197–200]. These coatings were fabricated based on sol–gel reaction, and their surface energy and morphology could be tuned through various combinations with different alkylsilanes. By using the facile sol–gel reaction between C8-TEOS and TMOS, the glassy hybrid coating with low critical surface energy was fabricated for releasing the settlement of algae and barnacles [47]. Unfortunately, the release efficiency of the coating for diatom stains was limited. The incorporation of 1 mol% of *n*-octadecyltrimethoxysilane (C18) with the long-chain alkyl group to the aforementioned hybrid coating slightly affected surface energies and elastic modulus but resulted in nano-scale topographies, which is responsible for the enhancement of fouling release ability toward sporelings *Ulva* and diatoms

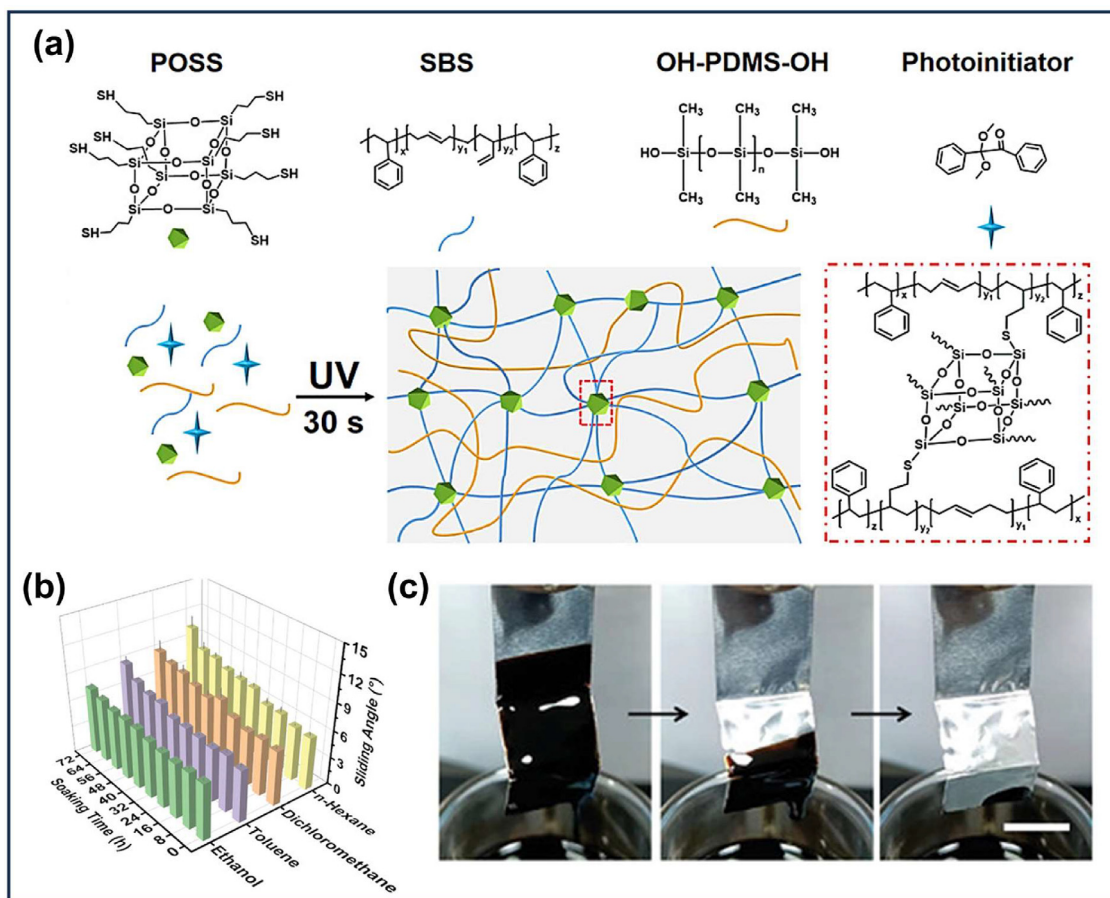


Fig. 19. (a) Schematic illustration of the fabrication of hybrid coatings through the thiol-ene click reaction between POSS and SBS under UV irradiation. (b) The ethanol sliding angle after soaking in various organic solvents for 72 h. (c) Photographs of the deformed coating surface in the anti-crude-oil-adhesion test. [91], Copyright 2022. Adapted with permission from John Wiley & Sons Inc.

Navicula [200]. However, these hybrid coatings exhibited poor flexibility and limited anti-biofouling ability under static conditions.

In recent years, to overcome these shortcomings, researchers have developed various hybrid coatings with biofouling resistance to enable them to exert excellent anti-biofouling performance under dynamic as well as static conditions. The current approaches can be classified into two categories, including physically mixing antifoulants (e.g., Cu_2O and enzymes) [9,201–203] and chemically attaching biofouling resistance moieties, including amphiphiles, zwitterions, quaternary ammonium salts, and antifoulants [204–207].

Generally, the antifoulants introduced by physical blending methods continuously leach out from the coatings, and their biofouling resistance efficiency remarkably degenerates when antifouling agents are completely leached out. In particular, both metallic copper antifoulants and natural antifoulants appeared in recent years may cause irreversible damage to the ecological environment and require evaluation of their impact on the environment before continuous use [208,209].

Chemically attaching biofouling resistant moieties onto hybrid coatings result in continuous anti-biofouling ability because of the non-leaching properties and minimizing the adverse effects on the ecological environment. Amphiphiles consist of polar and non-polar functionalities that prevent organisms from attaching in the first place [210]. Among them, PEG-based amphiphiles exert low adhesion to protein and cell through their steric excluded volume effects and hydration layer produced by hydrogen bonding, which have attracted considerable attention [211]. Our group introduced

amphiphilic telomers into low-surface-energy hybrid systems. The silane-bearing amphiphilic fluorocarbon acrylate-PEG telomers (FP) were reacted with TEOS and C8-TEOS through the sol-gel reaction to form a hybrid network [49]. We found that the surface of the coating with amphiphilic side chains could effectively deter the adhesion of marine bacteria *Pseudomonas* sp. (*P. sp.*) and diatoms *N. incerta* via self-enrichment of FP. Meanwhile, attributed to its low surface energy and low surface roughness, the hybrid coating exhibited excellent biofouling release ability. Notably, the introduction of FP remarkably increased its hardness and elastic modulus instead of degenerating these properties. Besides, these reactive amphiphilic telomers could chemically bond with epoxy-functionalized oligosiloxanes and function as anti-biofouling nano building blocks to prepare polymer-ceramic hybrid anti-biofouling coatings [84].

Chemically attached zwitterionic polymer coatings can effectively inhibit the adhesion of proteins and bacteria through strong electrostatically induced hydration [212,213]. Recently, our group prepared a fouling resistant epoxy-zirconium nanoparticle (ZP) through the sol-gel reaction of TPOZ, GPTMS, and sulfobetaine silane in which the zirconia core rendered the desired hardness, and the zwitterionic group enabled antibacterial capability. Next, amine-terminated hyperbranched polysiloxanes (HP) synthesized by KH550 were introduced into the ZP network through the epoxy-amine curing reaction as soft building blocks (Fig. 20a-c). It's also worth noting that the hyperbranched structure could cause more interaction sites with ZP and enabled the coating to undergo more elastic deformation. In this case, no crack appeared on the surface

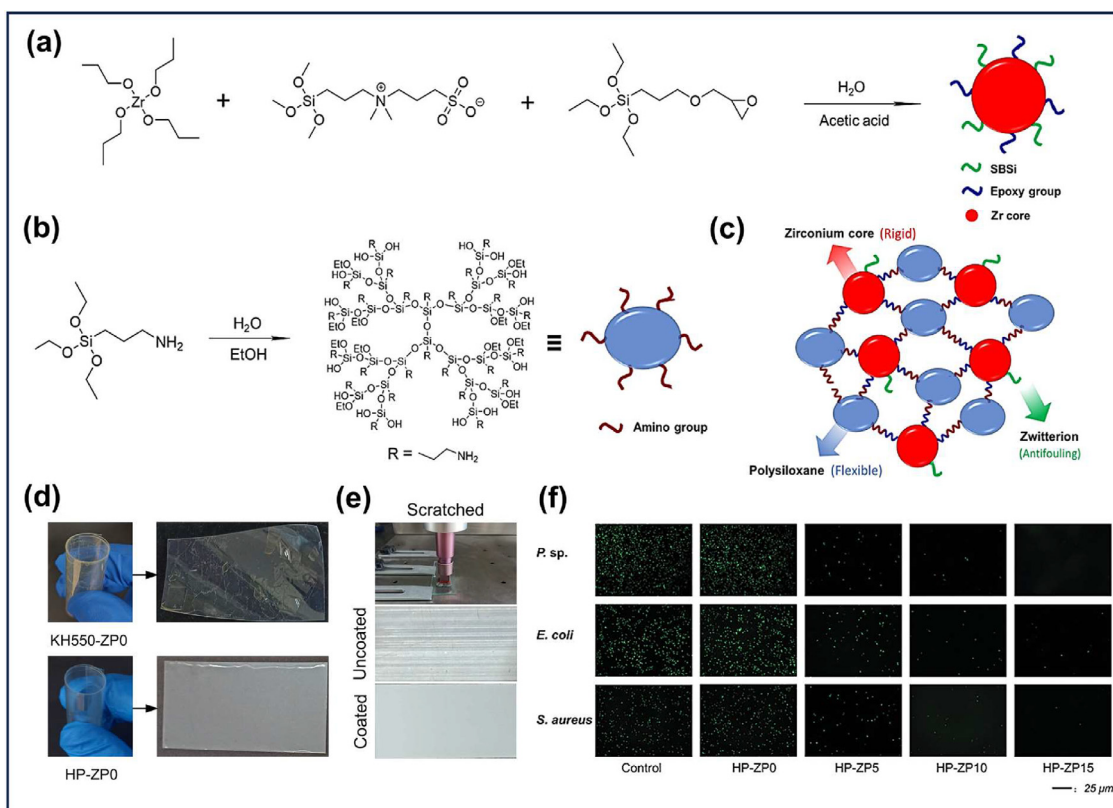


Fig. 20. Schematic illustration of (a) epoxy-zirconium particle, (b) amine-terminated hyperbranched polysiloxane, and (c) the hybrid network. (d) Photographs of the polymer-ceramic coatings in the rolling-up test. (e) Photographs of the uncoated and coated PET films after the steel-wool wear test. (f) Fluorescence microscopy images of *P. sp.*, *E. coli*, and *S. aureus* on the surface of hybrid coatings. [79], Copyright 2021. Adapted with permission from John Wiley & Sons Inc.

of HP-ZP0 after ten rolling-up cycles, whereas numerous cracks were observed on the surface of KH550-ZP0 after one rolling-up cycle (Fig. 20d). The resulting coating (10- μm thick) was transparent (> 99.5%), hard (7–9H pencil hardness), flexible (≤ 10 mm outward bending diameter), and scratch-resistant and exhibited excellent antibacterial capability against *Staphylococcus aureus* (*S. aureus*), *Escherichia coli* (*E. coli*), and *P. sp.* (Fig. 20e and f). The trade-off between flexibility, hardness, and anti-biofouling ability is critical for a coating [181]. This step-by-step hybrid strategy based on sol-gel reaction and epoxy-amine curing reaction shed new light on fabricating a new generation of anti-biofouling coatings.

Quaternary ammonium salts can cause bacterial membrane lysis by the contact-killing effect [214]. Based on the sol-gel reaction of 3-(trimethoxysilyl)propyldimethyloctadecyl ammonium chloride (TPDA) and TEOS, hybrid antibacterial coatings were developed and their surface hardness was more than twofold compared with that of uncoated polymethyl methacrylate (PMMA) [205]. Experimental results in a dynamic oral environment indicated that the mechanical strength and antibacterial activity of coated orthodontic devices were enhanced. These flexible hard hybrid antibacterial coatings have the potential to be used as coating materials for dental materials [206].

Chemically attaching antifoulants can effectively inhibit bacteria growth [215,216]. Antifoulants, such as borneol and isothiazolinone, have been used in hybrid coatings. For example, borneol acrylate was reacted with MOPMS and methyl methacrylate through free-radical polymerization for introduction into a hybrid network [217]. The effect of the borneol content on the hardness of coatings was examined, and the results revealed that the hardness decreased slightly with the increase in the borneol content. Although the results showed that borneol could exert antibacterial ability in coatings, the inhibition rates against *E. coli* and *Strepto-*

coccus mutans (*S. mutans*) were only 94.3% and 80.6%, respectively. Such inhibition rates were not suitable for application in harsh environments. Combining antifoulants with zwitterion could show synergistic biofouling resistance [204]. Furthermore, the authors found that introducing TEOS resulted in better anti-biofouling performance regardless of the type of antifoulants and zwitterionic groups. Although this phenomenon is notable, the related explanation is lacking.

Overall, chemically attaching biofouling resistant moieties, such as PEG, zwitterionic polymer, and quaternary ammonium salts into hybrid coatings results in continuous anti-biofouling ability. The non-leaching feature of these coatings minimizes the adverse effects on the ecological environment, which is the main development direction in the future. Meanwhile, considering that most of the biofouling resistance moieties are highly polar and hydrophilic, novel approaches are needed to maximize the anti-biofouling ability using a small amount of biofouling resistance moieties in order to minimize the damage of biofouling resistance moieties to the mechanical properties of the coating.

3.4. Self-healing

Endowing the materials with self-healing ability can repair damage and increase the longevity of materials. In biological systems, self-healing phenomena are common. A typical example is human skin, which can heal small cuts within hours. Inspired by biological systems, a large number of self-healing polymers including extrinsic self-healing polymers and intrinsic self-healing polymers have been developed [218]. However, most systems are based on soft materials, such as elastomers and hydrogels with low modulus and hardness [219–221]. Hard materials, such as polymer-ceramic hybrid coatings, are difficult to exhibit self-healing because

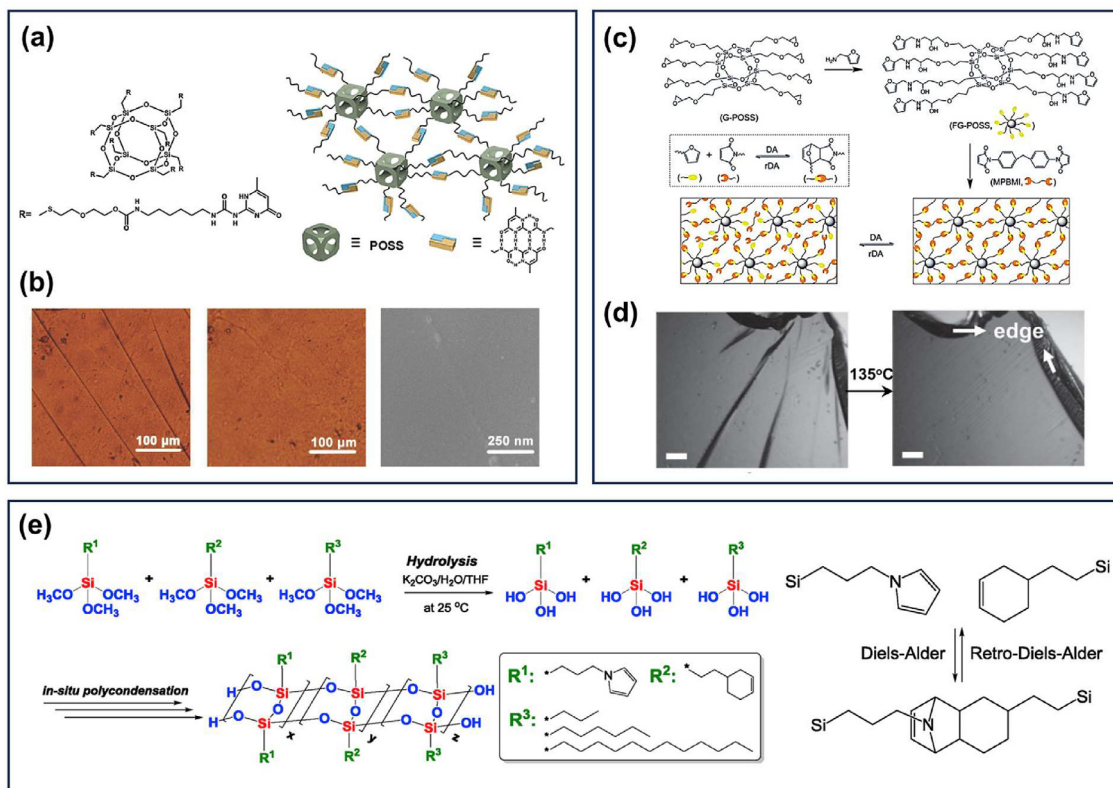


Fig. 21. (a) Schematic illustration of the UPy-POSS supramolecular coatings. (b) SEM images and optical microscope images of coating cracks were repaired at 80 °C for 2 min. [225], Copyright 2018. Adapted with permission from John Wiley & Sons Inc. (c) Schematic illustration of the fabrication of POSS hybrid materials based on the DA reaction. (d) Photographs of the crack-healing behavior. [227], Copyright 2013. Adapted with permission from Royal Society of Chemistry. (e) Synthesis of self-healing ladder-like polysilsesquioxane coatings based on the DA reaction. [95], Copyright 2017. Adapted with permission from Elsevier Science Ltd.

their highly cross-linking networks considerably restrict the mobility and entanglement of chain segments during self-healing [222].

To solve this problem, intrinsic self-healing polymer-ceramic hybrid coatings based on strong hydrogen bonds or dynamic covalent bonds have been developed [223,224]. 2-Ureido-4[1H]-pyrimidinone (UPy), as a quadruple hydrogen bond cross-linking motif, is widely used to prepare high-performance elastomers [218]. By grafting UPy groups onto POSS (UPy-POSS), UPy-POSS could form a robust 3D supramolecular network through self-complementary hydrogen bonds cross-linking (Fig. 21a) [225]. The supramolecular coating (~6.77 μm thick) had a Young's modulus of 4.5 ± 0.2 GPa, which was similar to PI (4.1 ± 0.1 GPa). The atomic oxygen resistance could be restored, and the micro-cracks on the coating could be healed after heating at 80 °C for 2 min (Fig. 21b). Therefore, the combination of strong hydrogen bonds and high grafting density provides an effective strategy for overcoming the limitation between mechanical strength and self-healing properties of self-healing polymer-ceramic hybrid coatings. Nevertheless, to obtain amorphous UPy-POSS supramolecular coating, the crystalline UPy powder needs to be hot-pressed and subsequently cooled. The preparation process is complex and coating substrates in a large area is difficult.

The Diels-Alder (DA) reaction has been used to develop many self-healing polymeric materials because of its thermally reversible feature and covalency robustness [226]. The introduction of the DA reaction into cross-linking points between inorganic components can endow the coating with thermally healable properties. For example, utilizing the DA reaction of furan-end functionalized POSS and bismaleimide, a self-healing hybrid coating with a compress modulus of 1.0 GPa and a yield strength of 118 MPa was prepared (Fig. 21c) [227]. Micro-cracks were healed after heating at 135 °C

for 30 min (Fig. 21d). However, the storage modulus and loss factor of the coating gradually decreased with the increase in heating cycles, which indicated that the retro-DA reaction became difficult.

The DA chemistry could also be introduced in ladder-like polysilsesquioxanes. A series of ladder-like polysilsesquioxanes consisting of inorganic Si backbone and organic periphery of diene, dienophile, and n-alkyl were reported (Fig. 21e) [95]. In this system, ladder-like polysilsesquioxanes had a well-defined structure and highly condensed network so that the thermal reversibility of the DA reaction was not affected. The effect of the alkyl chain length on the self-healing kinetics of coatings was investigated, and it was confirmed that a long alkyl chain length increased chain mobility. Besides, a long alkyl chain could plasticize the coating, resulting in a decrease in hardness and elastic modulus. The resulting coating could heal scratches and fully dissolve in a heated organic solvent after the decrosslinking retro-Diels-Alder reaction.

The combination of dynamic covalent bonds and covalent bonds in an appropriate ratio can maintain the cross-linking between polymer chains and simultaneously endow the network with dynamic characteristics. For example, UV-curable silanes with functional groups including acryloxypropyl, glycidoxypentyl, and cyclohexylepoxyethyl were reacted with diene-/dienophile-functionalized silane to prepare a multi-crosslinkable ladder-like polysilsesquioxanes [139]. After UV-curing, the pencil hardness of the coating increased from 6B to 3H. Moreover, the pencil hardness could be increased to 6H after DA cross-linking, which showed improved hardness compared with a single cross-linking method. These coatings (1.5-μm thick) exhibited a high hardness (6H pencil hardness) and elastic modulus (> 9 GPa). However, the films could only heal scratches with a thickness of < 10 μm, and its H/E^* value was low.

Considering the high proportion of inorganic components in hybrid coatings, the introduction of dynamic covalent bonds into inorganic networks is a promising approach for developing self-healing polymer–ceramic hybrid coatings. Actually, the Si–O–Si bond is reversible. In the presence of a strong base or acid, Si–O–Si bonds can undergo interesting chain exchange reactions [228,229]. This siloxane equilibration mechanism has been utilized to fabricate self-healing PDMS elastomers and stiff epoxy vitrimers [230,231].

Recently, the siloxane equilibration mechanism was extended and successfully applied to fabricate self-healing polymer–ceramic hybrid coatings. The introduction of triethylamine (TEA) into a highly condensed aminopropyl functionalized poly(silsesquioxane) network enabled silanol to form strong hydrogen bonds with TEA rather than aminopropyl, which accelerated the hydrolysis of the Si–O–Si network, this is, more free aminopropyl became available to attack silicon atoms upon contact with water [132]. Therefore, the Si–O–Si network was reversible when triggered by water. Soft polymeric micelles were uniformly dispersed in the Si–O–Si network and effectively improved flexibility. The resulting hybrid coating was hard yet flexible (a high H of ~ 0.83 GPa and a low E^* of ~ 5.89 GPa), and the scratches could be healed quickly under water vapor. However, with an increase in healing cycles, the healing rate gradually decreased because of TEA volatility.

Although various strong hydrogen bonds and dynamic covalent bonds have been reported to endow the self-healing ability with polymer–ceramic hybrid coatings, the stability of these systems needs to be emphasized when considering practical relevance. For instance, are the mechanical properties of the self-healing polymer–ceramic hybrid coating stable under high humidity environment? In addition, most of these systems are still brittle and require external stimuli to trigger the self-healing process. Future studies should simultaneously focus on improving the self-healing performance and stability of the coatings.

3.5. Other functions

In addition to the aforementioned functions, other functions, such as atomic oxygen resistance, anti-icing, anti-fogging, anti-corrosion, and flame-retardant, have been successfully incorporated into the polymer–ceramic hybrid coatings.

Atomic oxygen resistance. Upon exposure to atomic oxygen, normal organic polymeric materials degrade, which jeopardizes flight safety of spacecrafts in a low earth orbit environment (altitude range of 200–700 km) [232,233]. Atomic oxygen-resistant coating is the most effective approach, especially atomic oxygen-resistant polymer–ceramic hybrid coatings, which exhibits considerable potential for use in flexible membrane protection such as space membrane diffractive telescopes, and flexible solar cell arrays, among others. Using a heat-assisted hydrolysis and condensation reaction, perhydropolysilazane (PHPS) could be transformed into a nano-scale bicontinuous phase structure including PHPS and silica phases such that the PHPS provided flexibility and the silica provided hardness (Fig. 22a) [133]. Upon exposure to UV and atomic oxygen, the PHPS dominant phase self-evolved to the inorganic silica phases, exhibiting excellent atomic oxygen resistance (Fig. 22b). Besides, hybrid coatings based on POSS showed excellent atomic oxygen resistance by oxidating the surface to form a SiO₂ passivation layer [225]. Dynamic ester bonds formed by glutaric acid with epoxy resin and glycidoxypopyl POSS could ensure the recyclability of atomic oxygen-resistant coatings [234]. Similar to POSS, hyperbranched polysiloxanes have a high silicon content for forming SiO₂ passivation layers. Compared with POSS, hyperbranched polysiloxanes can be easily prepared at a low cost. The hyperbranched polysiloxane was synthesized by using a sol-gel reaction between PTMS and APMS [235]. The amino groups

of hyperbranched polysiloxanes could participate in the reaction between 4,4'-diaminodiphenyl ether (ODA) and pyromellitic dianhydride (PMDA), thereby chemically bonding to the PI. Upon exposure to atomic oxygen, the surface organic polyimide first degraded, whereas hyperbranched polysiloxanes formed a SiO₂ passivation layer. With the increase in the hyperbranched polysiloxane content, the erosion yield of the coating surface decreased. This method could regulate the compatibility of polyimide by regulating the organic functional group of organosilanes and integrating more silicon into the PI network without degrading other properties.

Anti-icing. At sub-zero temperatures, ice formation and adhesion on surfaces cause severe problems for infrastructure and facilities [153,236]. To solve this problem, polysilazane coatings incorporated with hydrophobic fluorine or PDMS have been reported [71,237]. Besides, the introduction of low-surface-energy perfluorinated silane and fillers via one-step sol-gel strategy can prepare flexible hard anti-icing coating. By modifying SiO₂ nanoparticles by 1H,1H,2H,2H-perfluorooctyltriethoxysilane (PFOTES) to reduce surface energy, PFOTES-SiO₂ nanoparticles could distribute uniformly in the hybrid precursor at a high weight percentage of 30% and impart the coating surface with hydrophobicity for anti-icing and self-cleaning applications [238]. Besides, low-surface-energy titanate nanotubes modified by PFOTES were incorporated into a hybrid network for anti-icing function and hardness improvement [239]. By optimizing the time of mechanical stirring, nanotube fillers were well-dispersed in the sol-gel precursor and formed a low-surface-energy smooth surface after curing, showing low ice-nucleation temperature and low adhesion to ice.

The step-by-step hybrid strategy can be used to prepare anti-icing coating [134]. Our group first synthesized epoxy-oligosiloxane nanoclusters and modified a part of them with PDMS to avoid macro-phase separation. Next, 4,4'-diaminodicyclohexylmethane was used to react not only with epoxy groups of oligosiloxane nanoclusters to provide coatings with a robust hybrid network but also with diglycerol decarbonate to form a non-isocyanate polyurethane network with a high density of polar groups (Fig. 22c). At the same time, the non-isocyanate polyurethane served as a plasticizer to improve flexibility. In particular, the hydrophobicity of coating improved after introducing PDMS, which is crucial for anti-icing (Fig. 22d). The resulting polymer–ceramic hybrid coating (200- μ m thick) exhibited strong adhesion to various substrates (2–8 MPa) and low adhesion to ice (~ 39 kPa) (Fig. 22e). Overall, current anti-icing polymer–ceramic hybrid coatings are mainly fabricated by low-surface-energy modification to reduce intermolecular forces and diffusion between the coating surface and ice. Furthermore, the low surface roughness characteristic avoids mechanical-interlocking-induced adhesion by ice.

Anti-fogging. Hybrid coatings with hydrophilicity and water uptake ability can be used for anti-fogging. These properties are tailored by a simple change of organosilanes. Through the sol-gel reaction of hydrophilic amino-functionalized trialkoxysilanes, hybrid coatings with high water uptake ability were reported for anti-fogging [240]. The organic–inorganic hybrid network enabled the coating with excellent elastic recovery ability after deformation and thus its scratch resistance was much better than that of the PVA film. However, when the residual silanol groups in the film were repeatedly exposed to high and low humidity, the condensation reaction occurred, which resulted in reduced water uptake and cracks on the coating surface, which severely limited its practical applications.

To overcome these shortcomings, the organically bridged trialkoxysilane was introduced into the hybrid network [241]. In this system, the hydrophilic ethylene glycol chain of bridged silane exerted space effect, which could prevent the remaining silanol

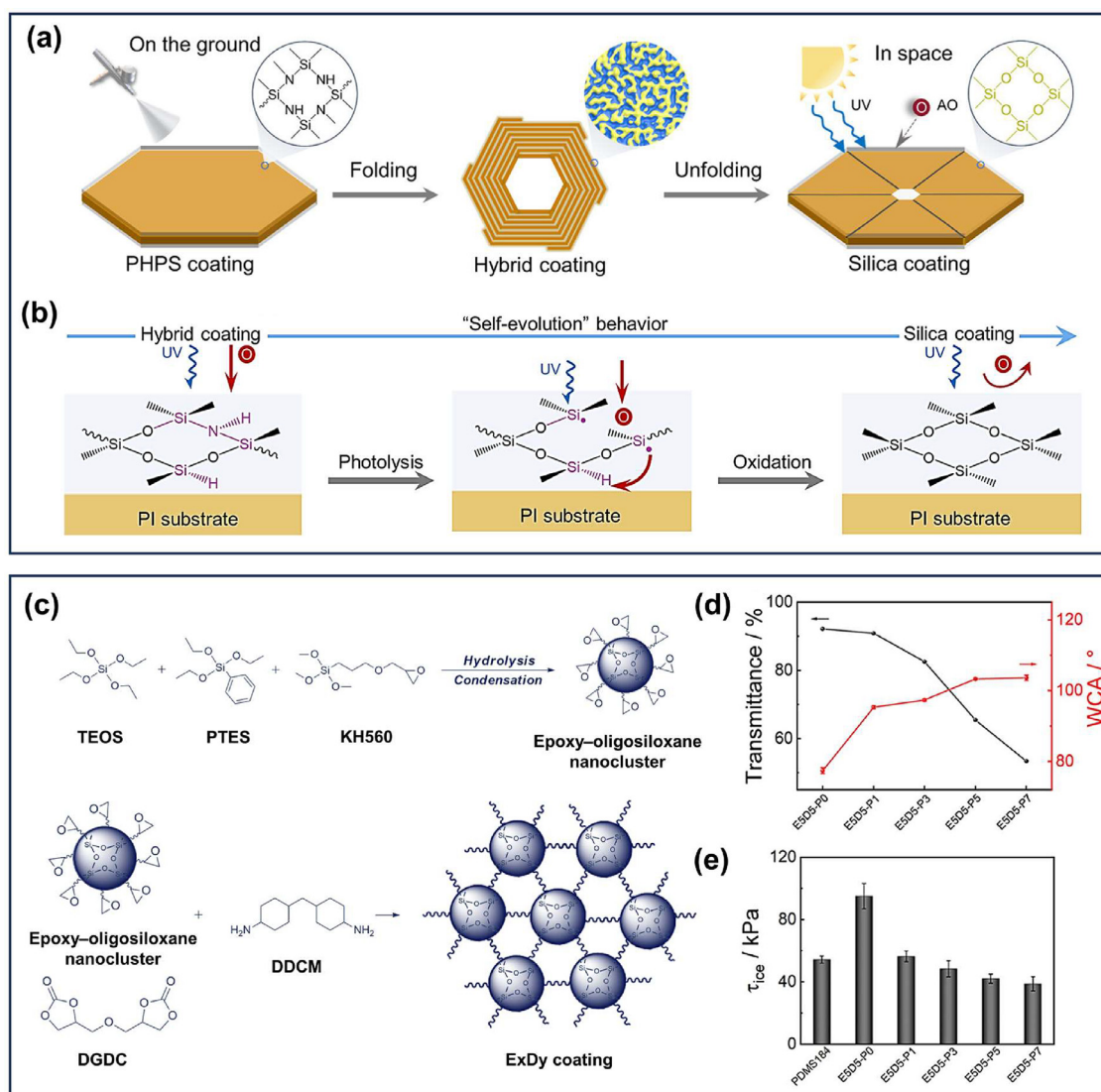


Fig. 22. (a) Schematic illustration of the fabrication of hybrid coatings on deployable membrane structures. (b) Schematic illustration of the chemical structure evolution of hybrid coatings under AO and UV synergistic radiation. [133], Copyright 2021. Adapted with permission from American Chemical Society. (c) Schematic illustration of the cross-linked network. Variations in (d) transmittance, water contact angle, and (e) ice adhesion strength (τ_{ice}) as functions of the PDMS content. [134], Copyright 2023. Adapted with permission from American Chemical Society.

groups from polycondensation when exposed to water vapor to achieve stable anti-fogging properties and simultaneously maintain scratch resistance. In addition, the hydrophilic hydroxyl group produced by epoxy-silane after immersion in the H_2SO_4 solution also provided hydrophilic and water uptake properties, which avoided the color change of coatings as the oxidation of amino groups.

Anti-corrosion. The corrosion of metals is a global problem and coating protection has become the most effective tool for anti-corrosion [70,242]. Conventional inorganic hard coatings used for anti-corrosion have inevitable problems, such as limited flexibility, high-temperature process, and easy cracking when forming thick films. By using the one-step sol-gel hybrid strategy, organic components can be readily introduced to improve toughness and obtain desirable functions. GPTMS/MTMS sol-gel coating could effectively delay corrosion on brass and bronze substrates [243]. Anti-corrosive hybrid coatings could also be prepared by the reaction between silane-functionalized epoxy resin and pre-hydrolyzed TEOS [244]. Hyperbranched polysiloxanes terminated by multiamine coupled with natural lacquer could considerably shorten the drying time and improve surface hardness and anti-

corrosive properties [245]. Besides, fluorine-modified polysilazanes were developed to form a low-surface-energy silicon-based coating for anti-corrosion [246]. The hardness and elastic recovery rate were superior to pure aluminum. The encapsulation of corrosion inhibitors within hybrid coatings is a promising approach for improving anticorrosive properties [247,248]. Besides, the introduction of ceramic powders into epoxy resin could improve corrosion resistance [249], and several formulations have been commercialized by large companies such as Freecom (USA) and Ceramic Polymer GmbH (Germany).

Flame-retardant. Endowing coatings with flame retardancy can considerably improve the safety of coated materials. In particular, with the growing popularity of large-size OLED screens and flexible electronics, polymer-ceramic hybrid coatings with flame retardancy have excellent application prospects. Phosphorus-containing organic amine flame retardants were covalently grafted to ladder-like polysilsesquioxanes by the epoxy-amine reaction and flame-retardant hybrid coatings were obtained after thermal curing [250]. Chemically grafted flame retardants could avoid producing opaque coatings and destroying the integrity of the hybrid network. The

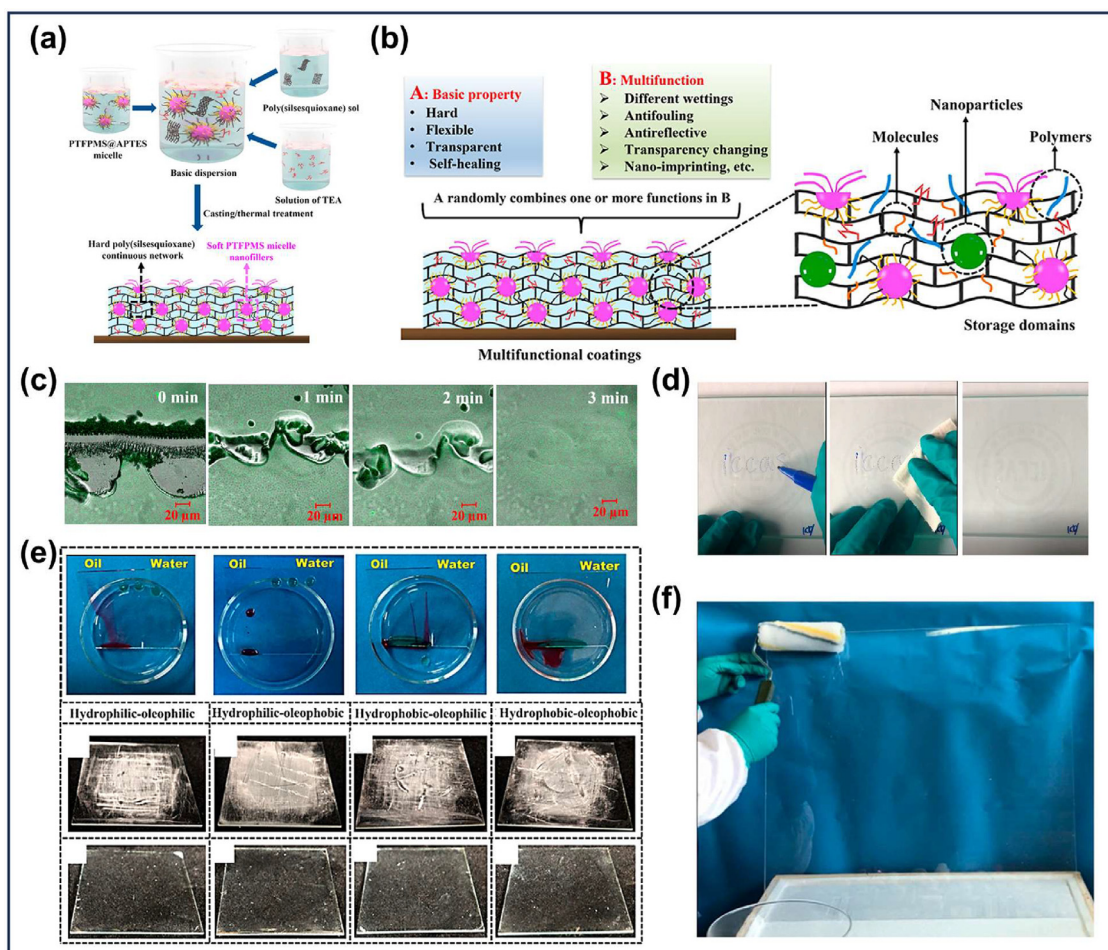


Fig. 23. Schematic illustration of (a) the preparation of hybrid networks and (b) the programmable incorporation of functional components to obtain multifunctional coatings. (c) Laser scanning confocal microscope images of the self-healing process of a scratch. (d) Photographs of the coating in the anti-graffiti test. (e) Photographs of coatings with various wettability before and after self-healing. (f) Photograph of large-scale preparation of coatings. [132], Copyright 2022. Adapted with permission from American Chemical Society.

effect of flame retardants on the flame retardancy of the coating was investigated, and the results revealed that when the content of flame retardants increased to 52.6 wt%, the coating sustainably burned for only 4 s after the flame was removed and the destroyed length was 16%. This phenomenon indicated that the coating had excellent flame retardancy. Besides, hyper-branched polysiloxanes containing phosphorus exhibited flame retardancy. The functional hyperbranched polysiloxane was synthesized through the co-polycondensation reaction among KH550, methyldiethanolamine, and triethyl phosphate [251]. The addition of phosphorus containing-hyperbranched polysiloxane remarkably improved flame retardancy and decreased the curing temperature of epoxy resins.

3.6. Multifunction

Considering that polymer-ceramic hybrid coatings with a single function cannot enhance the adaptability of coated materials in complex environments, imparting multifunctionality to polymer-ceramic hybrid coatings is desirable for satisfying high-performance requirements in industry. These coatings are highly crucial for the development of the next generation of protective coatings.

For example, integrating functions of self-healing and special wettability in a polymer-ceramic hybrid coating is desirable, but it

remains highly challenging. As discussed above, by combining the hard yet reversible Si–O–Si network and soft polymeric micelles using the one-step hybrid strategy, multifunctional hybrid coatings were developed (Fig. 23a and b) [132]. The prepared coating could heal scratches with an average width and depth of 50 μm after exposure to water vapor for 4 min (Fig. 23c). Based on the reversible hybrid network, additional functions could be introduced by physically mixing small amounts of compatible functional components. For instance, the introduction of low-surface-energy PDMS, fluorosilane, and stearic acid at appropriate content could endow the coating with excellent anti-graffiti ability (Fig. 23d). Furthermore, hydrophilicity and oleophilicity of coatings could be imparted by adding PVA. Hydrophilicity and oleophobicity of coatings could be achieved by adding PVA and fluorosilane. Hydrophobicity and oleophilicity of coatings could be realized by adding PDMS. Hydrophobicity and oleophobicity of coatings could be imparted by adding PDMS and fluorosilane (Fig. 23e). The coatings all exhibited excellent hardness, flexibility, and self-healing properties. This simple and programmable method allowed the coatings to be readily coated on various substrates on a large scale (Fig. 23f).

Simultaneously endowing a polymer-ceramic hybrid coating with transparency, anti-liquid adhesion, and anti-biofouling is desirable for various optical or display devices, especially devices in maritime industries and biomedical fields. Recently, our group

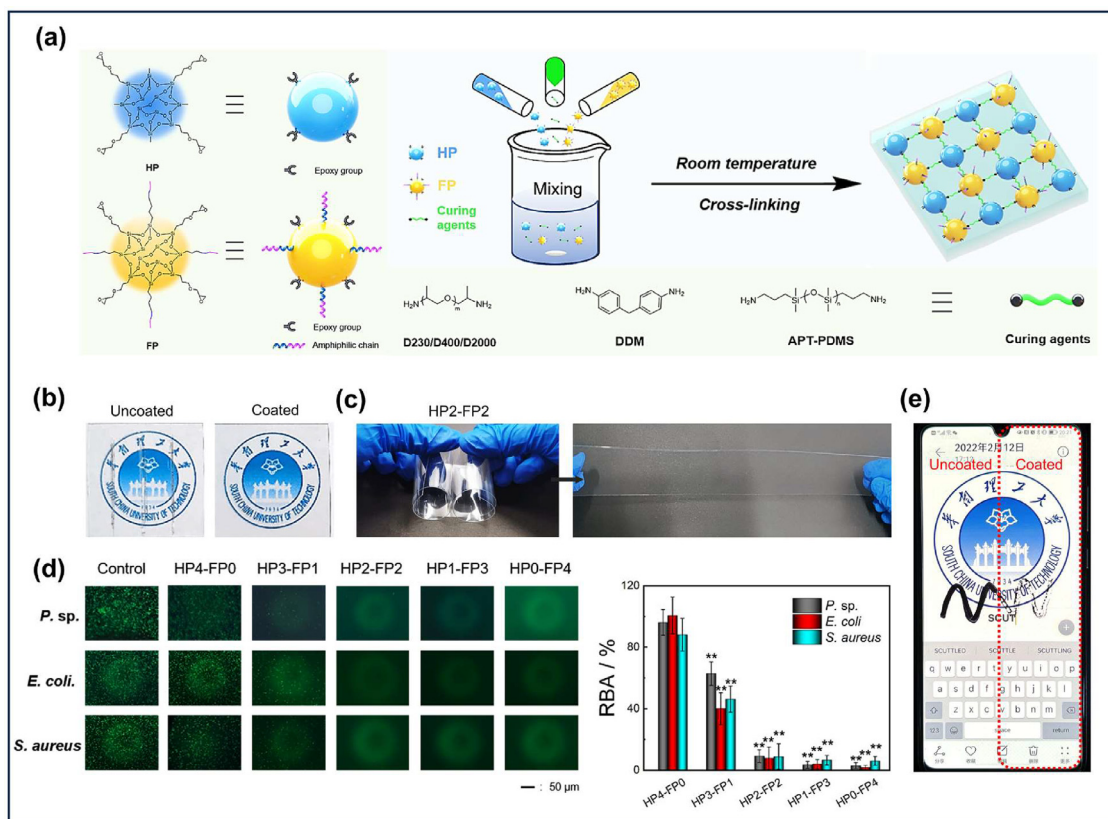


Fig. 24. (a) Schematic illustration of the fabrication of hybrid coatings. (b) Photographs of uncoated and coated PET films after the pencil scratching test. (c) Photograph of the hybrid coating in the rolling-up test. (d) Fluorescence microscopy images of *P. sp.*, *E. coli*, and *S. aureus* on the coated surface and the relative bacteria adhesion. (e) Photographs of the self-cleaning ability of the coated glass slide. [84], Copyright 2022. Adapted with permission from John Wiley & Sons Inc.

proposed a universal step-by-step hybrid strategy for fabricating multifunctional polymer–ceramic hybrid coatings based on the sol–gel chemistry and epoxy–amine curing system (Fig. 24a) [84]. Branched epoxy–oligosiloxane nanoclusters with rigid inorganic cores are used as the matrix, which can be crosslinked with amine-terminated curing agents to form transparent coatings. The linkage of the epoxy–amine renders reduction of the coating elastic modulus and improves the elastic recovery rate by introducing a flexible polymer chain. In this case, even if the hardness of the material was not sufficiently high, the lower effective modulus (high H/E^* value) and high elasticity caused the coating to have high wear resistance. The multifunctional hybrid coating was transparent, hard (6–7H pencil hardness) but flexible (10 mm outward bending diameter) (Fig. 24b and c). Furthermore, by using this strategy, the function of the coating was programmable. The anti-biofouling properties could be imparted by chemically bonding with amphiphilic telomer functionalized-oligosiloxane nanoclusters via the epoxy–amine reaction. The resulting coating exhibited considerably low bacterial adhesion against *P. sp.*, *E. coli*, and *S. aureus* (Fig. 24d). On the other hand, the anti-liquid adhesion properties could be imparted by introducing amine-terminated PDMS. By chemically bonding a small amount of PDMS to the matrix and brushing it onto the mobile phone screen, the coated screen exhibited superior anti-smudge ability (Fig. 24e). Furthermore, the fouling release ability of coatings could be enhanced by introducing PDMS. It should be noted that these coatings also showed outstanding durability. Even after 400 abrasions by steel wool, they still exhibited excellent self-cleaning and antibacterial performance. This simple and controllable strategy provides a novel pathway for achieving the industrialization application of multifunctional polymer–ceramic hybrid coatings.

4. Conclusions and outlook

The combination of ceramic-like hardness and polymer-like flexibility in a coating provides irreplaceable protection for numerous applications. In particular, based on the organic–inorganic hybrid approach, the functions of polymer–ceramic hybrid coatings can be easily tailored for many emerging equipment in the field of foldable displays, flexible and wearable electronics, maritime industries, and biomedical engineering. In this review, we have provided an overview of the advances and design strategies of functional polymer–ceramic hybrid coatings based on the organic–inorganic hybrid approach, including a one-step hybrid strategy and a step-by-step hybrid strategy. The corresponding synthetic method, functions, and mechanical properties improvement were introduced in detail and systematically demonstrated.

For one-step hybrid strategy based on the sol–gel reaction, the organic groups of the organosilane precursor can be tuned in a broad range to obtain the target function. However, an additional thermal process is required to increase the condensation degree and form a dense inorganic network. Furthermore, the functional moieties should be end-capped with silanes, Si–OH, or other coupling groups that can participate in the sol–gel reaction. Polymer–ceramic hybrid coatings derived from the one-step hybrid strategy of cyclosiloxanes or polysilazanes exhibit good mechanical properties and can be prepared under mild conditions, but there are few studies in the field of flexible hard coatings, and further research is required. Specifically, the step-by-step hybrid strategy exhibits many advantages, including structural tunability, mild reaction conditions, and easy operation. Notably, polysiloxanes with various 3D structures (branched, cage-like, ladder-like, and spherical) can be tailored based on sol–gel chemistry and used as building

blocks. These building blocks have tailor-designed active sites such as epoxy, amine, vinyl, acrylate, thiol, etc. to enable efficient organic reactions. After post-curing (e.g., photo and thermal) of the building blocks, the highly cross-linking hybrid network is obtained directly. This method can achieve molecular-scale hybridization between inorganic and functional organic polymer phases. Other modified nanomaterials, such as Si-CPDs and BN nanoplatelets can also be used as building blocks. Furthermore, the corporation of filler or structure reinforcement (3D structures, LBL assembly structure, and gradient structure) can control the physicochemical properties of hybrid coatings.

We also focus on how to endow polymer-ceramic hybrid coatings with transparency, anti-liquid adhesion, anti-biofouling, self-healing, atomic oxygen resistance, anti-icing, anti-fogging, anti-corrosion, flame-retardant or multifunction. The organic-inorganic hybrid approach provides a feasible pathway for fabricating functional polymer-ceramic hybrid coatings, enriching the classical methodology used to manufacture flexible hard coatings. Although these polymer-ceramic hybrid coatings vastly circumvent the limitation of simultaneously achieving superior flexibility and hardness in a coating and exerting their function, they still have several limitations and technical challenges in some fields of application.

i) Programmable integration of multifunction

The integration of various functions in a coating can enhance the adaptability of coated materials in complex environments, which is a new direction worthy of further research. However, the introduction of an additional function may interfere with the performance of the original function; therefore, some trade-offs must be made according to the corresponding application scenario. For example, transparent polymer-ceramic hybrid coatings with anti-biofouling ability are promising to find applications in maritime facilities and biomedical devices (e.g., underwater optical sensors and dental materials). However, the introduction of hydrophilic anti-biofouling moieties affects the mechanical properties of the coating. A trade-off exists between the anti-biofouling ability of the surface and the mechanical properties of the bulk. The smart design of anti-biofouling polymer-ceramic hybrid coatings combined with step-by-step hybrid and self-enrichment strategies is expected to break through this limitation. For self-healing polymer-ceramic hybrid coatings, a fast and efficient self-healing ability can be achieved in some intrinsic self-healing systems. However, most of these systems are still brittle and require external stimuli to trigger the self-healing process. Therefore, future studies should focus on realizing effective self-healing ability at mild conditions without comprising mechanical properties and original functions.

ii) Durability of polymer-ceramic hybrid coatings

Durability is critical for achieving excellent industrialization application of polymer-ceramic hybrid coatings. Although considerable progress has been achieved in the coexistence of hardness and flexibility of the coating, limited studies have been conducted on their durability, especially for surface functionalized coatings. Because when the surface functional molecules are destroyed under long-term physical or chemical corrosion, even if the coating bulk properties are maintained, its surface functions degenerate considerably. In particular, some coatings require long-term immersion in the liquid environment (e.g., ships and pipelines), and their durability differs considerably from that in the air environment. Therefore, the durability of the coatings should be evaluated under the corresponding application environmental conditions. Future studies should focus on the improvement of the durability of coatings for multi-scenario applications.

iii) Standardization and comparison of different polymer-ceramic hybrid coatings

Another major challenge plaguing the field is the lack of standardized test methods for assessing polymer-ceramic hybrid coatings. Because many authors use different wear pressures and bending diameters to evaluate their wear resistance and bending resistance separately, the accurate comparison between these systems becomes difficult. In particular, considering that the coatings with different functions have different application scenarios and serving environments, corresponding standards for reliable evaluation should be established. Additional challenges in this coating are the absence of predictive models by computation using artificial intelligence and machine learning, which can be instrumental in the design of polymer-ceramic hybrid coatings with desired properties.

Overall, functional polymer-ceramic hybrid coating is an ever-growing field, and we hope this review will encourage readers to broaden the application of polymer-ceramic hybrid coatings in foldable displays, flexible and wearable electronics, maritime industries, and biomedical engineering, among others.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Zhenqiang Zhang: Writing – original draft, Validation, Investigation. **Yinjie Huang:** Validation, Investigation. **Qingyi Xie:** Supervision. **Guojun Liu:** Writing – review & editing, Supervision. **Chunfeng Ma:** Writing – review & editing, Supervision, Conceptualization. **Guangzhao Zhang:** Supervision.

Data availability

Data will be made available on request.

Acknowledgments

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References

- [1] Choi MC, Kim Y, Ha CS. Polymers for flexible displays: from material selection to device applications. *Prog Polym Sci* 2008;33:581–630. doi:[10.1016/j.progpolymsci.2007.11.004](#).
- [2] Fang WF, Mu Z, He Y, Kong KR, Jiang K, Tang RK, Liu ZM. Organic-inorganic covalent-ionic molecules for elastic ceramic plastic. *Nature* 2023;619:293–9. doi:[10.1038/s41586-023-06117-1](#).
- [3] Novak BM. Hybrid nanocomposite materials-between inorganic glasses and organic polymers. *Adv Mater* 1993;5:422–33. doi:[10.1002/adma.19930050603](#).
- [4] Schottner G. Hybrid sol-gel-derived polymers: applications of multifunctional materials. *Chem Mater* 2001;13:3422–35. doi:[10.1021/cm011060m](#).
- [5] Bao MD, Zhu XD, He JW. Evaluation of toughness of hard coatings. *Surf Eng* 2013;22:11–14. doi:[10.1179/174329406x85056](#).
- [6] Sproul WD. New routes in the preparation of mechanically hard films. *Science* 1996;273:889–92. doi:[10.1126/science.273.5277.889](#).
- [7] Ritchie RO. The conflicts between strength and toughness. *Nat Mater* 2011;10:817–22. doi:[10.1038/nmat3115](#).
- [8] Fan W, Du T, Droce A, Jensen LR, Youngman RE, Ren X, Gurevich L, Bauchy M, Kristensen P, Xing B, Yu D, Smedskjaer MM. Resolving the conflict between strength and toughness in bioactive silica-polymer hybrid materials. *ACS Nano* 2022;16:9748–61. doi:[10.1021/acsnano.2c03440](#).
- [9] Ciriminna R, Bright FV, Pagliaro M. Ecofriendly antifouling marine coatings. *ACS Sustain Chem Eng* 2015;3:559–65. doi:[10.1021/sc500845n](#).

- [10] Ngoi KH, Wong JC, Chia CH, Jin KS, Kim H, Kim HC, Kim HJ, Ree M. Inorganic-organic nanocomposite networks: structure, curing reaction, properties, and hard coating performance. *Compos Sci Technol* 2022;218:109112. doi:10.1016/j.compscitech.2021.109112.
- [11] Zhu YW, Liu W, Ngai T. Polymer coatings on magnesium-based implants for orthopedic applications. *J Polym Sci* 2021;60:32–51. doi:10.1002/pol.20210578.
- [12] Musil J, Čiporová Z, Čerstvý R, Haviar S. Flexible hard (Zr, Si) alloy films prepared by magnetron sputtering. *Thin Solid Films* 2019;688:137216. doi:10.1016/j.tsf.2019.03.028.
- [13] Chen H, Zheng BC, Li YG, Wu ZL, Lei MK. Flexible hard TiAlSiN nanocomposite coatings deposited by modulated pulsed power magnetron sputtering with controllable peak power. *Thin Solid Films* 2019;669:377–86. doi:10.1016/j.tsf.2018.10.031.
- [14] Musil J, Sklenka J, Čerstvý R. Transparent Zr–Al–O oxide coatings with enhanced resistance to cracking. *Surf Coat Tech* 2012;206:2105–9. doi:10.1016/j.surfcoat.2011.09.035.
- [15] Wang ZY, Li XW, Wang X, Cai S, Ke PL, Wang AY. Hard yet tough V–Al–C–N nanocomposite coatings: microstructure, mechanical and tribological properties. *Surf Coat Tech* 2016;304:553–9. doi:10.1016/j.surfcoat.2016.07.061.
- [16] Penkov OV, Khadem M, Kim DE. Hard, flexible, and transparent nanolayered SiN(x)/BN periodical coatings. *ACS Appl Mater Interfaces* 2019;11:9685–90. doi:10.1021/acsami.8b22091.
- [17] Musil J, Blažek J, Fajfrlík K, Čerstvý R, Prokšová Š. Antibacterial Cr–Cu–O films prepared by reactive magnetron sputtering. *Appl Surf Sci* 2013;276:660–6. doi:10.1016/j.apsusc.2013.03.150.
- [18] Somiseti V, Allauddin S, Narayan R, Raju KVS. Flexible, hard, and tough bioinspired polyurethane thermosets from renewable materials: glycerol and 10-undecenoic acid. *J of Coat Technol and Res* 2017;15:199–210. doi:10.1007/s11998-017-9998-2.
- [19] Xu JJ, Asatekin A, Gleason KK. The design and synthesis of hard and impermeable, yet flexible, conformal organic coatings. *Adv Mater* 2012;24:3692–6. doi:10.1002/adma.201201197.
- [20] Wang XQ, Cao LJ, Hu Y, Chen YF, Jin T, Huang J, Zhang XH, Lin SD. Highly transparent, hydrophobic, hard and flexible coatings based on a novel melamine-formaldehyde resin synthesized by hydrophobic melamine. *Prog Org Coat* 2023;179:107487. doi:10.1016/j.porgcoat.2023.107487.
- [21] Fu JC, Yu HJ, Wang L, Lin L, Khan RU. Preparation and properties of UV-curable hyperbranched polyurethane acrylate hard coatings. *Prog Org Coat* 2020;144:105635. doi:10.1016/j.porgcoat.2020.105635.
- [22] Fu JC, Yu HJ, Wang L, Fahad S. Preparation and properties of UV-curable diamine-based polyurethane acrylate hard coatings. *Appl Surf Sci* 2020;533:147442. doi:10.1016/j.apsusc.2020.147442.
- [23] Li RA, Zhang KL, Chen GX, Su B, He MH. Stiff, self-healable, transparent polymers with synergistic hydrogen bonding interactions. *Chem Mater* 2021;33:5189–96. doi:10.1021/acs.chemmater.1c01242.
- [24] Ma S, Webster DC, Jabben F. Hard and flexible, degradable thermosets from renewable bioresources with the assistance of water and ethanol. *Macromolecules* 2016;49:3780–8. doi:10.1021/acs.macromol.6b00594.
- [25] Wang X, Wang Y, Bi S, Wang YG, Chen XG, Qiu LY, Sun JQ. Optically transparent antibacterial films capable of healing multiple scratches. *Adv Funct Mater* 2014;24:403–11. doi:10.1002/adfm.201302109.
- [26] Fu K, Tian Y, Zhu ZT, Zhang GX, Xie JJ, Zhang QY. Robust and highly transparent photocurable fluorinated polyurethane coating prepared via thiol-click reactions and what essentially influences omniphobic coating's anti-graffiti properties. *ACS Appl Polym Mater* 2022;4:8386–95. doi:10.1021/acspap.2c01311.
- [27] Gao P, Wang YL, Wang J, Wang FS, Ma W, Zhang ZZ, Men XH, Lu Y. Rational design of durable anti-fouling coatings with high transparency, hardness, and flexibility. *ACS Appl Mater Interfaces* 2022;14:29156–66. doi:10.1021/acsami.2c04279.
- [28] Son DH, Bae HE, Bae MJ, Lee S-H, Cheong IW, Park YI, Jeong J-E, Kim JC. Fast, localized, and low-energy consumption self-healing of automotive clearcoats using a photothermal effect triggered by NIR radiation. *ACS Appl Polym Mater* 2022;4:3802–10. doi:10.1021/acspap.1c01768.
- [29] Sheng YM, Wang MH, Zhang KP, Wu ZY, Chen YX, Lu X. An “inner soft external hard”, scratch-resistant, self-healing waterborne poly(urethane-urea) coating based on gradient metal coordination structure. *Chem Eng J* 2021;426:131883. doi:10.1016/j.cej.2021.131883.
- [30] Launey ME, Buehler MJ, Ritchie RO. On the mechanistic origins of toughness in bone. *Ann Rev Mater Res* 2010;40:25–53. doi:10.1146/annurev-matsci-070909-104427.
- [31] Yu YD, Kong KR, Tang RK, Liu ZM. A bioinspired ultratough composite produced by integration of inorganic ionic oligomers within polymer networks. *ACS Nano* 2022;16:7926–36. doi:10.1021/acsnano.2c00663.
- [32] Leyland A, Matthews A. Design criteria for wear-resistant nanostructured and glassy-metal coatings. *Surf Coat Tech* 2004;177-178:317–24. doi:10.1016/j.surfcoat.2003.09.011.
- [33] Musil J. Flexible hard nanocomposite coatings. *RSC Adv* 2015;5:60482–95. doi:10.1039/c5ra09586g.
- [34] Matthews A, Franklin S, Holmberg K. Tribological coatings: contact mechanisms and selection. *J Phys D-Appl Phys* 2007;40:5463–75. doi:10.1088/0022-3727/40/18/s07.
- [35] Zok FW, Miserez A. Property maps for abrasion resistance of materials. *Acta Mater* 2007;55:6365–71. doi:10.1016/j.actamat.2007.07.042.
- [36] Musil J, Sklenka J, Čerstvý R, Suzuki T, Mori T, Takahashi M. The effect of addition of Al in ZrO₂ thin film on its resistance to cracking. *Surf Coat Tech* 2012;207:355–60. doi:10.1016/j.surfcoat.2012.07.017.
- [37] Musil J, Jirout M. Toughness of hard nanostructured ceramic thin films. *Surf Coat Tech* 2007;201:5148–52. doi:10.1016/j.surfcoat.2006.07.020.
- [38] E. FJ. Compositions and methods for a high performance protective coating. US Patent US5936022A 1999.
- [39] Kim Y.S., Liang S.K., Choi D.W., Eui Y.K., Jung W.Y. Water-resistant and anti-corrosion ceramic coating material using nanoscale flakes. CN Patent CN201180005779 2015.
- [40] Graham T. XXXV.—On the properties of silicic acid and other analogous colloidal substances. *J Am Chem Soc* 1864;17:318–27. doi:10.1039/js8641700318.
- [41] Stöber Werner, Fink A. Controlled growth of monodisperse silica spheres in the micron size range. *J Colloid Interface Sci* 1968;26:62–9. doi:10.1016/0021-9797(68)90272-5.
- [42] Sumida K, Liang K, Reboul J, Ibarra IA, Furukawa S, Falcato P. Sol-gel processing of metal-organic frameworks. *Chem Mater* 2017;29:2626–45. doi:10.1021/acs.chemmater.6b03934.
- [43] Tsvetkova IN, Krasil'nikova LN, Khoroshavina YV, Galushko AS, Frantsuzova Yu V, Kychkin AK, Shilova OA. Sol-gel preparation of protective and decorative coatings on wood. *J Sol-Gel Sci Technol* 2019;92:474–83. doi:10.1007/s10971-019-04996-3.
- [44] Kim YH, Lee I, Lee H, Kang S-M, Lee Y, Kim S, Bae B-S. Sol-gel synthesized siloxane hybrid materials for display and optoelectronic applications. *J Sol-Gel Sci Technol* 2021. doi:10.1007/s10971-021-05491-4.
- [45] Sfameni Silvia, Rando G, Plutino MR. 'Perspective chapter: functional sol-gel based coatings for innovative and sustainable applications'. *Sol-gel method - recent advances. Sol-Gel Method - Recent Adv: IntechOpen* 2023. doi:10.5772/intechopen.110514.
- [46] Palmisano G, Bourhis EL, Ciriminna R, Tranchida D, Pagliaro M. ORMOSIL thin films: tuning mechanical properties via a nanochemistry approach. *Langmuir* 2006;22:11158–62. doi:10.1021/la061520w.
- [47] Tang Y, Finlay JA, Kowalke GL, Meyer AE, Bright FV, Callow ME, Callow JA, Wendt DE, Dettly MR. Hybrid xerogel films as novel coatings for antifouling and fouling release. *Biofouling* 2005;21:59–71. doi:10.1080/08927010500070935.
- [48] Dettly MR, Ciriminna R, Bright FV, Pagliaro M. Environmentally benign sol-gel antifouling and foul-releasing coatings. *Acc Chem Res* 2014;47:678–87. doi:10.1021/ar400240n.
- [49] Chen RZ, Xie QY, Zeng HH, Ma CF, Zhang GZ. Non-elastic glassy coating with fouling release and resistance performances. *J Mater Chem A* 2020;8:380–7. doi:10.1039/c9ta09794e.
- [50] Wang LM, McCarthy TJ. Covalently attached liquids: instant omniphobic surfaces with unprecedented repellency. *Angew Chem Int Ed* 2016;55:244–8. doi:10.1002/anie.201509385.
- [51] Wang Y, He JH. Fabrication of ultra-smooth hybrid thin coatings towards robust, highly transparent, liquid-repellent and antimicrobial coatings. *J Colloid Interface Sci* 2021;594:781–90. doi:10.1016/j.jcis.2021.03.077.
- [52] Chen XB, Zhou SX, You B, Wu LM. Mechanical properties and thermal stability of ambient-cured thick polysiloxane coatings prepared by a sol-gel process of organoalkoxysilanes. *Prog Org Coat* 2012;74:540–8. doi:10.1016/j.porgcoat.2012.01.021.
- [53] Catauro M, Bollino F, Papale F. Preparation, characterization, and biological properties of organic-inorganic nanocomposite coatings on titanium substrates prepared by sol-gel. *J Biomed Mater Res A* 2014;102:392–9. doi:10.1002/jbm.a.34721.
- [54] Rodič P, Iskra J, Milošev I. A hybrid organic-inorganic sol-gel coating for protecting aluminium alloy 7075-T6 against corrosion in Harrison's solution. *J Sol-Gel Sci Technol* 2014;70:90–103. doi:10.1007/s10971-014-3278-8.
- [55] Heo KJ, Lee DU, Shin JH, Park J, Lee BJ, Shin J, Jeong SB, Hwang GB, MacRobert AJ, Parkin IP, Jung JH, Choi DY. Transparent, robust, and photochemical antibacterial surface based on hydrogen bonding between a Si-Al and cationic dye. *ACS Appl Mater Interfaces* 2022;14:53285–97. doi:10.1021/acsami.2c16071.
- [56] Liu J, Wang Y, An L. Abnormal behavior of silica doped with small amounts of aluminum. *Sci Rep* 2016;6:35556. doi:10.1038/srep35556.
- [57] Masheder B, Urata C, Hozumi A. Transparent and hard zirconia-based hybrid coatings with excellent dynamic/thermo-responsive oleophobicity, thermal durability, and hydrolytic stability. *ACS Appl Mater Interfaces* 2013;5:7899–905. doi:10.1021/am401992h.
- [58] Ansari F, Ding Y, Berglund LA, Dauskardt RH. Toward sustainable multifunctional coatings containing nanocellulose in a hybrid glass matrix. *ACS Nano* 2018;12:5495–503. doi:10.1021/acsnano.8b01057.
- [59] Kim SH, Jang M, Yang H, Anthony JE, Park CE. Physicochemically stable polymer-coupled oxide dielectrics for multipurpose organic electronic applications. *Adv Funct Mater* 2011;21:2198–207. doi:10.1002/adfm.201002054.
- [60] Yoon MH, Kim C, Facchetti A, Marks TJ. Gate dielectric chemical structure-organic field-effect transistor performance correlations for electron, hole, and ambipolar organic semiconductors. *J Am Chem Soc* 2006;128:12851–69. doi:10.1021/ja063290d.
- [61] Zhang JN, Chen K, Sun X, Liu M, Hu X, He L, Huang ZR, Chai ZF, Xiao XS, Song YJ, Huang Q. MAX phase ceramics/composites with complex shapes. *ACS Appl Mater Interfaces* 2021;13:5645–51. doi:10.1021/acsami.0c22289.
- [62] Zhu H, Hiruta S, Demirci A, Kim S, Hoshino N, Akutagawa T, Mitsuishi M. Effects of hydride transfer ring-opening reaction on B(C₆F₅)₃ catalyzed poly-

- merization of D4H cyclosiloxane and dialkoxysilanes toward thermally stable silsesquioxane-siloxane hybrid materials. *Macromolecules* 2022;55:10134–44. doi:10.1021/acs.macromol.2c00948.
- [63] Zheng PW, McCarthy TJ. Rediscovering silicones: molecularly smooth, low surface energy, unfilled, UV/vis-transparent, extremely cross-linked, thermally stable, hard, elastic PDMS. *Langmuir* 2010;26:18585–90. doi:10.1021/la104065e.
- [64] Michalczyk MJ, Farneth WE, Vega AJ. High temperature stabilization of crosslinked siloxanes glasses. *Chem Mater* 2002;5:1687–9. doi:10.1021/cm00036a001.
- [65] Song YJ, Tang XZ, Liang YN, Lee HY, Liu M, Zhang LY, Bi SG, Mu CZ, Hu X. Bioinspired reinforcement of cyclosiloxane hybrid polymer. *Chem Commun* 2018;54:13415–18. doi:10.1039/c8cc06671j.
- [66] Song YJ, Chen XL, Liang YN, Zhang LY, Liu M, Hu X. Bio-inspired reinforcement of cyclosiloxane hybrid polymer via 'molecular stitching'. *Chem Eng J* 2019;368:573–6. doi:10.1016/j.cej.2019.02.184.
- [67] Yu CW, Li TH, Hu X, Liu M, Huang Q, He L, Song YJ. Anchoring effect of hyperbranched carborane in highly cross-linked cyclosiloxane networks toward high-performance polymers. *Macromolecules* 2023;56:4738–47. doi:10.1021/acs.macromol.3c00286.
- [68] Zhan Y, Grottenmüller R, Li W, Javadi F, Riedel R. Evaluation of mechanical properties and hydrophobicity of room-temperature, moisture-curable polysilazane coatings. *J Appl Polym Sci* 2021;138:50469. doi:10.1002/app.50469.
- [69] Guo X, Li PF, Liu GM, Tian Y, Zhang ZB, Xu CH, Jiang L. Nanocomposite: nano phased polymer dispersed in inorganic matrix via covalent bonds. *Nano Res* 2022;15:6582–9. doi:10.1007/s12274-022-4233-6.
- [70] Furtat P, Lenz-Leite M, Ionescu E, Machado RaF, Motz G. Synthesis of fluorine-modified polysilazanes via Si-H bond activation and their application as protective hydrophobic coatings. *J Mater Chem A* 2017;5:25509–21. doi:10.1039/c7ta07687h.
- [71] Guo X, Di Y, Liang QY, Li PH, Lv JY, Tian Y, Li QY, Jiang L, Xu CH, Zhang ZB. Inorganic-organic silica/PDMS nanocomposite antiadhesive coating with ultrahigh hardness and thermal stability. *ACS Appl Mater Interfaces* 2023;15:17245–55. doi:10.1021/acsami.3c00989.
- [72] Zhang HC, Ou JF, He YJ, Hu YT, Wang FJ, Fang XZ, Li W, Amirfazli A. Robust solid slippery coating for anti-icing and anti-sticking. *Colloid Surf A-Physicochem Eng Asp* 2024;682. doi:10.1016/j.colsurfa.2023.132853.
- [73] Lee AS, Choi S-S, Baek K-Y, Hwang SS. Hydrolysis kinetics of a sol-gel equilibrium yielding ladder-like polysilsesquioxanes. *Inorg Chem Commun* 2016;73:7–11. doi:10.1016/j.inoche.2016.09.004.
- [74] Sato Y, Hayami R, Gunji T. Characterization of NMR, IR, and Raman spectra for siloxanes and silsesquioxanes: a mini review. *J Sol-Gel Sci Technol* 2022;104:36–52. doi:10.1007/s10971-022-05920-y.
- [75] Ahmed N, Fan H, Dubois P, Zhang XW, Fahad S, Aziz T, Wan JT. Nano-engineering and micromolecular science of polysilsesquioxane materials and their emerging applications. *J Mater Chem A* 2019;7:21577–604. doi:10.1039/c9ta04575a.
- [76] Kang ES, Jung K, Park DH, Kang N, Ryu B. Thermo-optic characteristics in transparent glass fabric reinforced composite using inorganic-organic hybrid materials. *J Sol-Gel Sci Technol* 2012;62:333–7. doi:10.1007/s10971-012-2729-3.
- [77] Kim YH, Choi KM, Kim YH, Bae BS. Mechanically improved sol-gel derived methacrylate-siloxane hybrid materials with urethane linkage. *J Sol-Gel Sci Technol* 2018;89:111–19. doi:10.1007/s10971-018-4803-y.
- [78] Kim YH, Lim YW, Kim YH, Bae BS. Thermally stable siloxane hybrid matrix with low dielectric loss for copper-clad laminates for high-frequency applications. *ACS Appl Mater Interfaces* 2016;8:8335–40. doi:10.1021/acsami.6b01497.
- [79] Chen RZ, Zhang YS, Xie QY, Chen ZX, Ma CF, Zhang GZ. Transparent polymer-ceramic hybrid antifouling coating with superior mechanical properties. *Adv Funct Mater* 2021;31:2011145. doi:10.1002/adfm.202011145.
- [80] Jin JH, Yang S, Bae BS. Network structure-property relationship in UV-cured organic/inorganic hybrid nanocomposites. *Polym Chem* 2011;2:168–74. doi:10.1039/c0py00148a.
- [81] Im HG, Jang J, Jeon Y, Noh J, Jin J, Lee JY, Bae BS. Flexible transparent crystalline-ITO/Ag nanowire hybrid electrode with high stability for organic optoelectronics. *ACS Appl Mater Interfaces* 2020;12:56462–9. doi:10.1021/acsami.0c17130.
- [82] Kim DG, Jung B, Kim KH, Cho KY, Jeong YC. Highly robust and transparent flexible cover window films based on UV-curable polysilsesquioxane nano sol. *J Appl Polym Sci* 2020;137:49012. doi:10.1002/app.49012.
- [83] Choi GM, Jin J, Shin D, Kim YH, Ko JH, Im HG, Jang J, Jang D, Bae BS. Flexible hard coating: glass-like wear resistant, yet plastic-like compliant, transparent protective coating for foldable displays. *Adv Mater* 2017;29:1700205. doi:10.1002/adma.201700205.
- [84] Zhang YS, Chen ZX, Zheng H, Chen RZ, Ma CF, Zhang GZ. Multifunctional hard yet flexible coatings fabricated using a universal step-by-step strategy. *Adv Sci* 2022;9:2200268. doi:10.1002/adv.202200268.
- [85] Kim H, Bae JY, Kim YH, Kim YB, Bae BS. Low temperature curable epoxy siloxane hybrid materials for LED encapsulant. *J Appl Polym Sci* 2014;131:39968. doi:10.1002/app.39968.
- [86] Zhang KK, Huang SS, Wang JD, Liu GJ. Transparent organic/silica nanocomposite coating that is flexible, omniphobic, and harder than a 9H pencil. *Chem Eng J* 2020;396:125211. doi:10.1016/j.cej.2020.125211.
- [87] Bender DN, Hait S, Lichtenhan JD, Liu GJ. UV curing behavior of five heteroleptic POSS bearing methacrylate and glycidyl groups and evaluation of their potential for hard yet flexible coatings. *ACS Appl Poly Mater* 2022;4:1878–89. doi:10.1021/acsapm.1c01702.
- [88] Lin X, Li HT, Nie MX, Fu SR, Li Y, Zhang Q, Chen F, Han D, Fu Q. Engineering the properties of transparent hybrid coating toward high hardness, excellent flexibility, and multifunction. *ACS Appl Mater Interfaces* 2022;14:39432–40. doi:10.1021/acsami.2c13256.
- [89] Kim J-S, Lee S, Hwang YH, Kim Y, Yoo S, Bae B-S. Photo-curable sol-gel hybrid film as a dielectric layer by a thiol-ene reaction in air or N₂ for organic thin film transistors. *Electrochim Solid State Lett* 2012;15:G13–G15. doi:10.1149/2.021205esl.
- [90] Kim YH, Bae JY, Jin J, Bae BS. Sol-gel derived transparent zirconium-phenyl siloxane hybrid for robust high refractive index LED encapsulant. *ACS Appl Mater Interfaces* 2014;6:3115–21. doi:10.1021/am500315y.
- [91] Zheng WW, Huang JY, Zang XR, Xu XF, Cai WL, Lin ZQ, Lai YK. Judicious design and rapid manufacturing of a flexible, mechanically resistant liquid-like coating with strong bonding and antifouling abilities. *Adv Mater* 2022;34:2204581. doi:10.1002/adma.202204581.
- [92] Blattmann H, Mülhaupt R. Multifunctional POSS cyclic carbonates and non-isocyanate polyhydroxyurethane hybrid materials. *Macromolecules* 2016;49:742–51. doi:10.1021/acs.macromol.5b02560.
- [93] Seo J, Moon SW, Kang H, Choi BH, Seo JH. Foldable and extremely scratch-resistant hard coating materials from molecular necklace-like cross-linkers. *ACS Appl Mater Interfaces* 2019;11:27306–17. doi:10.1021/acsami.9b05738.
- [94] Lei X, Chen Y, Qiao M, Tian L, Zhang Q. Hyperbranched polysiloxane (HBPsi)-based polyimide films with ultralow dielectric permittivity, desirable mechanical and thermal properties. *J Mater Chem C* 2016;4:2134–46. doi:10.1039/c5tc03391h.
- [95] Jo YY, Lee AS, Baek KY, Lee H, Hwang SS. Thermally reversible self-healing polysilsesquioxane structure-property relationships based on Diels-Alder chemistry. *Polymer (Guildf)* 2017;108:58–65. doi:10.1016/j.polymer.2016.11.040.
- [96] Lee H, Kim S, Kim W, Kang SM, Kim YH, Jang J, Han SM, Bae BS. Highly transparent and resilient urethane-methacrylate siloxane composite for hard, yet stretchable protective coating. *Prog Org Coat* 2022;162:106567. doi:10.1016/j.porgcoat.2021.106567.
- [97] Jung K, Bae JY, Park SJ, Yoo S, Bae BS. High performance organic-inorganic hybrid barrier coating for encapsulation of OLEDs. *J Mater Chem* 2011;21:1977–83. doi:10.1039/c0jm02008g.
- [98] Yang S, Jin J, Kwak SY, Bae BS. Photocurable transparent cycloaliphatic epoxy hybrid materials crosslinked by oxetane. *J Appl Polym Sci* 2012;126:E380–6. doi:10.1002/app.36654.
- [99] Yang S, Kwak SY, Jin J, Kim JS, Choi Y, Paik KW, Bae BS. Thermally resistant UV-curable epoxy-siloxane hybrid materials for light emitting diode (LED) encapsulation. *J Mater Chem* 2012;22:8874–80. doi:10.1039/c2jm16355a.
- [100] Zhang YH, Yuan L, Guan QB, Liang GZ, Gu AJ. Developing self-healable and antibacterial polyacrylate coatings with high mechanical strength through crosslinking by multi-amine hyperbranched polysiloxane via dynamic vinyl-ous urethane. *J Mater Chem A* 2017;5:16889–97. doi:10.1039/c7ta04141a.
- [101] Niu S, Yan H. Novel hyperbranched polysiloxanes containing acetoacetyl groups synthesized through transesterification reaction. *Macromol Rapid Commun* 2015;36:739–43. doi:10.1002/marc.201400585.
- [102] Liu R, Yan HX, Zhang YB, Yang KM, Du S. Cyanate ester resins containing Si-O-C hyperbranched polysiloxane with favorable curing processability and toughness for electronic packaging. *Chem Eng J* 2022;433:133827. doi:10.1016/j.cej.2021.133827.
- [103] Choi SS, Lee AS, Hwang SS, Baek KY. Structural control of fully condensed polysilsesquioxanes: ladderlike vs cage structured polyphenylsilsesquioxanes. *Macromolecules* 2015;48:6063–70. doi:10.1021/acs.macromol.5b01539.
- [104] Kim YH, Choi GM, Bae JG, Kim YH, Bae BS. High-performance and simply-synthesized ladder-like structured methacrylate siloxane hybrid material for flexible hard coating. *Polymers (Basel)* 2018;10:449. doi:10.3390/polym10040449.
- [105] Kubisa P, Penczek S. Cationic activated monomer polymerization of heterocyclic monomers. *Prog Polym Sci* 1999;24:1409–37. doi:10.1016/S0079-6700(99)00028-3.
- [106] Aoshima S, Kanaoka S. A renaissance in living cationic polymerization. *Chem Rev* 2009;109:5245–87. doi:10.1021/cr900225g.
- [107] Tokar R, Kubisa P, Penczek S. Cationic polymerization of glycidol: coexistence of the activated monomer and active chain end mechanism. *Macromolecules* 1994;27:320–2. doi:10.1021/ma00080a002.
- [108] Penczek S, Cypryk M, Duda A, Kubisa P, Slomkowski S. Living ring-opening polymerizations of heterocyclic monomers. *Prog Polym Sci* 2007;32:247–82. doi:10.1016/j.progpolymsci.2007.01.002.
- [109] Kim KH, Jung B, Jeong YC. Foldable hard coating materials based on reaction-controlled polysilsesquioxane resin for flexible electronic devices. *Prog Org Coat* 2020;143:105639. doi:10.1016/j.porgcoat.2020.105639.
- [110] Na W, Lee AS, Lee JH, Hwang SS, Hong SM, Kim E, Koo CM. Lithium ion capacitors fabricated with polyethylene oxide-functionalized polysilsesquioxane hybrid ionogel electrolytes. *Electrochim Acta* 2016;188:582–8. doi:10.1016/j.electacta.2015.12.017.
- [111] Choi SS, Lee AS, Lee HS, Jeon HY, Baek KY, Choi DH, Hwang SS. Synthesis and characterization of UV-curable ladder-like polysilsesquioxane. *J Polym Sci Pol Chem* 2011;49:5012–18. doi:10.1002/pola.24942.
- [112] Lee AS, Choi SS, Song SJ, Baek KY, Hwang SS. Cationically photopolymerizable epoxy-functionalized thermoplastic polysilsesquioxanes: synthesis and properties. *RSC Adv* 2014;4:56532–8. doi:10.1039/c4ra02899c.

- [113] Lee AS, Jo YY, Jeon H, Choi SS, Baek KY, Hwang SS. Mechanical properties of thiol-ene UV-curable thermoplastic polysilsesquioxanes. *Polymer (Guildf)* 2015;68:140–6. doi:10.1016/j.polymer.2015.05.007.
- [114] Choi SS, Lee HS, Hwang SS, Choi DH, Baek KY. High photo- and electroluminescence efficiencies of ladder-like structured polysilsesquioxane with carbazole groups. *J Mater Chem* 2010;20:9852–4. doi:10.1039/c0jm02561e.
- [115] Li Z, Hu JF, Yang L, Zhang XQ, Liu XH, Wang Z, Li YW. Integrated POSS-dendrimer nanohybrid materials: current status and future perspective. *Nanoscale* 2020;12:11395–415. doi:10.1039/d0nr02394a.
- [116] Soldatov M, Liu H. Hybrid porous polymers based on cage-like organosiloxanes: synthesis, properties and applications. *Prog Polym Sci* 2021;119:101419. doi:10.1016/j.progpolymsci.2021.101419.
- [117] Wang JJ, Sun J, Zhou JF, Jin KK, Fang Q. Fluorinated and thermo-cross-linked polyhedral oligomeric silsesquioxanes: new organic-inorganic hybrid materials for high-performance dielectric application. *ACS Appl Mater Interfaces* 2017;9:12782–90. doi:10.1021/acsami.7b01415.
- [118] Zhou DL, Li JH, Guo QY, Lin X, Zhang Q, Chen F, Han D, Fu Q. Polyhedral oligomeric silsesquioxanes based ultralow-k materials: the effect of cage size. *Adv Funct Mater* 2021;31:2102074. doi:10.1002/adfm.202102074.
- [119] Im HG, Park GU, Park HY, Jin JH, Kang DJ. A robust transparent protective hard-coating material using physicochemically-incorporated silica nanoparticles and organosiloxanes. *Prog Org Coat* 2017;105:330–5. doi:10.1016/j.porgcoat.2017.02.004.
- [120] Kang DJ, Park GU, Im HG, Park HY, Jin J. Silica nanoparticle-embedded urethane acrylate nanohybrid thermosets for photo-patternable transparent hard coating. *Polymer (Guildf)* 2016;105:19–24. doi:10.1016/j.polymer.2016.10.008.
- [121] Chung PT, Chiou SH, Tseng CY, Chiang AS. Preparation and evaluation of a zirconia/oligosiloxane nanocomposite for LED encapsulation. *ACS Appl Mater Interfaces* 2016;8:9986–93. doi:10.1021/acsami.6b02082.
- [122] García-Heras M, Rincón JM, Romero M, Villegas MA. Indentation properties of ZrO₂-SiO₂ coatings on glass substrates. *Mater Res Bull* 2003;38:1635–44. doi:10.1016/s0025-5408(03)00187-9.
- [123] Suriano R, Ciapponi R, Griffini G, Levi M, Turri S. Fluorinated zirconia-based sol-gel hybrid coatings on polycarbonate with high durability and improved scratch resistance. *Surf Coat Tech* 2017;311:80–9. doi:10.1016/j.surfcoat.2016.12.095.
- [124] Pan KB, Liu CM, Zhu ZC, Feng TL, Tao SY, Yang B. Soft-hard segment combined carbonized polymer dots for flexible optical film with superhigh surface hardness. *ACS Appl Mater Interfaces* 2022;14:14504–12. doi:10.1021/acsami.2c00702.
- [125] Pan KB, Zhu ZC, Liu CM, Tao SY, Tang XD, Wei XY, Yang B. Flexible transparent hydrophobic coating films with excellent scratch resistance using Si-doped carbonized polymer dots as building blocks. *ACS Appl Mater Interfaces* 2023;15:26060–8. doi:10.1021/acsami.3c05078.
- [126] Chang CC, Lin ZM, Cheng LP. Preparation of superhydrophilic nanosilica/polycarbonate hard coatings on plastic substrate for antifogging and frost-resistant applications. *J Appl Polym Sci* 2019;136:48144. doi:10.1002/app.48144.
- [127] Mallakpour S, Naghdi M. Polymer/SiO₂ nanocomposites: production and applications. *Prog Mater Sci* 2018;97:409–47. doi:10.1016/j.pmatsci.2018.04.002.
- [128] Kim JH, Ko JH, Bae BS. Dispersion of silica nano-particles in sol-gel hybrid resins for fabrication of multi-scale hybrid nanocomposite. *J Sol-Gel Sci Technol* 2006;41:249–55. doi:10.1007/s10971-006-9001-7.
- [129] Lim SY, Shen W, Gao ZQ. Carbon quantum dots and their applications. *Chem Soc Rev* 2015;44:362–81. doi:10.1039/c4cs00269e.
- [130] Yu HJ, Shi R, Zhao YF, Waterhouse GI, Wu LZ, Tung CH, Zhang T. Smart utilization of carbon dots in semiconductor photocatalysis. *Adv Mater* 2016;28:9454–77. doi:10.1002/adma.201602581.
- [131] Liu JC, Xie Z, Shang YY, Ren JK, Hu RX, Guan B, Wang JX, Ikeda T, Jiang L. Lyophilic but nonwetttable organosilane-polymerized carbon dots inverse opals with closed-cell structure. *ACS Appl Mater Interfaces* 2018;10:6701–10. doi:10.1021/acsami.7b17936.
- [132] Hou Y, Zhu G, Cui J, Wu N, Zhao B, Xu J, Zhao N. Superior hard but quickly reversible Si-O-Si network enables scalable fabrication of transparent, self-healing, robust, and programmable multifunctional nanocomposite coatings. *J Am Chem Soc* 2022;144:436–45. doi:10.1021/jacs.1c10455.
- [133] Wang D, Ma JS, Li PF, Fan L, Wu YM, Zhang ZB, Xu CH, Jiang L. Flexible hard coatings with self-evolution behavior in a low earth orbit environment. *ACS Appl Mater Interfaces* 2021;13:46003–14. doi:10.1021/acsami.1c13807.
- [134] Zhang PL, Zhang GL, Pan JS, Ma CF, Zhang GZ. Non-isocyanate polyurethane coating with high hardness, superior flexibility, and strong substrate adhesion. *ACS Appl Mater Interfaces* 2023;15:5998–6004. doi:10.1021/acsami.2c22433.
- [135] Lee Y, Lee H, Im H-G, Jo W, Choi G-M, Kim T-S, Jang J, Bae B-S. Transparent and flexible hybrid cover window film: hard coating/substrate all-in-one composite film for reliable foldable display. *Compos Pt B-Eng* 2022;247:110336. doi:10.1016/j.compositesb.2022.110336.
- [136] Lee H, Lee HE, Wang HS, Kang SM, Lee D, Kim YH, Shin JH, Lim YW, Lee KJ, Bae BS. Hierarchically surface-textured ultrastable hybrid film for large-scale triboelectric nanogenerators. *Adv Funct Mater* 2020;30. doi:10.1002/adfm.202005610.
- [137] Kim YH, Choi GM, Shin D, Kim YH, Jang D, Bae BS. Transparent urethane-siloxane hybrid materials for flexible cover windows with ceramic-like strength, yet polymer-like modulus. *ACS Appl Mater Interfaces* 2018;10:43122–30. doi:10.1021/acsami.8b18141.
- [138] Leem JT, Seok WC, Yoo JB, Lee S, Song HJ. Hard coating materials based on photo-reactive silsesquioxane for flexible application: improvement of flexible and hardness properties by high molecular weight. *Polymers (Basel)* 2021;13:1564. doi:10.3390/polym13101564.
- [139] Jo YY, Lee AS, Baek KY, Lee H, Hwang SS. Multi-crosslinkable self-healing polysilsesquioxanes for the smart recovery of anti-scratch properties. *Polymer (Guildf)* 2017;124:78–87. doi:10.1016/j.polymer.2017.06.076.
- [140] Lai ZR, Liu GJ. Facile preparation of a transparent and rollable omniphobic coating with exceptional hardness and wear resistance. *ACS Appl Mater Interfaces* 2022;14:35138–47. doi:10.1021/acsami.2c10200.
- [141] Zhang KK, Huang SS, Wang JD, Liu GJ. Transparent omniphobic coating with glass-like wear resistance and polymer-like bendability. *Angew Chem Int Ed* 2019;58:12004–9. doi:10.1002/anie.201904210.
- [142] Bender DN, Zhang KK, Wang JD, Liu GJ. Hard yet flexible transparent omniphobic GPOSS coatings modified with perfluorinated agents. *ACS Appl Mater Interfaces* 2021;13:10467–79. doi:10.1021/acsami.0c23151.
- [143] Bu XY, Sun CF, Qiao CD, Gao X, Zhang YN, Ji XX, Tao FR, Gai LG, Liu LB. A bio-inspired slippery coating with mechanochemical robustness for anti-ice and anti-corrosion. *Appl Surf Sci* 2023;639:158143. doi:10.1016/j.apsusc.2023.158143.
- [144] Lin X, Nie MX, Liu H, Zhou DL, Fu SR, Zhang Q, Han D, Fu Q. Topology-enabled simultaneous enhancement of mechanical and healable properties in glassy polymeric materials using larger POSS. *Chem Mat* 2023;36:575–84. doi:10.1021/acs.chemmater.3c02790.
- [145] Jeong KM, Park SS, Nagappan S, Min GQ, Zhang YX, Qu MM, Zhang Y, Ha CS. Highly transparent, organic-inorganic hybrid UV-curable coating materials with amphiphobic characteristics. *Prog Org Coat* 2019;134:323–32. doi:10.1016/j.porgcoat.2019.05.029.
- [146] Zhu JL, Zheng JY, Hu Y, Wei J, Ma YC, Gou HL, Fan H. Facile fabricated transparent anti-smudge coating with high hardness and excellent flexibility from MTQ and branched silicone resins. *Prog Org Coat* 2023;185. doi:10.1016/j.porgcoat.2023.107907.
- [147] Velez M, Schuman TP, Day DE. Optical properties of optically transparent glass-ribbon composites. *J Compos Mater* 2013;48:3747–54. doi:10.1177/0021998313513204.
- [148] Krug Iii DJ, Asuncion MZ, Popova V, Laine RM. Transparent fiber glass reinforced composites. *Compos Sci Technol* 2013;77:95–100. doi:10.1016/j.compscitech.2012.12.010.
- [149] Lee D, Lim YW, Im HG, Jeong S, Ji S, Kim YH, Choi GM, Park JU, Lee JY, Jin J, Bae BS. Bioinspired transparent laminated composite film for flexible green optoelectronics. *ACS Appl Mater Interfaces* 2017;9:24161–8. doi:10.1021/acsami.7b03126.
- [150] Li YQ, Zhang L, Li CZ. Highly transparent and scratch resistant polysiloxane coatings containing silica nanoparticles. *J Colloid Interface Sci* 2020;559:273–81. doi:10.1016/j.jcis.2019.09.031.
- [151] Jin J, Ko JH, Yang S, Bae BS. Rollable transparent glass-fabric reinforced composite substrate for flexible devices. *Adv Mater* 2010;22:4510–15. doi:10.1002/adma.201002198.
- [152] Jang J, Im HG, Lim D, Bae BS. Preparation of high-performance transparent glass-fiber reinforced composites based on refractive index-tunable epoxy-functionalized siloxane hybrid matrix. *Compos Sci Technol* 2021;201:108527. doi:10.1016/j.compscitech.2020.108527.
- [153] Nistal A, Ruiz-González A, Choy K-L. Robust icephobic nanocomposite coatings with superior abrasion resistance. *Appl Mater Today* 2022;27:101480. doi:10.1016/j.apmt.2022.101480.
- [154] Qi XD, Yang L, Zhu JQ, Hou Y, Yang M. Stiffer but more healable exponential layered assemblies with boron nitride nanoplatelets. *ACS Nano* 2016;10:9434–45. doi:10.1021/acsnano.6b04482.
- [155] Bae G, Choi GM, Ahn C, Kim SM, Kim W, Choi Y, Park D, Jang D, Hong JW, Han SM, Bae BS, Jeon S. Flexible protective film: ultrahard, yet flexible hybrid nanocomposite reinforced by 3D inorganic nanoshell structures. *Adv Funct Mater* 2021;31. doi:10.1002/adfm.202010254.
- [156] Bender DN, Liu GJ. Endowment of omniphobicity and exceptional bendability to a wear-resistant POSS coating. *Chem Eng J* 2023;464:142702. doi:10.1016/j.cej.2023.142702.
- [157] Qi XD, Hou Y, Yang M. Scalable synthesis of multifunctional epidermis-like smart coatings. *Adv Funct Mater* 2019;29. doi:10.1002/adfm.201903984.
- [158] Qi XD, Zhang D, Ma ZB, Cao WX, Hou Y, Zhu JQ, Gan Y, Yang M. An epidermis-like hierarchical smart coating with a hardness of tooth enamel. *ACS Nano* 2018;12:1062–73. doi:10.1021/acsnano.7b05478.
- [159] Li Y, Chen SS, Li X, Wu MC, Qi SJ. Highly transparent, nanofiller-reinforced scratch-resistant polymeric composite films capable of healing scratches. *ACS Nano* 2015;9:10055–65. doi:10.1021/acsnano.5b03629.
- [160] Kim MS, Ahn Y, Kim BG. Improvement of scratch resistance in transparent hard surfaces through layer-by-layer coating. *Prog Org Coat* 2022;170:106991. doi:10.1016/j.porgcoat.2022.106991.
- [161] Meng XS, Zhou LC, Liu L, Zhu YB, Meng YF, Zheng DC, Yang B, Rao QZ, Mao LB, Wu HA, Yu SH. Deformation hard tissue with high fatigue resistance in the hinge of bivalve *Cristaria plicata*. *Science* 2023;380:1252–7. doi:10.1126/science.ade2038.
- [162] Inoue T, Oka SI, Hara T. Three-dimensional microstructure of robust claw of coconut crab, one of the largest terrestrial crustaceans. *Mater Des* 2021;206. doi:10.1016/j.matdes.2021.109765.
- [163] Shirai Y, Sasaki A, Sato S, Aoki D, Arimitsu K. Fabrication of an organic-inorganic hybrid hard coat with a gradient structure controlled by photoir-

- radiation. *ACS Appl Mater Interfaces* 2023;15:28563–9. doi:[10.1021/acsami.3c04399](https://doi.org/10.1021/acsami.3c04399).
- [164] Wang ZZ, Shi XM, Huang HB, Yao CM, Xie W, Huang C, Gu P, Ma XQ, Zhang ZQ, Chen LQ. Magnetically actuated functional gradient nanocomposites for strong and ultra-durable biomimetic interfaces/surfaces. *Mater Horiz* 2017;4:869–77. doi:[10.1039/c7mh00223h](https://doi.org/10.1039/c7mh00223h).
- [165] Wang Z, Wang K, Huang H, Cui X, Shi X, Ma X, Li B, Zhang Z, Tang X, Chiang MYM. Bioinspired wear-resistant and ultradurable functional gradient coatings. *Small* 2018;14:e1802717. doi:[10.1002/smll.201802717](https://doi.org/10.1002/smll.201802717).
- [166] Zou H, Wu SS, Shen J. Polymer/Silica nanocomposites: preparation, characterization, properties, and applications. *Chem Rev* 2008;108:3893–957. doi:[10.1021/cr068035q](https://doi.org/10.1021/cr068035q).
- [167] Vallet-Regí M, Colilla M, González B. Medical applications of organic-inorganic hybrid materials within the field of silica-based bioceramics. *Chem Soc Rev* 2011;40:596–607. doi:[10.1039/c0cs00025f](https://doi.org/10.1039/c0cs00025f).
- [168] Amini A, Khavari A, Barthelat F, Ehrlicher AJ. Centrifugation and index matching yield a strong and transparent bioinspired nacreous composite. *Science* 2021;373:1229–34. doi:[10.1126/science.abc0277](https://doi.org/10.1126/science.abc0277).
- [169] Ye HQ, Park E, Shin SC, Murali G, Kim D, Lee J, Kim IH, Kim SJ, Kim SH, Jeong YJ, In I. Photopatternable high-k polysilsesquioxane dielectrics for organic integrated devices: effects of UV curing on chemical and electrical properties. *Adv Funct Mater* 2023;33:2214865. doi:[10.1002/adfm.202214865](https://doi.org/10.1002/adfm.202214865).
- [170] Jiang B, Shi XR, Zhang T, Huang YD. Recent advances in UV/thermal curing silicone polymers. *Chem Eng J* 2022;435:134843. doi:[10.1016/j.cej.2022.134843](https://doi.org/10.1016/j.cej.2022.134843).
- [171] Song SS, Sun YJ, Lin YD, You B. A facile fabrication of light diffusing film with LDP/polycarbonates composites coating for anti-glare LED application. *Appl Surf Sci* 2013;273:652–60. doi:[10.1016/j.apsusc.2013.02.103](https://doi.org/10.1016/j.apsusc.2013.02.103).
- [172] Zhu HL, Fang ZQ, Wang Z, Dai JQ, Yao YG, Shen F, Preston C, Wu WX, Peng P, Jang N, Yu QK, Yu ZF, Hu LB. Extreme light management in mesoporous wood cellulose paper for optoelectronics. *ACS Nano* 2016;10:1369–77. doi:[10.1021/acsnano.5b06781](https://doi.org/10.1021/acsnano.5b06781).
- [173] Lim YW, Kwon OE, Kang SM, Cho H, Lee J, Park YS, Cho NS, Jin WY, Lee J, Lee H, Kang JW, Yoo S, Moon J, Bae BS. Built-In haze glass-fabric reinforced siloxane hybrid film for efficient organic light-emitting diodes (OLEDs). *Adv Funct Mater* 2018;28. doi:[10.1002/adfm.201802944](https://doi.org/10.1002/adfm.201802944).
- [174] Kim YH, Lee H, Kang SM, Bae BS. Two-step-enhanced stability of quantum dots via silica and siloxane encapsulation for the long-term operation of light-emitting diodes. *ACS Appl Mater Interfaces* 2019;11:22801–8. doi:[10.1021/acsami.9b06987](https://doi.org/10.1021/acsami.9b06987).
- [175] Kwak SY, Yang S, Kim NR, Kim JH, Bae BS. Thermally stable, dye-bridged nanohybrid-based white light-emitting diodes. *Adv Mater* 2011;23:5767–72. doi:[10.1002/adma.201103077](https://doi.org/10.1002/adma.201103077).
- [176] Jang J, Kim YH, Park S, Yoo D, Cho H, Jang J, Jeong HB, Lee H, Yuk JM, Park CB, Jeon DY, Kim YH, Bae BS, Lee TW. Extremely stable luminescent crosslinked perovskite nanoparticles under harsh environments over 1.5 years. *Adv Mater* 2021;33:2005255. doi:[10.1002/adma.202005255](https://doi.org/10.1002/adma.202005255).
- [177] Pei M, Lee AS, Hwang SS, Yang H. Ladder-like polysilsesquioxane dielectrics for organic field-effect transistor applications. *J Mater Chem C* 2017;5:10955–64. doi:[10.1039/c7tc02278f](https://doi.org/10.1039/c7tc02278f).
- [178] Shu PJ, Ai L, Kong YH, Ji H, Lu YH, Zhang J, Song WJ. UV-cured organic-inorganic composites for highly durable and flexible antireflection coatings. *Appl Surf Sci* 2022;584:152600. doi:[10.1016/j.apsusc.2022.152600](https://doi.org/10.1016/j.apsusc.2022.152600).
- [179] Zhang ZQ, Yu DF, Xu XB, Yang H, Wyman I, Wang JB, Wu X. Versatile snail-inspired superamphiphobic coatings with repeatable adhesion and recyclability. *Chem Eng Sci* 2021;230:116182. doi:[10.1016/j.ces.2020.116182](https://doi.org/10.1016/j.ces.2020.116182).
- [180] Zhang ZQ, Yu DF, Xu XB, Li HY, Mao TY, Zheng C, Huang JJ, Yang H, Niu ZH, Wu X. A polypropylene melt-blown strategy for the facile and efficient membrane separation of oil-water mixtures. *Chin J Chem Eng* 2021;29:383–90. doi:[10.1016/j.cjche.2020.03.033](https://doi.org/10.1016/j.cjche.2020.03.033).
- [181] Buddingh JV, Hozumi A, Liu GJ. Liquid and liquid-like surfaces/coatings that readily slide fluids. *Prog Polym Sci* 2021;123:101468. doi:[10.1016/j.progpolymsci.2021.101468](https://doi.org/10.1016/j.progpolymsci.2021.101468).
- [182] Pagliaro M, Ciriminna R, Palmisano G. Silica-based hybrid coatings. *J Mater Chem* 2009;19:3116–26. doi:[10.1039/b819615j](https://doi.org/10.1039/b819615j).
- [183] Urata C, Masheder B, Cheng DF, Hozumi A. How to reduce resistance to movement of alkane liquid drops across tilted surfaces without relying on surface roughening and perfluorination. *Langmuir* 2012;28:17681–9. doi:[10.1021/la303829p](https://doi.org/10.1021/la303829p).
- [184] Hikita M, Tanaka K, Nakamura T, Kajiyama T, Takahara A. Super-liquid-repellent surfaces prepared by colloidal silica nanoparticles covered with fluoroalkyl groups. *Langmuir* 2005;21:7299–302. doi:[10.1021/la050901r](https://doi.org/10.1021/la050901r).
- [185] Park J, Urata C, Masheder B, Cheng DF, Hozumi A. Long perfluoroalkyl chains are not required for dynamically oleophobic surfaces. *Green Chem* 2013;15:100–4. doi:[10.1039/c2gc36415h](https://doi.org/10.1039/c2gc36415h).
- [186] Ji ZF, Liu Y, Du FL. Facile synthesis of solvent-free and mechanically robust coating with self-cleaning property. *Prog Org Coat* 2020;149:105923. doi:[10.1016/j.porgcoat.2020.105923](https://doi.org/10.1016/j.porgcoat.2020.105923).
- [187] Xu XB, Zhang YC, Wen JX, Zhang ZQ, Xin T, Yu DF, Wu X, Yao QC, Fan JL, Peng XJ. Large-area, daily, on-site-applicable antiadhesion coatings formed via ambient self-crosslinking. *Chem Eng J* 2022;450:138156. doi:[10.1016/j.cej.2022.138156](https://doi.org/10.1016/j.cej.2022.138156).
- [188] Rabnawaz M, Liu GJ, Hu H. Fluorine-free anti-smudge polyurethane coatings. *Angew Chem Int Ed* 2015;54:12722–7. doi:[10.1002/anie.201504892](https://doi.org/10.1002/anie.201504892).
- [189] Chansomwong K, Kim YH, Lee H, Bae BS. Facile preparation of wear-resistant and anti-fingerprint hard coating with chemisorption of fluorosilane by simple wet coating. *J Sol-Gel Sci Technol* 2020;95:447–55. doi:[10.1007/s10971-020-05294-z](https://doi.org/10.1007/s10971-020-05294-z).
- [190] Yun HW, Choi GM, Woo HK, Oh SJ, Hong SH. Superhydrophobic, antireflective, flexible hard coatings with mechanically ultra-resilient moth-eye structure for foldable displays. *Curr Appl Phys* 2020;20:1163–70. doi:[10.1016/j.cap.2020.07.001](https://doi.org/10.1016/j.cap.2020.07.001).
- [191] Amini S, Kolle S, Petrone L, Ahanotu O, Sunny S, Sutanto CN, Hoon S, Cohen L, Weaver JC, Aizenberg J, Vogel N, Miserez A. Preventing mussel adhesion using lubricant-infused materials. *Science* 2017;357:668–73. doi:[10.1126/science.aa89777](https://doi.org/10.1126/science.aa89777).
- [192] Zhang ZQ, Xie QY, Zhang GL, Ma CF, Zhang GZ. Surface-enriched amphiphilic polysiloxane coating with superior antifouling ability and strong substrate adhesion. *ACS Appl Polym Mater* 2023;5:3524–33. doi:[10.1021/acscpm.3c00199](https://doi.org/10.1021/acscpm.3c00199).
- [193] Olsen M, Campos M, Lohning A, Jones P, Legget J, Bannach-Brown A, Mckirdy S, Alghafri R, Tajouri L. Mobile phones represent a pathway for microbial transmission: a scoping review. *Travel Med Infect Dis* 2020;35:101704. doi:[10.1016/j.tmaid.2020.101704](https://doi.org/10.1016/j.tmaid.2020.101704).
- [194] He XD, Wu HJ, Wang Y, Xiang YJ, Zhang K, Rao X, Kang ET, Xu LQ. Bimodal antimicrobial surfaces of phytic acid-prussian blue nanoparticles–cationic polymer networks. *Adv Sci* 2023;10:2300354. doi:[10.1002/advsc.202300354](https://doi.org/10.1002/advsc.202300354).
- [195] Pan JS, Ai XQ, Ma CF, Zhang GZ. Degradable vinyl polymers for combating marine biofouling. *Acc Chem Res* 2022;55:1586–98. doi:[10.1021/acs.accounts.2c00187](https://doi.org/10.1021/acs.accounts.2c00187).
- [196] Qiu HY, Feng K, Gapeeva A, Meurisch K, Kaps S, Li X, Yu LM, Mishra YK, Adelung R, Baum M. Functional polymer materials for modern marine biofouling control. *Prog Polym Sci* 2022;127:101516. doi:[10.1016/j.progpolymsci.2022.101516](https://doi.org/10.1016/j.progpolymsci.2022.101516).
- [197] McMaster DM, Bennett SM, Tang Y, Finlay JA, Kowalke GL, Nedved B, Bright FV, Callow ME, Callow JA, Wendt DE, Hadfield MG, Detty MR. Antifouling character of 'active' hybrid xerogel coatings with sequestered catalysts for the activation of hydrogen peroxide. *Biofouling* 2009;25:21–33. doi:[10.1080/08927010802431298](https://doi.org/10.1080/08927010802431298).
- [198] Finlay JA, Bennett SM, Brewer LH, Sokolova A, Clay G, Gunari N, Meyer AE, Walker GC, Wendt DE, Callow ME, Callow JA, Detty MR. Barnacle settlement and the adhesion of protein and diatom microfouling to xerogel films with varying surface energy and water wettability. *Biofouling* 2010;26:657–66. doi:[10.1080/08927014.2010.506242](https://doi.org/10.1080/08927014.2010.506242).
- [199] Sokolova A, Cilz N, Daniels J, Staflieni SJ, Brewer LH, Wendt DE, Bright FV, Detty MR. A comparison of the antifouling/foul-release characteristics of non-biocidal xerogel and commercial coatings toward micro- and macrofouling organisms. *Biofouling* 2012;28:511–23. doi:[10.1080/08927014.2012.690197](https://doi.org/10.1080/08927014.2012.690197).
- [200] Detty MR, Ciriminna R, Bright FV, Pagliaro M. Xerogel coatings produced by the sol-gel process as anti-fouling, fouling-release surfaces: from lab bench to commercial reality. *ChemNanoMat* 2015;1:148–54. doi:[10.1002/cnma.201500056](https://doi.org/10.1002/cnma.201500056).
- [201] Regina VR, Sohoel H, Lokanathan AR, Bischoff C, Kingshott P, Revsbech NP, Meyer RL. Entrapment of subtilisin in ceramic sol-gel coating for antifouling applications. *ACS Appl Mater Interfaces* 2012;4:5915–21. doi:[10.1021/am301554m](https://doi.org/10.1021/am301554m).
- [202] Marceaux S, Bressy C, Perrin F-X, Martin C, Margailan A. Development of polyorganosilazane-silicone marine coatings. *Prog Org Coat* 2014;77:1919–28. doi:[10.1016/j.porgcoat.2014.06.020](https://doi.org/10.1016/j.porgcoat.2014.06.020).
- [203] Xu LQ, Neoh K-G, Kang E-T. Natural polyphenols as versatile platforms for material engineering and surface functionalization. *Prog Polym Sci* 2018;87:165–96. doi:[10.1016/j.progpolymsci.2018.08.005](https://doi.org/10.1016/j.progpolymsci.2018.08.005).
- [204] Tan JY, Liang X, Yang JL, Zhou SX. Sol-gel-derived hard coatings from tetraethoxysilane and organoalkoxysilanes bearing zwitterionic and isothiazolinone groups and their antifouling behaviors. *J Mater Chem B* 2021;10:406–17. doi:[10.1039/d1tb02069b](https://doi.org/10.1039/d1tb02069b).
- [205] Park S, Kim JY, Choi W, Lee MJ, Heo J, Choi D, Jung S, Kwon J, Choi SH, Hong J. Ladder-like polysilsesquioxanes with antibacterial chains and durable siloxane networks. *Chem Eng J* 2020;393:124686. doi:[10.1016/j.cej.2020.124686](https://doi.org/10.1016/j.cej.2020.124686).
- [206] Park S, Jeong HJ, Moon JH, Jang EY, Jung S, Choi M, Choi W, Park K, Ahn HW, Hong J. Polysilsesquioxane with potent resistance to intraoral stress: functional coating material for the advanced dental materials. *Appl Surf Sci* 2022;578:152085. doi:[10.1016/j.apsusc.2021.152085](https://doi.org/10.1016/j.apsusc.2021.152085).
- [207] Shang D, Sun XY, Shen XY, Hang JZ, Jin LJ, Shi LY. Effects of PEG-TMS on the stability and antifouling performances of hydrocarbon-modified amphiphilic xerogel coatings. *Prog Org Coat* 2018;121:142–50. doi:[10.1016/j.porgcoat.2018.04.029](https://doi.org/10.1016/j.porgcoat.2018.04.029).
- [208] Luo H, Yin XQ, Tan PF, Gu ZP, Liu ZM, Tan L. Polymeric antibacterial materials: design, platforms and applications. *J Mater Chem B* 2021;9:2802–15. doi:[10.1039/d1tb00109d](https://doi.org/10.1039/d1tb00109d).
- [209] Leonardi AK, Ober CK. Polymer-based marine antifouling and fouling release surfaces: strategies for synthesis and modification. *Annu Rev Chem Biomol Engineer* 2019;10:241–64. doi:[10.1146/annurev-chembioeng-060718-030401](https://doi.org/10.1146/annurev-chembioeng-060718-030401).
- [210] Callow JA, Callow ME. Trends in the development of environmentally friendly fouling-resistant marine coatings. *Nat Commun* 2011;2:244. doi:[10.1038/ncomms1251](https://doi.org/10.1038/ncomms1251).
- [211] Hu P, Xie QY, Ma CF, Zhang GZ. Silicone-based fouling-release coatings for marine antifouling. *Langmuir* 2020;36:2170–83. doi:[10.1021/acs.langmuir.9b03926](https://doi.org/10.1021/acs.langmuir.9b03926).
- [212] Ma CF, Zhou H, Wu B, Zhang GZ. Preparation of polyurethane with zwitterionic side chains and their protein resistance. *ACS Appl Mater Interfaces* 2011;3:455–61. doi:[10.1021/am101039q](https://doi.org/10.1021/am101039q).

- [213] Chen W-L, Cordero R, Tran H, Ober CK. 50th anniversary perspective: polymer brushes: novel surfaces for future materials. *Macromolecules* 2017;50:4089–113. doi:[10.1021/acs.macromol.7b00450](https://doi.org/10.1021/acs.macromol.7b00450).
- [214] Li WL, Thian ES, Wang M, Wang ZY, Ren L. Surface design for antibacterial materials: from fundamentals to advanced strategies. *Adv Sci* 2021;8:2100368. doi:[10.1002/advsc.202100368](https://doi.org/10.1002/advsc.202100368).
- [215] Xie QY, Zeng HH, Peng QM, Bressy C, Ma CF, Zhang GZ. Self-stratifying silicone coating with nonleaching antifoulant for marine anti-biofouling. *Adv Mater Interfaces* 2019;6:1900535. doi:[10.1002/admi.201900535](https://doi.org/10.1002/admi.201900535).
- [216] Xie QY, Ma CF, Liu C, Ma JL, Zhang GZ. Poly(dimethylsiloxane)-based polyurethane with chemically attached antifoulants for durable marine antibiofouling. *ACS Appl Mater Interfaces* 2015;7:21030–7. doi:[10.1021/acsami.5b07325](https://doi.org/10.1021/acsami.5b07325).
- [217] Cheng QL, Guo XW, Hao XJ, Shi ZS, Zhu S, Cui ZC. Fabrication of robust antibacterial coatings based on an organic-inorganic hybrid system. *ACS Appl Mater Interfaces* 2019;11:42607–15. doi:[10.1021/acsami.9b15031](https://doi.org/10.1021/acsami.9b15031).
- [218] Wang SY, Urban MW. Self-healing polymers. *Nat Rev Mater* 2020;5:562–83. doi:[10.1038/s41578-020-0202-4](https://doi.org/10.1038/s41578-020-0202-4).
- [219] Zhang ZP, Qian L, Cheng JF, Xie QY, Ma CF, Zhang GZ. Room-temperature self-healing polyurea with high puncture and impact resistances. *Chem Mat* 2023;35:1806–17. doi:[10.1021/acs.chemmater.2c03782](https://doi.org/10.1021/acs.chemmater.2c03782).
- [220] Song Y, Liu Y, Qi T, Li GL. Towards dynamic but supertough healable polymers through biomimetic hierarchical hydrogen-bonding interactions. *Angew Chem Int Ed* 2018;57:13838–42. doi:[10.1002/anie.201807622](https://doi.org/10.1002/anie.201807622).
- [221] Chen R, Xu XB, Yu DF, Liu MH, Xiao CH, Wyman I, Wang ZP, Yang H, Wu X. Temperature-regulated flexibility of polymer chains in rapidly self-healing hydrogels. *NPG Asia Mater* 2019;11:1–15. doi:[10.1038/s41427-019-0123-0](https://doi.org/10.1038/s41427-019-0123-0).
- [222] Song PA, Wang H. High-performance polymeric materials through hydrogen-bond cross-linking. *Adv Mater* 2020;32:1901244. doi:[10.1002/adma.201901244](https://doi.org/10.1002/adma.201901244).
- [223] Li BR, Cao PF, Saito T, Sokolov AP. Intrinsically self-healing polymers: from mechanistic insight to current challenges. *Chem Rev* 2023;123:701–35. doi:[10.1021/acs.chemrev.2c00575](https://doi.org/10.1021/acs.chemrev.2c00575).
- [224] Sun P, Qin B, Xu JF, Zhang X. Supramonomers for controllable supramolecular polymerization and renewable supramolecular polymeric materials. *Prog Polym Sci* 2022;124. doi:[10.1016/j.progpolymsci.2021.101486](https://doi.org/10.1016/j.progpolymsci.2021.101486).
- [225] Wang XH, Li YX, Qian YH, Qi H, Li J, Sun JQ. Mechanically robust atomic oxygen-resistant coatings capable of autonomously healing damage in low earth orbit space environment. *Adv Mater* 2018;30:1803854. doi:[10.1002/adma.201803854](https://doi.org/10.1002/adma.201803854).
- [226] Liu XK, Li YX, Fang X, Zhang ZQ, Li SH, Sun JQ. Healable and recyclable polymeric materials with high mechanical robustness. *ACS Mater Lett* 2022;4:554–71. doi:[10.1021/acsmaterialslett.1c00795](https://doi.org/10.1021/acsmaterialslett.1c00795).
- [227] Xu ZG, Zhao Y, Wang XG, Lin T. A thermally healable polyhedral oligomeric silsesquioxane (POSS) nanocomposite based on Diels-Alder chemistry. *Chem Commun* 2013;49:6755–7. doi:[10.1039/c3cc43432j](https://doi.org/10.1039/c3cc43432j).
- [228] Zheng P, McCarthy TJ. A surprise from 1954: siloxane equilibration is a simple, robust, and obvious polymer self-healing mechanism. *J Am Chem Soc* 2012;134:2024–7. doi:[10.1021/ja2113257](https://doi.org/10.1021/ja2113257).
- [229] Yi B, Wang S, Hou CS, Huang X, Cui JX, Yao X. Dynamic siloxane materials: from molecular engineering to emerging applications. *Chem Eng J* 2021;405:127023. doi:[10.1016/j.cej.2020.127023](https://doi.org/10.1016/j.cej.2020.127023).
- [230] Schmolke W, Perner N, Seiffert S. Dynamically cross-linked polydimethylsiloxane networks with ambient-temperature self-healing. *Macromolecules* 2015;48:8781–8. doi:[10.1021/acs.macromol.5b01666](https://doi.org/10.1021/acs.macromol.5b01666).
- [231] Wu X, Yang X, Yu R, Zhao XJ, Zhang Y, Huang W. A facile access to stiff epoxy vitrimers with excellent mechanical properties via siloxane equilibration. *J Mater Chem A* 2018;6:10184–8. doi:[10.1039/c8ta02102c](https://doi.org/10.1039/c8ta02102c).
- [232] Yu CY, Ju PF, Wan HQ, Chen L, Li HX, Zhou HD, Chen JM. POSS-grafted PAI/MoS₂ coatings for simultaneously improved tribological properties and atomic oxygen resistance. *Ind Eng Chem Res* 2019;58:17027–37. doi:[10.1021/acs.iecr.9b02439](https://doi.org/10.1021/acs.iecr.9b02439).
- [233] Tang DH, Li HG, Gu HY, Lv SB, Ma JY, Zhang YZ, Song LX. Porous silica coating with excellent atomic oxygen protection performance and flexibility. *Surf Coat Tech* 2022;447:128840. doi:[10.1016/j.surfcoat.2022.128840](https://doi.org/10.1016/j.surfcoat.2022.128840).
- [234] Zhang Y, Xu H, Wang H, Wu R, Jin BQ, Wu JR. Atomic oxygen resistance vitrimers with high strength, recyclability, and thermal stability. *ACS Appl Mater* 2022;4:5152–60. doi:[10.1021/acsapm.2c00659](https://doi.org/10.1021/acsapm.2c00659).
- [235] Lei XF, Chen Y, Zhang HP, Li XJ, Yao P, Zhang QY. Space survivable polyimides with excellent optical transparency and self-healing properties derived from hyperbranched polysiloxane. *ACS Appl Mater Interfaces* 2013;5:10207–20. doi:[10.1021/am402957s](https://doi.org/10.1021/am402957s).
- [236] Zheng HL, Liu GJ, Nienhaus BB, Buddingh JV. Ice-shedding polymer coatings with high hardness but low ice adhesion. *ACS Appl Mater Interfaces* 2022;14:6071–82. doi:[10.1021/acsami.1c23483](https://doi.org/10.1021/acsami.1c23483).
- [237] Kuang J, An X, Huang C, Chen H, Wei C. Design and preparation of durable anti-icing polysilazane coatings with abrasion and UV resistance. *J Appl Polym Sci* 2022;140:53416. doi:[10.1002/app.53416](https://doi.org/10.1002/app.53416).
- [238] Wu XH, Chen Z. A mechanically robust transparent coating for anti-icing and self-cleaning applications. *J Mater Chem A* 2018;6:16043–52. doi:[10.1039/c8ta05692g](https://doi.org/10.1039/c8ta05692g).
- [239] Wu XH, Tang YX, Silberschmidt VV, Wilson P, Chen Z. Mechanically robust transparent anti-icing coatings: roles of dispersion status of titanate nanotubes. *Adv Mater Interfaces* 2018;5:1800773. doi:[10.1002/admi.201800773](https://doi.org/10.1002/admi.201800773).
- [240] Maeda T, Hamada T, Tsukada S, Katsura D, Okada K, Ohshita J. Antifogging hybrid materials based on amino-functionalized polysilsesquioxanes. *ACS Appl Polym Mater* 2021;3:2568–75. doi:[10.1021/acsapm.1c00155](https://doi.org/10.1021/acsapm.1c00155).
- [241] Maeda T, Sugimoto T, Hamada T, Katsura D, Mineoi S, Ohshita J. Highly durable antifogging materials based on polysilsesquioxane with double hydrophilic groups: effect of bridged tetraethylene glycol chains in polysilsesquioxane films. *ACS Appl Polym Mater* 2022;4:7599–606. doi:[10.1021/acsapm.2c01247](https://doi.org/10.1021/acsapm.2c01247).
- [242] Umoren SA, Solomon MM. Protective polymeric films for industrial substrates: a critical review on past and recent applications with conducting polymers and polymer composites/nanocomposites. *Prog Mater Sci* 2019;104:380–450. doi:[10.1016/j.pmatsci.2019.04.002](https://doi.org/10.1016/j.pmatsci.2019.04.002).
- [243] Bescher E, Mackenzie JD. Sol-gel coatings for the protection of brass and bronze. *J Sol-Gel Sci Technol* 2003;26:1223–6. doi:[10.1023/A:1020724605851](https://doi.org/10.1023/A:1020724605851).
- [244] Bakhshandeh E, Jannesari A, Ranjbar Z, Sobhani S, Saeb MR. Anti-corrosion hybrid coatings based on epoxy-silica nano-composites: toward relationship between the morphology and EIS data. *Prog Org Coat* 2014;77:1169–83. doi:[10.1016/j.porgcoat.2014.04.005](https://doi.org/10.1016/j.porgcoat.2014.04.005).
- [245] Chen YX, Zhang GL, Zhang GZ, Ma CF. Rapid curing and self-stratifying lacquer coating with antifouling and anticorrosive properties. *Chem Eng J* 2021;421:129755. doi:[10.1016/j.cej.2021.129755](https://doi.org/10.1016/j.cej.2021.129755).
- [246] Cheng YH, Wu CT, Hu LH. Dual functional low surface energy coating of anti-corrosion/fouling via crosslinking polysilazane preceramic precursor incorporated with fluorine. *Prog Org Coat* 2023;177:107409. doi:[10.1016/j.porgcoat.2023.107409](https://doi.org/10.1016/j.porgcoat.2023.107409).
- [247] Khranov AN, Voevodin NN, Balbyshev VN, Donley MS. Hybrid organo-ceramic corrosion protection coatings with encapsulated organic corrosion inhibitors. *Thin Solid Films* 2004;447–448:549–57. doi:[10.1016/j.tsf.2003.07.016](https://doi.org/10.1016/j.tsf.2003.07.016).
- [248] Raps D, Hack T, Wehr J, Zheludkevich ML, Bastos AC, Ferreira MGS, Nuyken O. Electrochemical study of inhibitor-containing organic-inorganic hybrid coatings on AA2024. *Corros Sci* 2009;51:1012–21. doi:[10.1016/j.corsci.2009.02.018](https://doi.org/10.1016/j.corsci.2009.02.018).
- [249] Adnan MM, Dalod ARM, Balci MH, Glaum J, Einarsrud MA. In situ synthesis of hybrid inorganic-polymer nanocomposites. *Polymers (Basel)* 2018;10:1129. doi:[10.3390/polym10101129](https://doi.org/10.3390/polym10101129).
- [250] Peng CH, Huang A, Ma XJ, Zhong JH, Chen GR, Luo WA, Zeng BR, Yuan CH, Xu YT, Dai LZ. Transparent and flame-retardant hybrid protective coating with high surface hardness, yet foldability. *Prog Org Coat* 2023;175:107346. doi:[10.1016/j.porgcoat.2022.107346](https://doi.org/10.1016/j.porgcoat.2022.107346).
- [251] Zhang YB, Yan HX, Feng GP, Liu R, Yang KM, Feng WX, Zhang SY, He C. Non-aromatic Si, P, N-containing hyperbranched flame retardant on reducing fire hazards of epoxy resin with desirable mechanical properties and lower curing temperature. *Compos Pt B-Eng* 2021;222:109043. doi:[10.1016/j.compositesb.2021.109043](https://doi.org/10.1016/j.compositesb.2021.109043).